ARINC

SOFTWARE DATA LOADER USING ETHERNET INTERFACE

ARINC REPORT 615A-2

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FOREWORD

Activities of AERONAUTICAL RADIO, INC. (ARINC)

and the

Purpose of ARINC Reports and Specifications

Aeronautical Radio, Inc. is a corporation in which the United States scheduled airlines are the principal stockholders. Other stockholders include a variety of other air transport companies, aircraft manufacturers and non-U.S. airlines.

Activities of ARINC include the operation of an extensive system of domestic and overseas aeronautical land radio stations, the fulfillment of systems requirements to accomplish ground and airborne compatibility, the allocation and assignment of frequencies to meet those needs, the coordination incident to standard airborne communications and electronics systems and the exchange of technical information. ARINC sponsors the Airlines Electronic Engineering Committee (AEEC), composed of airline technical personnel. The AEEC formulates standards for electronic equipment and systems for the airlines. The establishment of Equipment Characteristics is a principal function of this Committee.

It is desirable to reference certain general ARINC Specifications or Report which are applicable to more than one type of equipment. These general Specifications and Reports may be considered as supplementary to the Equipment Characteristics in which they are referenced. They are intended to set forth the desires of the airlines pertaining to components and general design, construction and test criteria, in order to insure satisfactory operation and the necessary interchangeability in airline service. The release of a Specification or Equipment Characteristics should not be construed to obligate ARINC or any airline insofar as the purchase of any components or equipment is concerned.

An ARINC Report (Specification or Characteristic) has a twofold purpose, which is:

(1) To indicate to the prospective manufacturers of airline electronic equipment the considered opinion of the airline technical people, coordinated on an industry basis, concerning requisites of new equipment, and

(2) To channel new equipment designs in a direction which can result in the maximum possible standardization of those physical and electrical characteristics which influence interchangeability of equipment without seriously hampering engineering initiative.

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1.0 INTRODUCTION

1.1 Purpose of this Report

This document provides general and specific design guidance for the development of software data loading equipment for all types of aircraft. It defines a Portable Data Loader (PDL) for carry-on applications and an Airborne Data Loader (ADL) for installation on the airplane. The primary function of the data loader is uploading data and operational software programs to airborne computers. A secondary function of the data loader is downloading data from airborne computers. This document describes the desired capability of data loading equipment and the standards necessary to ensure interchangeability of the standardized mass storage medium. Form and fit interchangeability is specified and expected for the ADL, but not specified for the PDL.

This document defines data loaders designed to load new generation avionics equipment over a high-speed interface. Avionics equipment, which is loadable over the ARINC 429 data bus, will continue to be loaded by data loaders defined by ARINC Report 615. However, the ARINC Report 615A definition should not preclude the combination of both data loading types in one physical data loader should the market desire such a data loader.

Equipment manufacturers should note that this document aims to encourage them to produce maintenance-free high performance equipment rather than that of minimum weight and dimension. They are at liberty to accomplish this objective by means of design techniques they consider to be the most appropriate, as their customers are interested primarily in the end result rather than the means employed to achieve it.

1.2 Document Conventions

ARINC documents are voluntary standards intended to ensure interoperability of equipment independent of the manufacturer or the airframe installation. Some data loader functions defined in this document must be implemented in order to meet a minimum level of compatibility between loaders (and other tools) designed to this standard, loadable software designed to this standard, and software transport media designed to this standard.

In this document, “should” is used to define a capability that must be implemented for the unit to meet the minimum level of compatibility intended by this standard and “does/is/will” are used to express a statement of fact based on other requirements. “May” is used to express an optional capability. In cases where these capabilities are implemented, it “should” be implemented in a specific manner defined in this standard. Otherwise an incompatibility may exist with the aircraft or other interfacing equipment.

1.3 Function of the Equipment

The data loaders (both PDL and ADL) should be able to transfer data between a selected airborne computer and a standard removable media. These transfers should occur via an Ethernet data bus defined in ARINC Specification 646. The transfer of data from the data loader to the airborne computer is referred to as “uploading.” The transfer of data from the airborne computer to the data loader is referred to as “downloading.” Both upload and download capability is desired for the PDL and the ADL.

Airline operators desire a data loader that is capable of supporting the new generation of avionics equipment that requires high-speed loading, due to the increased size of operational programs and data being loaded. The data loader should support new generation avionics equipment that is foreseeable over the operational life of the airplane. Some airline operators may install both new generation and legacy generation equipment requiring loading by ARINC 615A and ARINC 615. The ARINC 615A definition should not preclude the combination of both data loading types in one physical data loader should the market desire such a data loader.

Protocol file names are case sensitive. Refer to Section 5.1. | c-1

COMMENTARY

The airline user community set the desired operational capability for data loading. The following specifications should be viewed as minimum capabilities. Airlines welcome additional improvements providing they are cost-effective.

The goal is to load any LRU within 15 minutes.

Multiple units should be loadable with one operator action.

Hard-wired selection switches are undesirable.

Operator interaction should be limited to the installation of the physical media, selection of file(s) and verification that the operation is complete.

1.4 Unit Description

In this document all references to the “data loader” should be interpreted as meaning both the PDL and the ADL.

1.4.1 PDL Description

The PDL should be a portable unit housing all of the components, electronic circuitry, controls, etc., incident to the functioning of the system within an easily transportable instrument case. All control and status indications should be available on the front panel of the PDL. This unit is intended for use on all avionics equipment with embedded software.

The PDL is expected to perform the data loading functions defined in this document. However, some airline operators may desire a more elaborate portable device that is capable of performing additional functions such as retrieval of maintenance data from ARINC 604 BITE systems or ARINC 624 Onboard Maintenance Systems. ARINC Report 644A provides guidelines for the development of a Portable Multi-Purpose Access Terminal (PMAT) that provides additional functionality.

1.4.2 ADL Description

The ADL should be consistent with on-board equipment features including standard aircraft instrument panel mounting features, cable connection, easy access, environmental performance and provide for straightforward

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1.0 INTRODUCTION

1.4.2 ADL Description (cont'd)

operation. The design should provide adequate protection of the medium from contaminating environments.

1.5 Interchangeability

1.5.1 Media Interchangeability

Interchangeability of the formatted medium is desired regardless of the manufacturing sources of the data loader or the media.

1.5.2 Unit Interchangeability

Interchangeability of data loader equipment is desired regardless of the manufacturing source. The medium and drive should support the format features described in Section 4.

1.6 Interoperability

Avionics equipment intending to comply with this standard should be designed to meet the electrical and functional interface characteristics of this standard.

Interoperability between ARINC 615A and ARINC 615 data loaders is desirable. On airplanes where there is a mix of ARINC 615A and ARINC 615 loadable avionics equipment, the load port should support either data loader or a combined ARINC 615A/615 data loader.

The standards contained herein may be applied to other equipment capable of data loading, though not specifically dedicated to the data loading function. When the ADL function is incorporated into other systems located on the airplane, those systems should contain the digital interfaces and protocols specified in this document and support the standardized media types.

c-1 Software loads should be compliant with ARINC Report 665, Loadable Software Standards.

1.7 Reliability

The anticipated use of the data loader demands the utmost attention to the need for reliability in all phases of design, production, installation and operation of the equipment.

COMMENTARY

> The designer may be surprised to find no elaborate requirements for reliability. The airlines are in a most fortunate position in this regard because they have found the pressures of the "marketplace" exert a truly meaningful influence upon the design and production quality control necessary to achieve high equipment reliability.

1.8 Maintainability

The equipment design should be modular in concept and should be assembled to facilitate shop repair. The stacking of individually replaceable components should be avoided. The spacing of components should facilitate shop repair and construction should permit module or component removal without interference. All modules or subassemblies should be interchangeable with other sub-units bearing the same part number. Parts replacement should only be necessary when the part fails to meet the minimum operational requirements.

1.9 Regulatory Approval

The equipment should meet all applicable Federal Aviation Administration (FAA), European Joint Aviation Authority (JAA), Federal Communications Commission (FCC), and any other airline regulatory requirements. This document does not and cannot set forth the specific requirements that equipment must meet to assure approval. Such information should be obtained from the regulatory agencies themselves.

1.10 Related Documents

The latest version of the following documents apply:

RTCA DO-160, Environmental Conditions and Test Procedures for Airborne Equipment

RTCA DO-178, Software Considerations in Airborne Systems and Equipment Certification

EUROCAE ED-12, Software Considerations in Airborne Systems and Equipment Certification

EUROCAE ED-14, Environmental Conditions and Test Procedures for Airborne Equipment

European Computer Manufacturer's Association (ECMA), Standard 267 – 120 mm DVD-Read Only Disk c-1

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2.0 INTERCHANGEABILITY STANDARDS

2.1 Introduction

This section sets forth the physical criteria, input and output interfaces, and power supply characteristics desired for the Portable Data Loader (PDL) and Airborne Data Loader (ADL).

Manufacturers should note that, although this Report does not preclude the use of different interface and interwiring features, the practical problem of redesigning to accommodate some special system could very well make the use of that other design prohibitively expensive for the customer. They should recognize, therefore, the practical advantages of developing equipment in accordance with the standards set forth in this document.

Avionics equipment intending to meet these interfaces standards must comply with Sections 5 and 6 of this document concerning the data loading protocol and bus interface.

2.2 Form Factor and Connectors

2.2.1 Physical Size

2.2.1.1 PDL Physical Size

The PDL should be designed to be a portable flight line piece of test equipment. Although no restrictions have been made on the actual size of the PDL, the users desire that this unit be as small as possible. Shock absorbing bumpers or rails could be an attractive feature to users concerned about durability. All accessory components such as cables and disks should be carried as an integral part of the PDL completely contained within the packaging unit.

2.2.1.2 ADL Physical Size

The ADL should be designed for commercial aircraft instrument panel mounting height and width dimensions found in MS25212. Panel height should not exceed 4.5 inches maximum including power supply. Any vibration/shock isolation features needed to meet this application should be contained within the enclosure and not require any special alignment for installation. ADL length should be limited to 7.0 inches maximum from rail mounting surface of front panel to the rear of the chassis excluding connector protrusion.

COMMENTARY

Although MS25212 limits instrument depth to 6.5 inches, the ADL may exceed this value to a limit of 7.0 inches. Though undesirable, the manufacturers believe that this depth is necessary to house the disk drive. If you can make it shorter, by all means do it.

2.2.2 Data Loader Weight

2.2.2.1 PDL Weight

The packaged weight of the PDL should not exceed 17.6 pounds (8 kg). The weight should be distributed as evenly as possible so as to provide for convenient transportation by one person using only one hand.

2.2.2.2 ADL Weight

The packaged weight of the ADL should not exceed 6.6 pounds (3 kg).

2.2.3 Controls and Indicators

2.2.3.1 PDL Controls and Indicators

The PDL operator's interface should be via a control panel, which is an integral part of the unit. The control panel should be designed for ease of use and should contain all switches necessary to control system functions and all indicators necessary to determine system and functional status.

2.2.3.2 ADL Annunciator Lights

The ADL should contain a method of displaying the status of the current operation. Any indicators should not distract the flight deck crew during flight.

COMMENTARY

There are conflicting views as to whether or not operational status indicators should appear on the panel of the ADL. Some indicated that with the MCDU interface, ADL status indicators are redundant and could be distracting to operators. For applications where the MCDU interface is not implemented, however, status indicators were felt to be necessary for operator usage. The solution to this problem recommended herein is to mount the status indicators behind a movable panel (e.g., the disk drive access door) so that they will be visible only with the door open.

2.2.4 Drive/Interface Mechanisms

The drive mechanism and media loading features are defined in Section 4 of this document.

2.2.5 Connector

2.2.5.1 PDL Connector

The placement of signals on the connector pins should be backward compatible with ARINC Report 615 to allow manufacturers to supply a combined ARINC 615A/615 data loader in one physical package if desired.

The interface connector mounted on the PDL should be a MS27508E-18A-53P or equivalent. The connector should be mounted on the front (control) panel of the PDL. Refer to Attachment 1 for connector pinout definition.

2.2.5.2 ADL Connector

The placement of signals on the connector pins should be backward compatible with ARINC Report 615 to allow manufacturers to supply a combined ARINC 615A/615 data loader in one physical package if desired.

The ADL interface connector should be an MS27508E-18A-53P mounted on the rear panel of the unit. Refer to Attachment 1 for connector pinout definition.

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2.0 INTERCHANGEABILITY STANDARDS

2.2.6 Data Bus Interface

The interface to the software data loader is an Ethernet interface defined by ARINC Specification 646.

COMMENTARY

Data loader and target hardware devices are expected to support 10BaseT connections. If a data loader or device supports higher speed connections (e.g., 100BaseT), then it must be able to automatically fallback to a lower speed when connected to a slower device.

2.3 Power Circuitry

2.3.1 Primary Power Input

The data loader should be able to operate normally in the range of 104 to 122 Vac (RMS), 380 to 420 Hz. The data loader should not be damaged by a power input in the range of 0 to 97 Vac (RMS) for a period of up to ten seconds. Additional guidance on power input is provided in ARINC Report 609, "Design Guidance for Aircraft Electrical Power Systems."

The data loader should operate with no degradation in its specified performance through a power interruption of 20 millisecond duration.

When a power interruption occurs for a period equal to or greater than 20 milliseconds, the unit should either continue to operate with no degradation, or if the data transfer process is affected, the data loader should reinitialize in accordance with the normal power-up sequence.

2.3.1.1 PDL Primary Power Input

The PDL should be designed to operate from aircraft-supplied 115 Vac, 400 Hz power. Power dissipation should not exceed 30 watts.

COMMENTARY

Provision of other power options such as 115 Vac, 50 to 60 Hz; 230 Vac, 50 to 60 Hz and +28 Vdc are open to the marketplace.

The data loader will interface with numerous types of equipment in various installation configurations. In order to eliminate adverse effects that may result from voltage and phase differences in multiple power sources, the data loader interwiring installer is urged to provide power to the data loader from the same power source as the equipment being loaded.

However, this may not always be practical, therefore, the data loader manufacturers are urged to build their equipment to be insensitive to such voltage and phase differences.

For units requiring heaters, power dissipation should not exceed 100 watts during heater operation. The selection of a disk drive needing a heater to extend low temperature performance is discouraged. In the event a heater is used, precautions should be taken to prevent fires or ignition of explosive gases.

COMMENTARY

It seems sensible that PDL units delivered to customers whose operations are limited to mild climates would not need a heater. Thus, equipment manufacturers are encouraged to advise the user as to the desirability, or undesirability, of this feature for his particular application.

2.3.1.2 ADL Primary Power Input

The ADL design should operate from 115 Vac, 400 Hz power supplied from the aircraft power bus. Power dissipation should not exceed 10 watts. The ADL power supply will be compatible with a RTCA DO-160, Section 16, Category A power source.

COMMENTARY

Provision of other power options such as 115 Vac, 50 to 60 Hz; 230 Vac, 50 to 60 Hz and +28 Vdc are open to the marketplace.

As a permanently installed peripheral device to on-board computer systems, the ADL must consider size, weight, and heat dissipation.

2.3.2 Power Control Circuitry

The data loader should include automatic control features which prevent inadvertent upset of previously recorded data in the event of normal aircraft power cycling or power interrupts. The PDL should have a guarded power switch designed to prevent inadvertent power removal from the unit during the data transfer operation. The ADL should not require a separate power switch.

2.3.3 Internal Circuit Protection

2.3.3.1 PDL Internal Circuit Protection

The PDL should contain over-current protection for the electronic circuits. If an internal heater is employed, heater power should be supplied through a separate circuit with appropriate over-current protection. The current rating of each circuit protection device should be clearly identified. If a fuse is employed in the design, spare fuses should be provided in an easily accessible storage location in the PDL.

2.3.3.2 ADL Internal Circuit Protection

The ADL should include internal circuit protection features, which do not require the use of fuses. The ADL should sustain operation without damage or disturbance to previously recorded data when operated from a RTCA DO-160, Section 16, Category A power source, including voltage spikes per Section 17, Category A. The power supply should include overvoltage, undervoltage and overload protection.

2.4 Environmental Conditions

The design of the data loader should be suitable for airline operation while meeting specified performance during and/or after exposure to the minimum environmental conditions listed in Attachment 5. Although the ADL can be expected to operate in a Category X environment per

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2.0 INTERCHANGEABILITY STANDARDS

RTCA DO-160, Sections 10, 11, 12, and 14, measures should be taken to limit the potential for contaminant entry through the media access opening.

COMMENTARY

The document used for the environmental test of airborne equipment is RTCA Document DO-160, "Environmental Conditions and Test Procedures for Airborne Equipment." Although the PDL is not airborne equipment and RTCA DO-160 is not directly applicable, the PDL will be operating in an environment very similar to that incurred by airborne equipment. Therefore, many of the test procedures in RTCA DO-160 for the data loader can be used exactly as provided. Other RTCA DO-160 test procedures are useful as guidance material for determining the test procedures needed to validate PDL operation in environmental conditions differing from those of the airborne unit.

2.4.1 Temperature

The data loader should be able to withstand a non-operating temperature range of -55° C to +85° C. The medium is expected to withstand non-operating temperatures within the range of -30° C to +70° C.

2.4.1.1 PDL Temperature

The PDL should be designed to operate in the temperature range of -15° C to +55° C.

COMMENTARY

Use of an internal fan for air circulation or a heater for warming is discouraged for the PDL. If used, manufacturers are urged to use fans that generate a minimal amount of noise.

2.4.1.2 ADL Temperature

The ADL should be designed to operate over the temperature range of RTCA DO-160, Category A1 equipment (-15° C to +55° C, +70° C short term) without the need for internal heaters or cooling air. Medium tracking compatibility which includes writing and reading at opposite temperature extremes without excessive data loss or upset of previously recorded data should be provided by the drive design.

2.4.2 Pressure

Both the ADL and PDL should be designed to operate at any pressure altitude ranging from -1000 feet to 15,000 feet. Further, the ADL should operate at up to 40,000 feet for as much as 15 minutes.

COMMENTARY

Although 40,000 feet is specified here, the marketplace will be expecting performance up to "the typical ceiling of the aircraft."

2.4.3 Rain

The PDL should exhibit no physical deterioration due to exposure to a moderate rain with the cover properly installed. The PDL should operate with no degradation in its specified performance immediately after removal from the rain exposure.

2.4.4 Solvent Resistance

The PDL control panel and interface cable should be resistant to the following fluids which may come in contact with the equipment during normal airline service:

a. Engine oil per MIL-L-7808, MIL-L-23699, and P&WA Specification 521B, Type I and Type 2

b. Hydraulic test fluid

c. Jet engine fuel

d. Windshield rain repellent

e. De-icing fluid

2.4.5 Vibration

The ADL should function without error, damage to medium, or upset of previously recorded data while operating in either write or read modes during vibration requirements of RTCA DO-160, Section 8, Category K, fixed wing aircraft. The unit should meet these requirements when instrument panel, console, or equipment rack (non-isolated) mounted in any of the normal mounting planes.

2.4.6 Shock/Handling

2.4.6.1 PDL Shock/Handling

The PDL should be capable of withstanding a drop of 30 inches in which the point(s) of impact can be corners or any surface of the PDL case. The PDL should not sustain any damage that would preclude meeting the operation and environmental conditions of this Report.

COMMENTARY

The importance of building an abuse resistant unit cannot be overly stressed. Unfortunately, an extremely well built and sturdy unit could easily become too heavy to be a truly portable unit.

2.4.6.2 ADL Shock/Handling

The ADL should function without unrecoverable error, damage to medium, or upset of previously recorded data while operating in either write or read modes during a shock impact to any of the major axis as described in RTCA DO-160, Section 7 (6 Gs, 11 msec). The ADL should withstand normal bench handling shock without damage. The ADL should remain attached to its mounting when subjected to the crash safety requirements of RTCA DO-160, Section 7.

2.4.7 Electromagnetic Compatibility

2.4.7.1 PDL Electromagnetic Compatibility

The PDL should be designed to operate in the presence of strong electromagnetic fields. The supplier should use

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2.4.7.1 PDL Electromagnetic Compatibility (cont'd)

design and fabrication techniques which will minimize the effect of such fields on the normal operation of the unit.

2.4.7.2 ADL Electromagnetic Compatibility

The ADL power supply should support operation as Category A equipment per RTCA DO-160, Sections 18, 19, 20 and 21. Under no circumstances should electromagnetic interference result in permanent damage or upset of previously recorded data.

2.4.8 Dust

The data loader should exhibit no physical deterioration and should operate with no degradation to its specified performance during and following exposure to a dirty atmosphere. The diskette access opening should be fitted with a cover door and gasket to minimize the opportunity for dust and other contaminants to enter the mechanism.

2.4.9 Explosive Atmosphere

The data loader should be designed to minimize the potential for ignition of explosive gases. RTCA DO-160, Section 9, Environment III, Category X applies.

2.4.10 Fungus Resistance

The data loader design should minimize the use of fungus nutrient materials. RTCA DO-160, Section 13, Category X environment applies.

2.5 Grounding

The data loader interface connector shell should be electrically bonded to chassis ground. The ADL panel mounting features should include bonding surfaces which maintain chassis ground continuity between the ADL chassis and the aircraft mating surfaces.

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3.0 UNIT DESIGN

3.1 Introduction

This section provides data loader design guidance that enhances the usability of the data loader in real life operating conditions and environments. Key design goals are:

a. To make the controls easily operable in all environmental conditions, such as temperature extremes, lighting extremes and operator weather protection clothing.

b. To make the data loading operation intuitive and unambiguous by providing task and context appropriate feedback in the form of messages or indicators, and providing continuous feedback about the progress of the load.

c. To provide data loader fault reporting in an easily understandable form, with the intent of aiding the operator in isolating the source of a data loading problem.

d. To minimize shop maintenance time.

3.2 Controls

3.2.1 Media Ejector

The data loader should have a media ejection capability that allows the operator to remove the media cartridge without a special tool. The ejector implementation should operate independent of power being applied to the unit. Insertion and ejection of the media should be mechanized in a manner, which provides the operator straightforward operation and positive assurance that the media has been properly installed or ejected. The design should be accomplished in a way that minimizes inadvertent ejection.

3.2.2 Selector Controls

As a minimum, the data loader should provide controls to perform the following functions:

a. Mode Selection

b. Source Medium Selection

c. Load Selection

d. Load Destination Selection

e. Accept/Continue

f. Cancel/Abort

g. Target Hardware Identification

c-2 The Target Hardware Identification function should be implemented via FIND. Refer to Attachment 3.

3.3 Indicators/Displays

The data loader should have a display as part of the integral control panel. The display should be easily readable in both direct sunlight and low ambient light conditions. Indicators may also be included. A display/indicator test should be performed automatically during the data loader self-test procedure. The display/indicators on the ADL should be hidden from view during flight. The display/indicators should provide the capabilities described in the following sections.

As a minimum, the display should be capable of displaying 15 alphanumeric upper case characters, plus the hyphen (-). All letters and numbers must be distinguishable from each other (e.g., “2” versus “Z” or “0” versus “Q”).

Indicators/displays with a limited life span (e.g., incandescent lamps) should be easily replaceable without disassembly of the unit.

COMMENTARY

Since the portable data loader is expected to operate in a wide range of temperatures, manufacturers are urged to select appropriate indicators/displays for the unit. The indicators/displays should operate over the entire operating temperature range specified.

The ADL indicators should be out of sight during flight operation to avoid distracting the crew. The indicators should, however, be easily accessible. It is acceptable to place indicators behind the media access door.

At all times, the operator should be able to determine the following:

a. Loader is on and operating (activity and/or progress indication)

b. Loader mode of operation (upload, download, configuration verification, etc.)

c. In the event of a failure, (Loader, Target, Media, etc.) it should be apparent that the failure exists. The appropriate actions to resolve the failure should be annunciated.

d. Pending operator action (e.g., select target, insert next disk, etc.)

e. Status of current operation (e.g., Ready, Complete, In Progress)

3.4 Legends and Instructions

All function selectors and indications on the data loader control panel should be clearly and unambiguously annotated. The size of the lettering and the display (if provided) of the PDL should be such that an operator with normal vision can read the information from a distance of 24 inches. The size of the lettering and the display (if provided) of the ADL should be such that an operator with normal vision can read the information from a distance of 18 inches.

COMMENTARY

The actual procedures for operation of the data loader may vary from user to user. It is expected that users will have their own set of instructions. However, the manufacturers are urged to provide as much help as

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3.0 UNIT DESIGN

3.4 Legends and Instructions (cont'd)

possible to an operator without a manual at hand. It is expected that a basic set of instructions mounted on the equipment will be provided. The detail of these instructions should be of sufficient detail to preclude incorrect or hazardous use of the equipment.

3.5 Self-Load Capability

The data loader should be loadable from its local drives. As an option, it may be loadable from the Ethernet interface. Its software configuration should be displayed in the same manner as other ARINC 615A-compatible units. This includes conformance to ARINC Report 665, Loadable Software Standards.

c-1

3.5.1 Equipment Data Base

The data loader should be able to load the airplane-specific data base of loadable target hardware and a fleet-specific list of equipment names and their logical addresses.

3.5.2 Operational Software

The data loader may include the capability to update its own operational program.

3.6 Special Physical Provisions

3.6.1 Media Access Opening

The media access opening should be designed to preclude damage to the data loader or the media cartridge by an attempt to insert the media cartridge improperly.

COMMENTARY

The users desire a media access opening designed to prevent damage to the data loader and the media cartridge when the unit is exposed to the environmental conditions described in Section 2.4.

3.6.2 Cable Size and Terminations

The PDL should provide space for storing the standard interface cable. The standard PDL interface cable should be 10 feet in length (3 meters) and be terminated with a connector type MS27473T-18A-53S at the data loader end. It should use connector type MS27473T-18A-53P (or MS27473T-18B-53P) at the aircraft/airborne computer end. See Attachment 1 for interwiring assignments.

3.6.3 Internal Heater

Internal heater elements are discouraged, but may be supplied as part of the PDL. If provided, they should be thermostatically controlled.

3.7 Testing

The data loader should be capable of two modes of internal testing:

a. Self-test exercised automatically at the time of data loader power-up

b. Maintenance diagnostic test

3.7.1 Autonomous Self-Test

The data loader should have the capability to exercise a self-test. The self-test sequence should validate the ability of the unit to perform its intended function. The self-test should be executed at the time of power-up and whenever so directed by receipt of a self-test command.

As a minimum the self-test should involve an indicator test, an instruction set exercise for the processor used, a RAM memory check, and a check on PROM-based program content. If the unit uses programmable input/output (I/O), then a test should be run to validate the I/O.

3.7.2 Maintenance Diagnostic Test

A maintenance test media capable of a thorough test of the data loader should be provided by the supplier. When the maintenance test media is installed, the data loader should run a specialized test of the unit.

The maintenance diagnostic tests should, as a minimum, distinguish between failures of the media drive, media controller, interface electronics and media. The data loader should include signal wrap-around capability at the Ethernet interface to allow as nearly a complete validation as is feasible with a minimum of additional test equipment. The test coverage should be 90% or better.

The method used to annunciate maintenance test failures is not defined herein, and may be implemented at the manufacturer's discretion.

COMMENTARY

The maintenance media is intended for detailed troubleshooting in the shop. Alternatively, it may be used for engineering analysis on-board the aircraft. The internal Self-Test of the data loader is expected to be adequate for all routine line maintenance functions.

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4.0 SOFTWARE TRANSPORT MEDIA TYPE

4.1 Media Interchangeability

Three different media types should be provided:

3.5 inch floppy disc

PC Card

Optical Disc with a diameter of either 12 or 8 centimeters (cm).

c-1

A mix of any three media types for a single data load is not supported by the data loader.

COMMENTARY

A 3.5 inch floppy disc may be used when the data will fit on a single disc. Larger data loads may consider the use of an optical disc or a PC Card device. In this way, airlines will handle one medium.

Use of the floppy disc media for concurrent disk loads is not recommended.

c-1

Implementers are not recommended to perform concurrent parallel loads with floppy discs, to prevent the operator from swapping discs.

The media should be compliant with ARINC Report 665.

4.1.1 3.5 Inch Floppy Disk

The storage medium will be a 3.5 inch double-sided high-density (HD) magnetic diskette with 1.44 Megabytes of storage capacity. Interchangeability of the formatted medium is desired regardless of the manufacturing sources of the data loader or the diskettes.

4.1.1.1 Drive Mechanism

The drive mechanism and media loading features should operate reliably when mounted normally in any of the three primary axes, and exposed to typical commercial aircraft environments.

This disk drive mechanism should accept a 3.5 inch double-sided, high-density (HD) magnetic disk cartridge. The mechanism should be able to read and write to the formatted medium.

4.1.1.2 Physical Format

The data on the magnetic double-sided disk should be modified frequency modulation (MFM) encoded on 80 track/cylinder locations providing an unformatted storage capacity of 2 million bytes. The medium includes the standard features associated with the 3.5 inch hard-shell diskette including form factor, write protect, access shutter, chucking hub, etc. The MFM encoding, track preamble/postamble, gaps, sector ID field and data field, erasure direction, etc., should be in accordance with the ANSI standard format for high density (2 megabyte) 3.5 inch floppy disk media. Physical diskette recorded format characteristics are listed below for reference. They include the following:

Number of Cylinders	80
Number of Tracks	160
Number of Surfaces	2
Recording Density	17434 BPI (Side 1, TK 79)
Recording Method	MFM
Recording Range	22.428 mm to 41.961 mm (Side 0)
	21.666 mm to 40.970 mm (Side 1)
Track Pitch	0.1875 mm TK-TK

4.1.1.3 Logical Format

The logical format for the 3.5 inch floppy disk is defined in ARINC Report 615A.

4.1.2 PC Card Device

The storage medium will be a PC Card Standard. The size of this card may be type 1, type 2 or type 3 (as defined in the PC Card definition). Technology of this card is not specified herein. (Flash disk, rotational hard disc and other technologies are acceptable.) Card technology and size are dependent on the market. The PC Card should be compatible with the PC Card Slot Standard. PC Card memory extension is not supported by this standard. The use of the PC Card for uploads is not supported by this standard. c-1

COMMENTARY

The airlines do not want PC Cards to be used for uploads to avionics equipment. However, for non-commercial operations, the operator could perform functions which are not defined in this report. This might include erasure and “zeroizing” media, and other functions viewed to be desirable.

4.1.2.1 PC Card Interface

The AT Attachment (ATA), a storage interface standard developed by the National Committee on Information Technology Standards (NCITS) and accredited by the American National Standards Institute (ANSI), is utilized as the disc drive interface on most personal and mobile computers. Hardware drives and software drivers required for this interface should ensure compatibility with the PC Card ATA specification. The PC Card interface should accept type 1, 2 and 3 PC Cards. c-2

To insure PC Card compatibility, the information in the Touples of the CIS (Cartridge Information Section) should be used to build a common table of access characteristics. Care should be taken in determining the power of the cartridge: some use 3.3 volts power, others use 5 volts power. Power information of the cartridge should be given by its hardware wiring and/or by a specific Touple from CIS. c-2

4.1.2.2 Logical Format

The logical format for the PC Card device is defined in ARINC Report 615A.

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4.0 SOFTWARE TRANSPORT MEDIA TYPE

4.1.3 Optical Medium

The storage media should be a re-writable 12 cm Digital Video Disc (DVD-RAM), read-only Digital Video Disc (DVD-ROM) or Compact Disc (CD); and re-writable 8 cm DVD-RAM, DVD-ROM or CD-ROM disc. Interchangeability of the storage media is essential, regardless of the manufacturing sources of the data loader or the disc

a. 12 cm Optical Disc

The 12 cm optical disc may be recorded in the methods described below. The data loader space requirements permit access for only a single disc drive. The disc may be inverted to access additional media. The various methods of recording and their corresponding capacities are as follows:

1. Single side, single substrate Compact Disc physical format provides data density of 650 Megabytes (MB).

2. Single side, single substrate in DVD physical format provides data density of 4.7 Gigabyte (GB).

3. Double substrate, double side in DVD physical format provides data density of 9.4 GB.

c-1

4. Single substrate, double layer with space between layers in DVD physical format provides data density of 8.5 GB.

5. Double substrate, quadruple layer with space between each layer in DVD physical format provides data density of 17.0 GB.

b. 8 cm Optical Disc

The 8 cm re-writable DVD-RAM, DVD-ROM or CD-ROM optical disc provides storage capacities of 1.4 GB, 1.4 GB or 180 MB per side, respectively.

4.1.3.1 Drive Mechanism

The drive mechanism should be the 12 cm DVD type. The drive should provide read capability and should not provide write capability. The drive mechanism and media loading features should operate reliability when mounted in any of three primary axes. The optical drive should be operated only for data loading during ground operations. The drive mechanism is not intended to be operated in flight. The drive mechanism should have adequate vibration resilience to operate when exposed to the typical ground handling environments of commercial aircraft when blocked or taxing.

The disc drive mechanism should accept both 12 and 8 cm optical disc. The mechanism should be able to read only and support any DVD-RAM, DVD-R, DVD-ROM, CD-R/RW, and CD-ROM formatted medium.

4.1.3.2 Physical Format

Phillips – Sony Yellow Book defines the physical format for Compact Discs used for recording computer data.

Phillips – Sony Orange Book defines the physical format for Recordable Compact Discs.

ECMA – 267, Sections 4 and 5 define the DVD optical disc physical format

4.1.3.3 Logical Format

The logical format for the 12 and 8 cm media was introduced in ARINC Report 665. c-1

4.1.3.4 Disc Replication

COMMENTARY

Commercial replicating equipment is readily available for 12 cm optical disc media. Those organizations wishing to use 8 cm discs should assess mass replication before committing to this size.

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5.0 LOAD PROTOCOL

5.1 Introduction

This section defines the necessary Software Load Protocol functions for the Data Loader and on the device being loaded (hereafter called the Target Hardware). The correct implementation of the protocol will guarantee to an equipment supplier that the device will be able to be loaded by any Data Loader that respects the Standard.

This protocol takes into account some specifics of the loading process and of the Target Hardware. In general it expects the loading process to follow the scenarios defined in Attachment 2. The protocol also expects the Target Hardware to have the following attributes:

A means to ensure that the software presented by the Data Loader has been loaded correctly before responding that the load is complete.

A TFTP service that can handle the data loading protocol.

The ability to enter a "data loading mode" if necessary, when the data loader requests a transfer.

The ability to report part numbers of the loaded software.

The basic philosophy of the load protocol is that the operator is responsible for determining which loads are presented to the Target Hardware, the Target Hardware is responsible for determining which files are required to satisfy the load request, and the Data Loader simply acts as a file server. The Data Loader and the Target Hardware exchange all data through TFTP (Trivial File Transfer Protocol). This includes both loadable data and protocol overhead data.

The loadable data is contained in files of the load, while the communications protocol data is contained in special files called protocol files. Only the protocol files are described in this section. Protocol file names should be composed of all upper case alphanumeric characters. Software Loads are defined in and should be compliant with ARINC Report 665. c-1

5.2 Scope

The Load Protocol specification defines a means to:

a. Upload Target Hardware memories.

b. Download Target Hardware memories.

c. Get configuration information from Target Hardware

d. Interrupt at any time any of the three previous operations (interruption requested by the operator or by the Target Hardware).

e. Obtain subscriber information (such as MAC address, IP address, and Target Hardware identifier.)

To this end this specification:

a. Defines the protocols for uploading, downloading and Information operations, and interruption services.

b. Describes the messages exchanged between the Data Loader Protocol and the end user.

c. Defines the protocol files exchanged between the Data Loader and the Target Hardware.

d. Defines the Simple Network Identification Protocol (SNIP), which should be used to determine which Target Hardware are present on the network and to obtain minimal information to establish communications. c-1

5.3 ARINC 615A Data Loading Architecture

5.3.1 General Description

The FIND protocol allows the data loader to dynamically identify the subscribers present on the network and to obtain for each of them the communication parameters such as MAC address, IP address, target hardware identifier, and position. This protocol is defined in Attachment 3 of ARINC Report 615A. The FIND protocol is based on UDP packets, therefore MAC addresses are supported by the applicable data link layer fields. In the same way, the applicable IP layer fields support IP addresses. c-2 c-2

5.3.2 TFTP Protocol

All the exchanges between the Data Loader and the Target Hardware are based on file transfers. Therefore, several types of files will be defined as follows:

Files of the load described in ARINC Report 665

Protocol files

The TFTP protocol is described in documents named RFC (Request for Comments) which are under the responsibility of the IAB (Internet Architecture Board). The references of this RFC are given below.

This section gives the option used by TFTP and the error messages concerning the transfer, this makes it possible to optimize the transfer of the file and the interoperability between any Data Loader and targets.

There is no maximum file size associated with each file. c-1

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5.0 LOAD PROTOCOL

c-1 5.3.2.1 TFTP Reference Documents

The applicable TFTP RFCs are:

RFC	Title	Category
1123	Requirements for internet Hosts Application and support	Standard
1350	TFTP protocol (Revision 2)	Standard
2347	TFTP option extension	Standards track, Updates 1350
2348	TFTP Blocksize Option	Standards track, Updates 1350
2349	TFTP Timeout Interval and Transfer Size Option	Standards track, Updates 1350
1785	TFTP option negotiation analysis	Informational, Updates 1350

5.3.2.2 TFTP Options

The TFTP options may be implemented to give more efficiency to the file transfer. The option mechanism provides the ability to negotiate the value of the optional field and the ability to detect that the peer computer does not support this specific option. In this case, default standard values must be used. In any case the transfer
c-2 should not fail due to a non-implemented option. After
c-1 the end of the last TFTP option, no more UDP or IP data should be present. For all TFTP used in the ARINC Report 615A protocol (that is, protocol files as well as data files), the only mode used is the "octet" mode (binary). The options are described in the following section.

5.3.2.2.1 Blocksize

c-1 Blocksize is specified by "blksize" in the frame. It indicates the number of bytes in a block. The specification for Blocksize is given in RFC 2348.

If the option is used, valid value range of the Blocksize is between 8 and 65464, as defined in the RFC 2348. The principle of TFTP option negotiation allows only the reduction of the value provided by the initiator of the TFTP session. Therefore, it is recommended that initiator gives the maximum value it can support. If one subscriber does not want IP segmentation, the maximum value of the Blocksize should be 1428.

This value is given by:

1500 bytes Max Ethernet Frame

c-2 minus 60 bytes IP layer with option

minus 8 bytes UDP layer

minus 4 bytes TFTP header

c-1 Total = 1500 – 60 – 8 – 4 = 1428

If the option is not used, the TFTP standard value of the block is 512 bytes, as defined in the RFC 1350.

5.3.2.2.2 Transfer Size of the File

Transfer Size File, specified by "tsize" in the frame, indicates the size of the file, which will be transferred. RFC 2349 provides the specification of File Transfer Size. c-1

The value is not specified.

5.3.2.2.3 Timeout Interval

Timeout interval, specified by "timeout", indicates the number of seconds to wait before re-transmitting the last packet. RFC 2349 provides the specification for Timeout Interval. c-1 c-2

The default value should be 2 seconds.

5.3.2.3 TFTP Adaptation to Data Loading Protocol c-1

5.3.2.3.1 TFTP Super-set

ARINC Report 615A protocol is based on standard TFTP protocol. However, the Load Protocol Layer requires extensions to the standard TFTP protocol to handle issues such as target hardware mode changes and operator interruptions. In order to maintain the original TFTP packet types, the flexibility of the error packet is exploited to implement ARINC Report 615A specific messages. TFTP error packets containing a zero for the error code value are defined by their error message text. By defining unique ARINC Report 615A messages as specific text in the error message text field they have meaning for ARINC Rep[ort 615A applications, but look like errors in a normal TFTP session. The TFTP super-set is defined in sections 5.3.2.3.4 Definition of the Waiting Message and 5.3.2.3.6 Definition of the Abort Message. c-2 c-1

5.3.2.3.2 Dedicated Port Number

Concerning Target Hardware, Upload operation implies a specific process, such as: erase, then program, and then check physical devices as flash memory. For this reason, suppliers of Target hardware may want to separate communication between Data-Load application and other file-transfer application. To enable such a functional switch in a simple way, a recommended solution is to use a different well-known port number. All TFTP used in the ARINC Report 615A protocol should use the well-known port number 59 (decimal), registered as "Any Private File Service." This approach implies the use of specific TFTP implementation or TFTP package allowing parameterization of the well-known, port number. c-1

5.3.2.3.3 Error Message General Description

If an error occurs during the transfer of a file, an error packet is sent to indicate the type of error. This packet is identified by the op code value of 05, followed by an error code value defined as follows:

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5.0 LOAD PROTOCOL

Error Code Value	Meaning
0	Not defined, see error message
1	File not found
2	Access violation
3	Disc full or allocation exceeded
4	Illegal TFTP operation
5	Unknown transfer ID (port number)
6	File already exists
7	No such user
8	TFTP option refused
"Error message" in NETASCII	Gives clearly the error message with ASCII character string

The error code value should be used to describe the error. The error message field should be used with the error code value of 0. The error message field may be used with error code values other than 0.

c-1

The error code value 3 should be used by the data loader to inform the target hardware that the data loader cannot open a new TFTP process because no more resources are available

c-2

The error code value 3 should also be used by the target hardware to inform the data loader that the target hardware cannot open a new TFTP process because no more resources are available.

c-1

5.3.2.3.4 Definition of the Waiting Message

For the Load Protocol to simplify the exchange between the Data Loader and the Target Hardware, a waiting message is implemented via the Error Message facility. It uses the Error Code 0. Either the Data Loader or the Target Hardware can initiate a wait message.

The format of the waiting message is: "WAIT:x" in NETASCII, where x equals the waiting time in seconds.

The unit receiving the wait message should abort the current TFTP transfer and initiate a new one after the specified time.

To clarify the format of this error, an example of the waiting error Protocol Data Unit (PDU) with a waiting time of 50 seconds is:

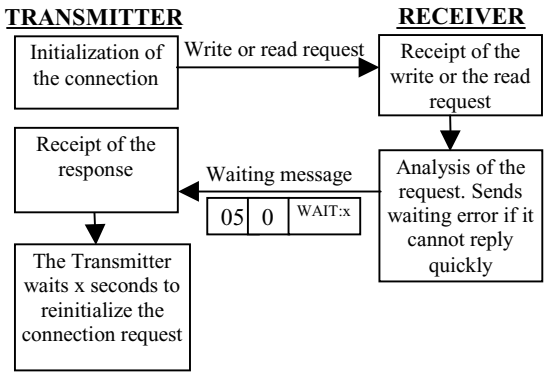
OpCode	Error Code Value	Error Message (text)	ASCII String Termination
0005_{16}	0000_{16}	"WAIT:50"	00_{16}

The maximum value for the wait delay is 65535 seconds.

c-1

5.3.2.3.5 Utilization of the Waiting Message

This waiting message is used only when a connection request (Read or Write) is received by a computer and when it cannot accept this connection because it is busy and/or it cannot reply to this request quickly. This is illustrated as follows:



5.3.2.3.6 Definition of the Abort Message

c-1

To abort the process in progress, the Data Loader Protocol sends an abort message in response to a TFTP request from the Target Hardware. It uses the Error Code value of 0 of the TFTP. The format of this message is the following: "ABORT:XXXX" in NETASCII. The XXXX string corresponds to the representation as four hexadecimal characters of the status code corresponding to the abort request by the Data Loader. This way, the Target Hardware is able to know the origin of the abort and memorize information if necessary, for example, to switch between operator abort or application abort. The format of this error is as follows:

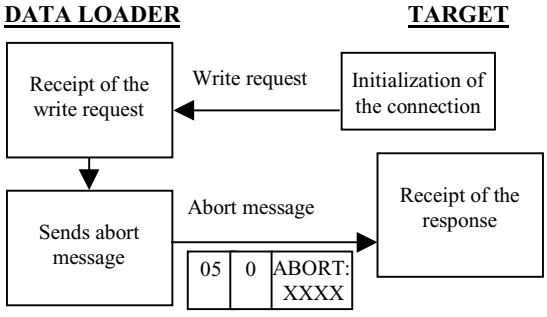
OpCode	Error Code Value	Error Message (text)	ASCII String Termination
0005_{16}	0000_{16}	"ABORT:XXXX"	00_{16}

c-1

NOTE: XXXX = 4 ASCII characters representing the HEXADECIMAL value of the Load Status Code (Table 6.4.10).

5.3.2.3.7 Utilization of the Abort Message

This abort message is used when a [Abort_request] message is received by the Data Loader Protocol; this mechanism can be used by the Data Loader Protocol itself in case of detection of problems. This is illustrated as follows:



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5.0 LOAD PROTOCOL

5.3.2.3.8 TFTP Option Implementation

COMMENTARY

The following sections define how to implement the currently known options for equipment claiming conformance to this specification. It is important to understand that equipment may implement these options for increased functionality, but must operate with peers that only implement the minimal option values. It is also important that equipment accept any option request, but ignore options that it does not implement.

It is expected that special functionality can be implemented in the future through option negotiation. Since all unknown options should be ignored, equipment can ask for options, but only peers that contain matching functionality will respond. As an example, the desire to zeroize media after loading could be implemented as an option.

5.3.2.3.8.1 Blocksize Option Implementation

c-2 The Data Loader should implement the TFTP block size option, including the implementation of Internet Protocol (IP) fragmentation/re-assembly. This capability should be available on any data loader for the target hardware to use.

Target hardware may, or may not, use the TFTP block size option. The target hardware designer has the responsibility to select adequate capability to ensure adequate performance versus additional complexity or technical needs, such as memory.

5.3.2.3.8.2 File Transfer Size

This option should not be used in the ARINC Report 615A protocol, because the size of the significant files is already present in the Header File.

5.3.2.3.8.3 Timeout Interval

This option should not be used in the ARINC Report 615A protocol. If it is present on a TFTP request message, it should be ignored. Attachment 4 defines a simple and reliable protocol to clearly define time-outs and retries on error. The selected solution is based on static and fixed time-outs and maximal retries numbers. Reliable networks should ensure low level of transmission error.

5.3.2.3.9 TFTP Block Number

c-2 Concerning the TFTP block number, which is an unsigned 16-bit counter, the first data block should correspond to block number one. A "WRQ" (Write c-1 ReQuest) is acknowledged with a "ACK" (ACKnowledge) packet having a block number of zero. If overflow occurs due to the number of transmitted blocks, c-2 the next value after 65535 should be one.

5.3.2.3.10 File Transfer Size

c-1 This option should not used in the ARINC Report 615A protocol.

5.3.2.3.11 Timeout Interval

Timeout Interval option should not be used in the ARINC Report 615A protocol, since timeout is defined in ARINC Report 615A, Attachment 4. c-1

5.4 Operations

5.4.1 Introduction

The Load Protocol defines a set of messages to manage communications between the LSA and the LDA. This standard defines the messages used for communication between the Data Loader Protocol and the Load Server Application.

Complementary messages may be implemented on the target hardware side, but the implementation is left to the equipment manufacturer.

COMMENTARY

The layered approach (Load Protocol and Load Application) is used to define the protocol. The target-hardware or data-loader designer may decide not to implement its equipment this way. Regardless, the protocol files operations defined in this standard must be implemented.

These messages are used in the operations defined below:

a. Information Operation: The data loader gets information on the configuration of the Target Hardware.

b. Uploading Operation: The data loader uploads files to the Target Hardware.

c. Downloading Operation: The data loader downloads files from the Target Hardware.

d. Registration of all present target hardware through the use of the FIND protocol defined in Attachment 3. c-1 c-2

These operations and their associated messages are independent and serial in nature. In other words, the data loading system should be able to perform the operations in any order, but a single operation needs to be completed before a second operation can be initiated for a specific target hardware. Only the interruption messages are defined for all operations, so that any operation may be aborted.

The data loader should implement these four operations listed above. c-1

The target hardware should implement FIND operations and may implement one, two or three other operations.

Before each operation, data loader may run the FIND request and register all the answers, in compliance with the FIND definition in Attachment 3. c-2

An answer to an ARINC 615A operation request that is not supported should be performed by a complete and correct initialization step, with the correct TFTP transfer of the first file. This file should contain an Operation c-2

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5.0 LOAD PROTOCOL

Acceptance Status Code with the value of 0X1002, which means "operation not supported by the Target."

c-1 If the Target Hardware does not implement all the operations, it should implement the initialization step of all operations in order to be able to answer to an operation request that it is not supported.

The messages exchanged between the LSA and the Data Loader protocol on the Data Loader are defined in Section 6.

c-1 The different Operations and their associated messages are discussed in the following sections.

5.4.2 "Information Operation"

This operation is used during on ground maintenance operations, with the intent to recover information on the configuration of the Target Hardware and Loadable Software Airplane parts (i.e. identifiers and Part numbers of the hardware and the software).

The Data Loader should implement this operation. The Target Hardware may implement or not this operation.

In this operation, two steps are defined.

a. Initialization step: In the first step the Data Loader LSA initializes the Information Operation. This step is used to inform the Target Hardware LDA of this operation and to determine if it is operational and only this.

   The access to this mode will be achieved through a [TH_Information_Initialization] message. The answer will be [Information_Initialization_ Response]. It indicates the acceptance or the refusal of this request to the LSA. If the Target Hardware refuses the request then the LSA notifies the operator and aborts the Information Operation.

b. Transfer step: If the Initialization step is accepted, the c-1 target will send the file <THW_ID_POS>.LCL which contains Target Hardware information to the Data Loader Protocol. In addition, the Target Hardware periodically sends a file to indicate the status of the process.

The messages are as follows:

[TH_Information]. It reports the requested Target Hardware information and completes the Information Operation.

[Information_Status]. It gives the evolution and status of this operation. The LSA should inform the user of the current status and continue waiting for the [TH_Information] message.

5.4.3 "Uploading Operation"

This operation is used during on ground maintenance operations to upload files from the Data Loader to the Target Hardware.

The Data Loader should implement this operation. The Target Hardware may implement or not this operation.

In this operation, three steps are defined:

a. Initialization step: In the first step the Data Loader LSA initializes the Uploading Operation. This makes it possible to inform the Target Hardware LDA of a request for this operation and to determine if it is operational.

   The access to this mode will be accomplished through a [TH_Uploading_Initialization] message. The answer will be [Uploading_Initialization_Response]. It indicates the acceptance or the denial of this request to the LSA. If the Target Hardware refuses the request then the LSA notifies the operator and aborts the Upload Operation.

b. List transfer step: If the Initialization step is accepted, the Data Loader LSA will initiate the load list transfer by issuing the <THW_ID_POS> message to c-1 the Data Loader Protocol. The Data Loader Protocol sends the list of loads which are to be potentially uploaded via the file <THW_ID_POS >.LUR. This c-1 list is analyzed by the Target Hardware and its sends its status via the file <THW_ID_POS>.LUS. If one c-1 of the load is not accepted by the Target Hardware, the complete list should be rejected.

   The messages will be [Upload_ Information_Status]. It gives the evolution and status of this operation. The LSA should inform the user of the current status.

c. Transfer step: The Target Hardware recovers the loadable files by performing TFTP reads of the desired files of the Load.

   The message [Upload_Information_Status] indicates the evolution and status of the uploading to the LSA.

In the case of a File not being available, the message [File_Not_Available] warns the operator.

The message [Upload_Information_Status] indicates also the completion success or failure of the uploading operation.

5.4.4 "DownLoad Operation"

This operation is used during on ground maintenance operations to download files from the Target Hardware to the Data Loader.

Two modes are considered: Media defined mode and Operator defined mode.

a. In Media defined download Mode, a predefined list of Downloadable files is sent to the Target Hardware.

b. In Operator defined download mode, the Target Hardware sends the list of potentially downloadable files and the operator selects in it the files which are to be Downloaded.

The Data Loader should implement the two modes. The Target Hardware may implement either none, one or both modes.

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5.0 LOAD PROTOCOL

5.4.4.1 Media Defined Download Mode

Initialization step: In the first step the LSA initializes the Downloading Operation.

c-1 | The <THW_ID_POS >.LNR is located on the media. If more than one file with LNR extension exist on the media, the operator should select one of these files.

This makes it possible to inform the Target Hardware LDA of this operation and to know if it is operational.

The access to this mode will be achieved through a [Downloading_Disk_Initialization] message. The answer will be a [Downloading_Initialization_Response]. It indicates the acceptance or the denial of this request to the LSA. If the Target Hardware refuses the request then the LSA notifies the operator and aborts the Download Operation.

List Transfer step: In this step the Data Loader Protocol sends the list of downloadable files via the
c-1 | <THW_ID_POS>.LNR. The messages will be [Downloading_Information_Status]. It gives the evolution and status of this operation to the LSA.

Transfer step: In this step, the list of downloadable files
c-1 | via the <THW_ID_POS >.LNR is analyzed by the Target Hardware and it sends the files defined in it.

The messages will be:

[Downloading_File_Receipt]. It informs the end user that the transfer is completed.

[Downloading_Information_Status] It gives the evolution and status of this operation to the LSA.

5.4.4.2 Operator Defined Download Mode

Initialization step: In the first step the LSA initializes the Downloading operation.

This makes it possible to inform the Target Hardware LDA of this operation and to know if it is operational. After this, the Target Hardware sends the list of files which can be downloaded.

The access to this mode will be achieved through a [Downloading_Operator_Initialization] message. The answer will be a [Downloading_Initialization_ Response]. It indicates the acceptance or the denial of this request to the Data Loader.

List Transfer step: If this operation is accepted, the Target Hardware sends the list of potentially downloadable data
c-1 | files in the <THW_ID_POS>.LNL by the [Downloading File_List_Receipt] message. After this, the operator selects the downloadable files and the Data Loader sends
c-1 | this list of selected files via the <THW_ID_POS>.LNA.

The messages will be:

[Downloading_File_List_Receipt] it gives the list of potentially downloadable data files.

[File_Selection]. It indicates the selected files to the Target Hardware.

[Downloading_Information_Status] messages give the evolution

Transfer step: At the receipt of <THW_ID_POS>.LNA, | c-1
this file is analyzed by the Target Hardware and it sends the specified file.

The messages will be:

[Downloading_File_Receipt] message, it informs the end user that the transfer is completed.

[Downloading_Information_Status] messages give the evolution and status of this operation.

5.4.5 Common Services

5.4.5.1 Addresses and Data Loadable Equipment Recovery

In order to recover network configuration (MAC addresses, IP addresses, Target Hardware names, etc.), the Data Loader and the Target Hardware should use the Simple Network Identification Protocol (SNIP) defined in Attachment 3.

5.4.5.2 Status Messages

To know the evolution of the operation progress, specific files are used. They are periodically sent by the Target Hardware to the Data Loader.

These status files are also used:

as a “Heart Beat.”

in order to abort an operation.

in order to close any operation upon completion.

The names of these files are:

<THW_ID_POS>.LCS for Information operation.

<THW_ID_POS>.LUS for Uploading operation. c-1

<THW_ID_POS>.LNS for Downloading operation.

5.4.5.3 Transfer Interruptions

Both the data loader and target hardware have the ability to interrupt the transfer phase of Uploading, Downloading and Information operations.

The data loader initiates an interruption by responding to a TFTP write request for one of the operation status files <THW_ID_POS>.LCS, <THW_ID_POS>.LUS or <THW_ID_POS>.LNS with an Abort Message. The c-2
Abort message defined in Section 5.3.2.3.6 contains a status code reporting whether the operator or the data loader initiated the abort. The operator abort is initiated by a [Abort_Request] message to the Data Loader protocol from the Load Server Application. On receipt of the Abort Message the target hardware aborts the operation in progress and responds with an operation status file echoing the status code from the Abort Message in its “operation” status code.

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5.0 LOAD PROTOCOL

5.4.5.3 Transfer Interruptions (cont'd)

c-2 The target hardware initiates an interruption by setting the "operation" status code in the operation status file to 1003_{16} indicating an abort by the target and transmits the status file. On receipt of the status file the data loader will abort the operation in progress.

After receiving the status file with the "operation" status code set to indicate an abort (1003_{16}, 1004_{16}, 1005_{16}) the data loader should display informative text about the interruption per the table in Section 6.4.10.

--`,,`-`-`,,`,,`,`,,`---

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6.0 LOAD PROTOCOL DEFINITION

6.1 Summary of Messages

Operation	Step	Message	LSA DLP
	Initialization		
		[TH_Information_Initialization]	
		[Information_Initialization_Response]	
	Transfer		
		[TH_Information]	
		[Information_Status]	
		[Abort_Request]	
Uploading	Initialization		
		[TH_Uploading_Initialization]	
		[Uploading_Initialization_Response]	
	List transfer		
		[Load_List]	
		[Upload_Information_Status]	
		[Abort_Request]	
	Transfer		
		[File_Not_Available]	
		[Upload_Information_Status]	
		[Abort_Request]	
Downloading Media defined mode	Initialization		
		[Downloading_Disk_Initialization]	
		[Downloading_Initialization_Response]	
	List transfer		
		[Downloading_Information_Status]	
		[Abort_Request]	
	Transfer		
		[Downloading_File_Receipt]	
		[Abort_Request]	
		[Downloading_Information_Status]	
Downloading Operator defined mode	Initialization		
		[Downloading_Operator_Initialization]	
		[Downloading_Initialization_Response]	
	List transfer		
		[Downloading_File_List_Receipt]	
		[File_Selection]	
		[Downloading_Information_Status]	
		[Abort_Request]	
	Transfer		
		[Downloading_File_Receipt]	
		[Downloading_Information_Status]	
		[Abort_Request]	

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6.0 LOAD PROTOCOL DEFINITION

6.2 Description of Messages

These messages are passed between the LSA and the Data Loader Protocol on the Data Loader. The Target Hardware will require a complementary set of messages between the Target Hardware Protocol and the LDA. The implementation of the Target Hardware messages is left up to the equipment manufacturer, as it is not relevant to the standard as long as the protocol file operations are followed.

6.2.1 [TH_Information_Initialization] Message

6.2.1.1 When Generated

This message is generated by the Data Loader LSA to initialize the Information operation and to obtain the configuration of the Target Hardware.

6.2.1.2 Effect on Receipt

c-1 | The Data Load Protocol layer issues a TFTP read for the file <THW_ID_POS>.LCI and evaluated the response from the Target Hardware. If the Target Hardware completes the TFTP Read then the [Information_Initialization_Response] message is generated giving the acceptance or the refusal to initialize this operation.

6.2.1.3 Protocol File Used

c-1 | The Data Load Protocol layer will generate a TFTP read of the file <THW_ID_POS>.LCI from the Target Hardware.

6.2.2 [Information_Initialization_Response] Message

6.2.2.1 When Generated

This message is generated by the Data Loader Protocol following a [TH_Information_ Initialization] message.

6.2.2.2 Effect on Receipt

On receipt of the message, the LSA will have the result of the Initialization step.

6.2.2.3 Protocol File Used

c-1 | This message will be issued when the file <THW_ID_POS>.LCI is recovered by the Data Loader Protocol (TFTP Read).

6.2.3 [TH_Information] Message

6.2.3.1 When Generated

This service is generated following a [Information_Initialization_Response] message and if the operation is accepted by the Target Hardware.

6.2.3.2 Effect on Receipt

On receipt of the file, the Data Loader LSA will update the Target Hardware configuration report.

6.2.3.3 Protocol File Used

This message will be issued when the file <THW_ID_POS>.LCL is recovered by the Data Loader Protocol (TFTP Write). | c-1

6.2.4 [Information_Status] Message

6.2.4.1 When Generated

This message is generated following a [Information_ Initialization_Response] message and if the operation is accepted by the Target Hardware. It makes it possible to know the status of the Target Hardware.

It is useful particularly when the delay time to recover the <THW_ID_POS>.LCL file is long. | c-1

6.2.4.2 Effect on Receipt

The End User is informed of the evolution and status of the initialization operation.

6.2.4.3 Protocol File Used

A TFTP write of the file <THW_ID_POS>.LCS will be periodically performed by the Target Hardware. | c-1

6.2.5 [TH_Uploading_Initialization] Message

6.2.5.1 When Generated

This message is generated by the Data Loader LSA to initialize the Uploading operation.

6.2.5.2 Effect on Receipt

The Data loader Protocol layer issues a TFTP read for the file <THW_ID_POS>.LUI and evaluates the response from the Target Hardware and completes the TFTP Read then the [Uploading_Initialization_Response] message is generated giving the acceptance or the refusal to initialize this operation. | c-1

6.2.5.3 Protocol File Used

The Data Loader Protocol layer will generate a TFTP read of the file <THW_ID_POS>.LUI from the Target Hardware. | c-1

6.2.6 [Uploading_Initialization_Response] Message

6.2.6.1 When Generated

This message is generated by the Data Loader Protocol following a [TH_Uploading_Initialization] message.

6.2.6.2 Effect on Receipt

On receipt of the message, the LSA will have the result of the initialization step.

6.2.6.3 Protocol File Used

This message will be issued when the file <THW_ID_POS>.LUI is recovered by the Data Loader Protocol (TFTP Read). | c-1

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6.0 LOAD PROTOCOL DEFINITION

6.2.7 [Load_List] Message

6.2.7.1 When Generated

This message is generated by the Data Loader LSA in order to propose a list of Loads to be updated to the Target Hardware.

6.2.7.2 Effect on Receipt

The Target Hardware will decide which Loads will be updated

6.2.7.3 Protocol File Used

The Data Loader Protocol will generate a TFTP write of the file <THW_ID_POS>.LUR on the Target Hardware. c-1 In fact the TFTP write is started when the <THW_ID_POS >.LUS is received this makes it possible to synchronize the Data Loader and the Target Hardware in the list transfer step.

6.2.8 [File_Not_Available] Message

6.2.8.1 When Generated

This message is generated by the Data Loader Protocol following a TFTP read of the file <name_of_the file> (<Data_File_Name> or <HEADER_FILE>.LUH) and if this file is not available in the Data Loader, so that the Data Loader Protocol must indicate to the Target Hardware that it cannot answer the request immediately. The Target Hardware must try again after a delay time given by the Data Loader Protocol.

6.2.8.2 Effect on Receipt

On receipt of this message, the Data Loader LSA will give a feed back about the delay to the Operator.

6.2.8.3 Protocol File Used

The answer to the TFTP Read of the file <name_of_the file> performed by the Target Hardware is a TFTP error with

Error code = 0 and
Error Message = WAIT:<DELAY TIME >

6.2.9 [Upload_Information_Status] Message

6.2.9.1 When Generated

This service is generated after a [Upload_ Initialization_Response] message and if the operation is accepted by the Target Hardware. It makes to possible to know the status of the Target Hardware.

6.2.9.2 Effect on Receipt

The Operator is informed of the evolution and status of the Uploading operation.

6.2.9.3 Protocol File Used

c-1 A TFTP write of the file <THW_ID_POS>.LUS will be periodically performed by the TH.

6.2.10 [Downloading_Disk_Initialization] Message

6.2.10.1 When Generated

This message is generated by the Data Loader LSA to initialize the Downloading in automatic mode.

6.2.10.2 Effect on Receipt

The Data Loader Protocol issues a TFTP read of the file <THW_ID_POS>.LND and evaluates the response from c-1 the Target Hardware. If the Target Hardware completes the TFTP Read the [Downloading_Initialization_ Response] message is generated giving the acceptance or the refusal to initialize this operation.

6.2.10.3 Protocol File Used

The Data Loader Protocol layer will generate a TFTP read of the file <THW_ID_POS>.LND from the Target c-1 Hardware.

6.2.11 [Downloading_Initialization_Response] Message

6.2.11.1 When Generated

This message is generated by the data Loader Protocol following a [Downloading_Disk_Initialization] or a [Downloading_Operator_Initialization] message.

6.2.11.2 Effect on Receipt

On receipt of the message, the LSA will have the result of the initialization step.

6.2.11.3 Protocol File Used

This message will be issued when the file <THW_ID_POS>.LND or the file c-1 <THW_ID_POS>.LNO is recovered by the Data Loader Protocol (TFTP Read).

6.2.12 [Downloading_File_Receipt] Message

6.2.12.1 When Generated

This message is generated by the Data Loader Protocol for each data file received.

6.2.12.2 Effect on Receipt

The Data Loader LSA informs the operator of the receipt of the data file

6.2.12.3 Protocol File Used

Not applicable.

6.2.13 [Downloading_Information_Status] Message

6.2.13.1 When Generated

This service is generated after a [Downloading_Initialization_Response] message and if the operation is accepted by the Target Hardware. It makes it possible to know the status of the target Hardware.

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6.0 LOAD PROTOCOL DEFINITION

6.2.13.2 Effect on Receipt

The operator is informed of the evolution and status of the Downloading operation

6.2.13.3 Protocol File Used

c-1 | A TFTP write of the file <THW_ID_POS>.LNS will be periodically performed by the TH.

6.2.14 [Downloading_Operator_Initialization] Message

6.2.14.1 When Generated

This message is generated by the Data Loader LSA to initialize the Downloading in manual mode.

6.2.14.2 Effect on Receipt

c-1 | The Data Loader Protocol issues a TFTP read of the file <THW_ID_POS>.LNO and evaluates the response from the Target Hardware. If the Target Hardware completes the TFTP Read the [Downloading Initialization Response] message is generated giving the acceptance or the refusal to initialize this operation.

6.2.14.3 Protocol File Used

c-1 | The Data Loader Protocol layer will generate a TFTP read of the file <THW_ID_POS>.LNO from the Target Hardware.

6.2.15 [Downloading_File_List_Receipt] Message

6.2.15.1 When Generated

c-1 | This message is generated by the Data Loader Protocol on receipt of the file <THW_ID_POS>.LNL.

6.2.15.2 Effect on Receipt

The Data Loader LSA informs the operator of the receipt of the data file. This file is used by the operator to select the data files which will be Downloaded.

6.2.15.3 Protocol File Used

c-1 | The Target Hardware will generate a TFTP write of the file <THW_ID_POS>.LNL on the Data Loader.

6.2.16 [File_Selection] Message

6.2.16.1 When Generated

This message is generated by the Data Loader Protocol to transfer the choice of Data files to the Target Hardware.

6.2.16.2 Effect on Receipt

The Target Hardware knows the Data files which are to be downloaded by Data Loader.

6.2.16.3 Protocol File Used

c-1 | The Data Loader Protocol will generate a TFTP write of the file <THW_ID_POS>.LNA on the Target Hardware.

6.2.17 [Abort_Request] Message

6.2.17.1 When Generated

This message is generated by the Data Loader LSA to trigger an abort of an operation (Upload or Download or Information).

6.2.17.2 Effect on Receipt

When the Data Loader Protocol receives from the target a request to open a TFTP to write the file <THW_ID_POS>.LCS, .LUS or .LNS. It refuses the TFTP and sends an abort Error message to the target which stops the operation in progress. After this, the target sends the file <THW_ID_POS>.LCS, .LUS or .LNS with the status code 1004_{16} (if aborted by the data loader) or 1005_{16} (if cancelled by the operator) to inform the data loader that the abort or cancellation has been taken into account. | c-1 | c-2

6.2.17.3 Protocol File Used

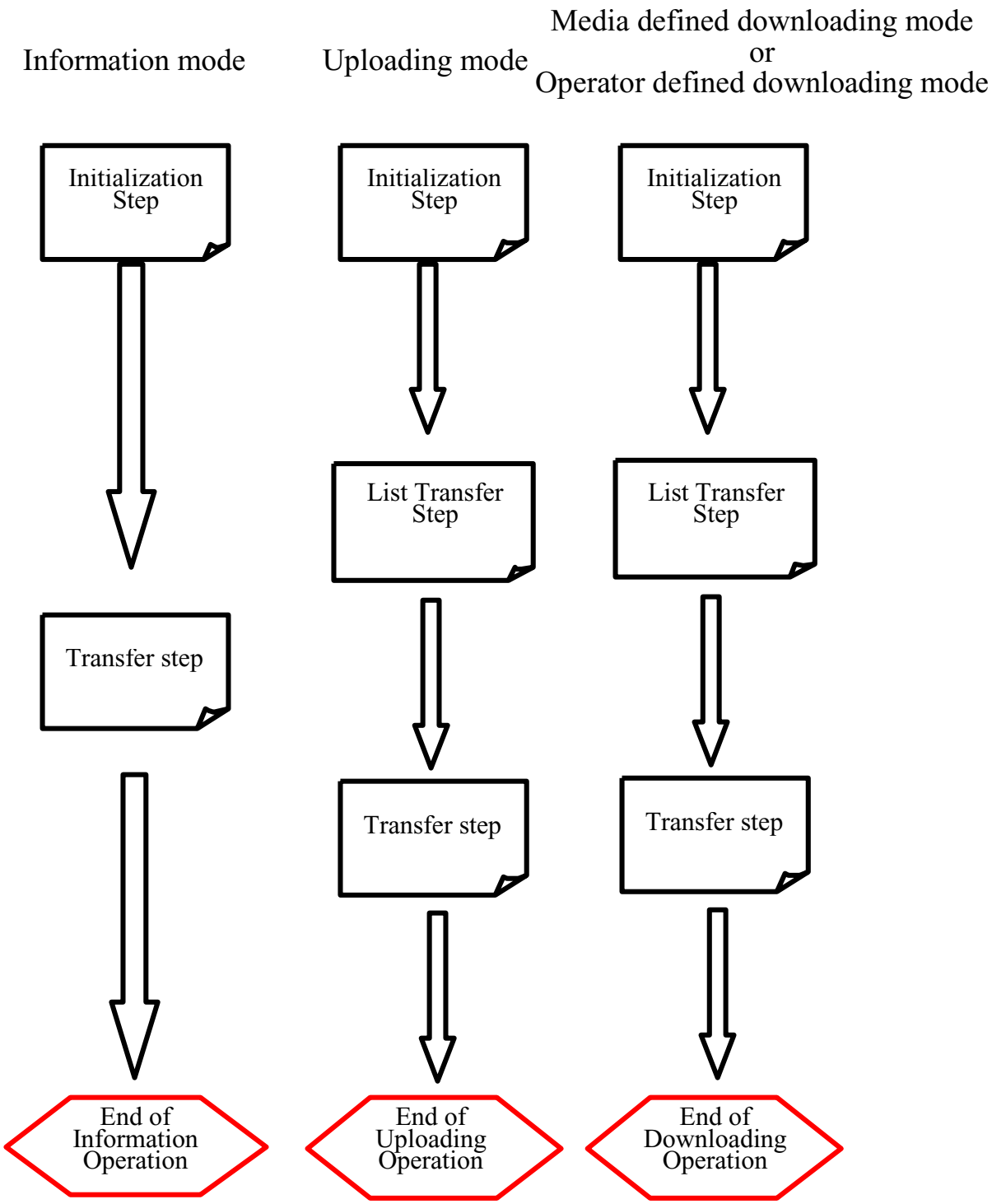
The Data Loader Protocol will send an Abort error message on receipt of a TFTP write of the file <THW_ID_POS>.LCS, <THW_ID_POS>.LUS or <THW_ID_POS>.LNS. | c-1

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6.0 LOAD PROTOCOL DEFINITION

6.3 Message Sequence Chart

6.3.1 Information/Uploading/Downloading Operation: Summary Description

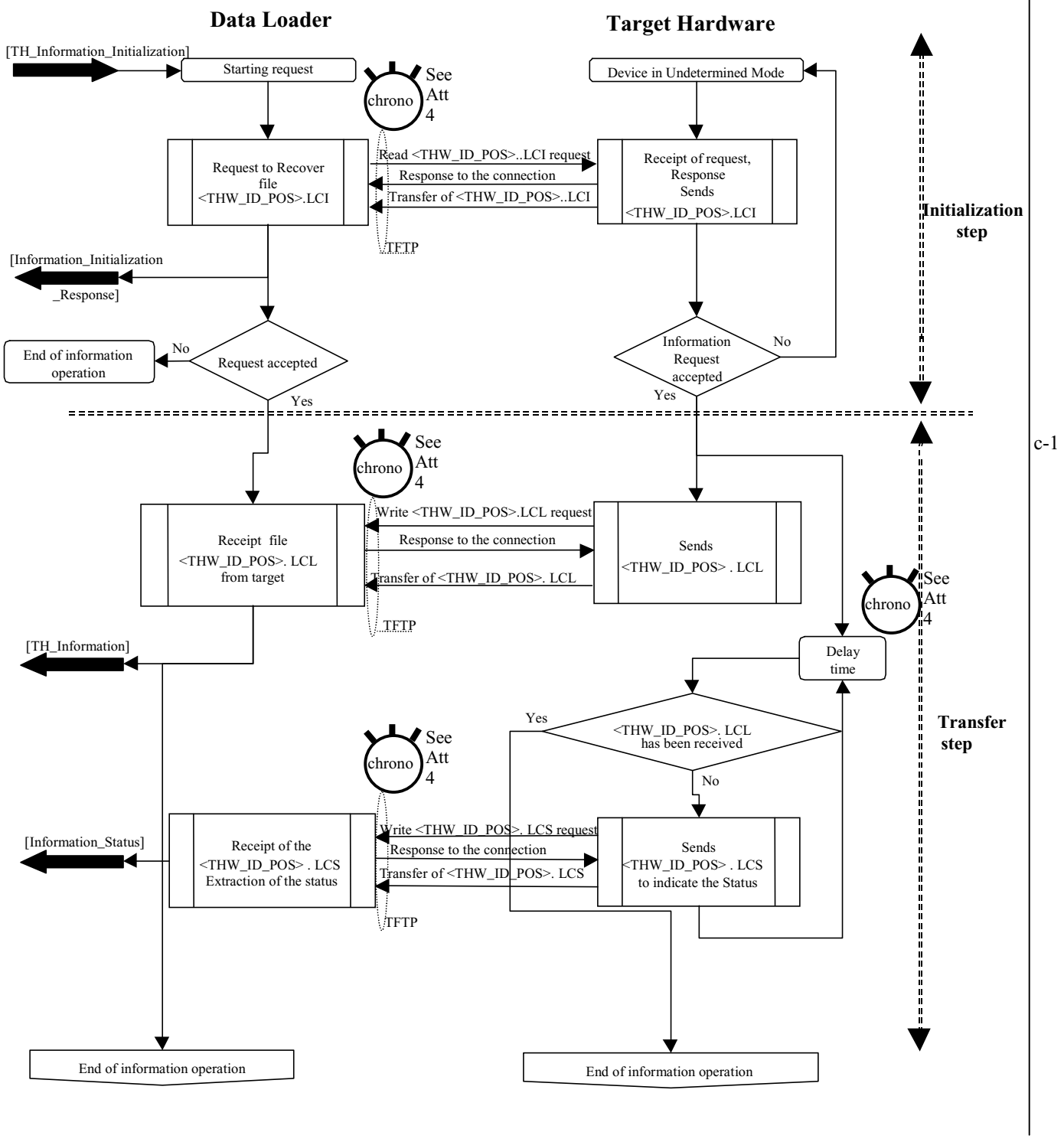


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6.0 LOAD PROTOCOL DEFINITION

6.3.2 Information Operation: Initialization and Transfer Step

The files used are: <THW_ID_POS>.LCI, <THW_ID_POS>.LCL and <THW_ID_POS>.LCS



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6.0 LOAD PROTOCOL DEFINITION

6.3.2 Information Operation: Initialization and Transfer Step (cont'd)

Explanation of Diagram in 6.3.2:

Steps following initialization step may have variable duration time. So, to check that both side (data loader and target hardware) are always in data loading operational mode and in coherent internal state, status files are frequently sent by the target hardware to the data loader, allowing each one to control each other.

c-1

Any invalid structure is considered as a fatal error.

As soon as the transfer step phase begins:

c-2 Status files should be sent within a maximum delay in compliance with the DLP time-out requirement defined in Attachment 4. Periodic transmission is not mandatory, just required within the maximum delay.

c-1 Any fatal error detected by the target should induce production of a status file containing adequate status code and status description, in compliance with time-out requirements.

c-2 A status file may be sent immediately by the target hardware to support the abort message defined in Section 5.3.2.3.6 and the interrupt service defined in Section 6.3.6.

Note that "Information Operation" represents a particular case. The transmission of the THW_ID_POS.LCL file may be immediate or may take some time.

In case of a fast response in information mode (i.e., end of transmission of the THW_ID_POS.LCL file before DLP time-out expiration), the target may not send the status file. In this case, the operation will be completed at the end of transmission of the THW_ID_POS_.LCL file.

c-1 In the other case, if at least one status file is produced, the target hardware should send status files in compliance with the DLP time-out requirements. The activity will be completed only after transmission of a status file with a global status code "Operation Completed" 0003_{16} or fatal error. This status file is the last file sent.

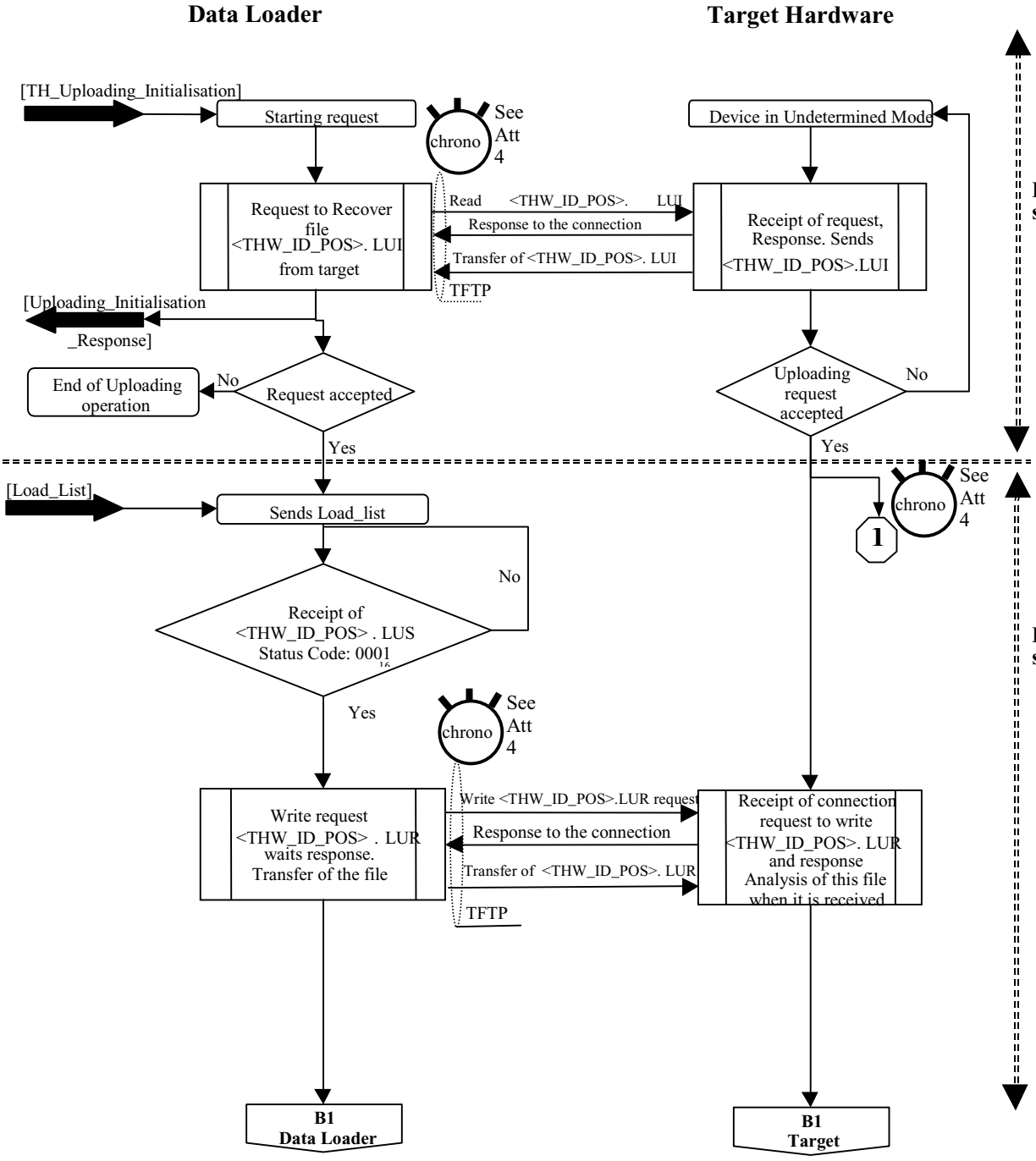
The Exception Timer field, in the status file, is used when the target needs to re-boot to switch from operational mode to data loading mode. This mode is required to run data loading operations. During the switch phase, the target hardware probably should not be able to produce status files. Using this field, the target hardware may inform the data loader that during the "exception timer" wait, it will be silent (no status file) but that the information operation is still in process. If the target hardware responds sooner, the data loader should continue normal operation without waiting for the exception timer to elapse. The target hardware supplier should minimize the "Exception Timer" value to minimize the silent phase. The data loader should receive a new status file before this delay expiration, otherwise the information operation should be aborted.

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6.0 LOAD PROTOCOL DEFINITION

6.3.3 Uploading Operation: Initialization, List Transfer and Transfer

The files used are: <THW_ID_POS>.LUI, <THW_ID_POS>.LUR and <THW_ID_POS>.LUS

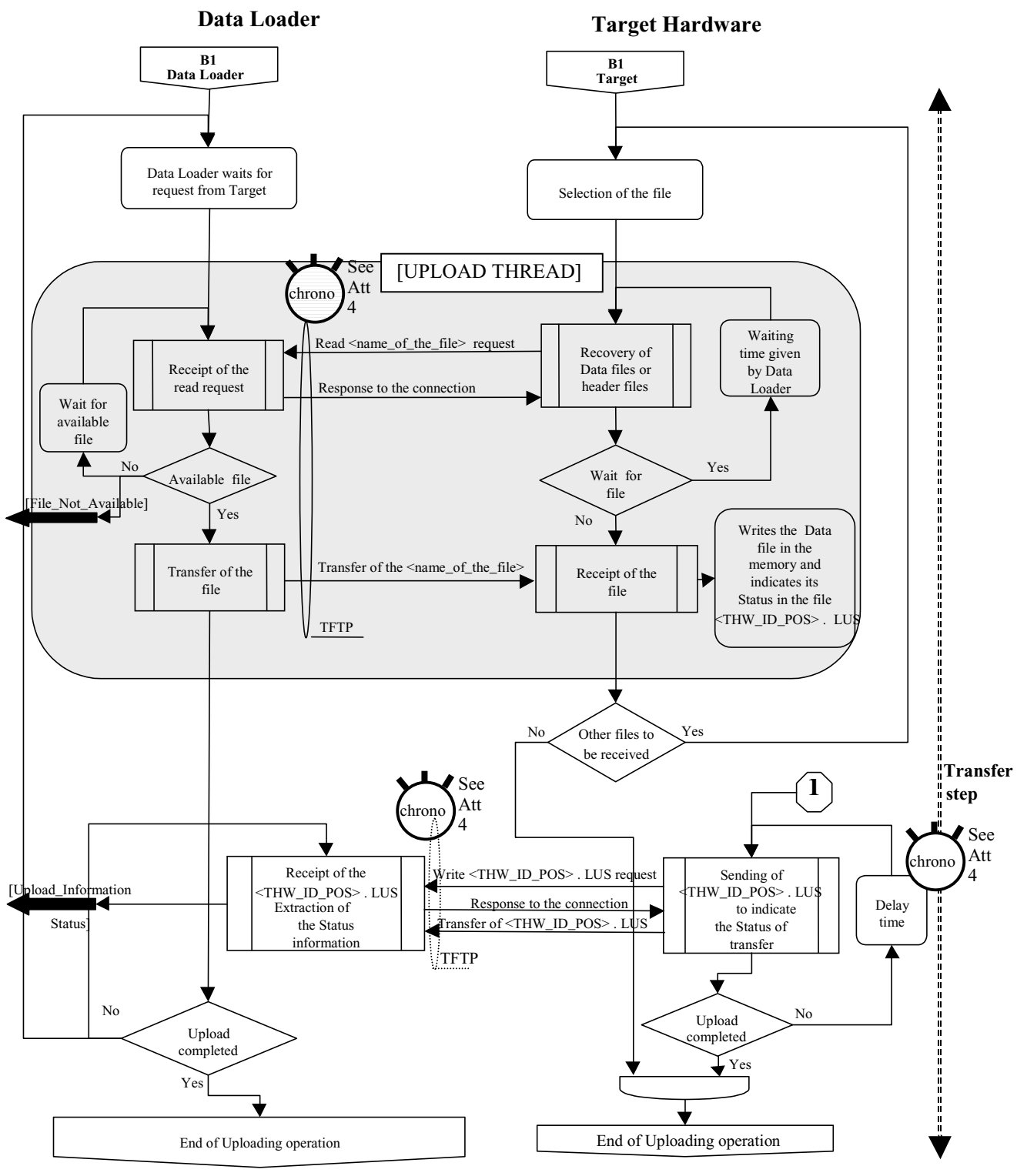


The Data Loader and Target go to file transfer step

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6.0 LOAD PROTOCOL DEFINITION

6.3.3 Uploading Operation: Initialization, List Transfer and Transfer (cont'd)



c-1

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6.0 LOAD PROTOCOL DEFINITION

Explanation of diagrams in 6.3.3:

Steps following the initialization step may have variable duration time. So, to check that both sides (data loader and target hardware) are always in data loading operational mode and in a coherent internal state, status files are frequently sent by the target hardware to the data loader, allowing each one to control each other. c-1

Any invalid structure is considered as a fatal error, including .LUR or .LUH file structure.

As soon as the list transfer step phase begins:

Status files should be sent within a maximum delay in compliance with the DLP time-out requirement defined in Attachment 4. Periodicity transmission is not mandatory, just required within the maximum delay. c-2

Any fatal error detected by the target should induce production of a status file containing adequate status code and status description, in compliance with time-out requirements. c-1

A status file may be sent immediately by the target hardware to support the abort message defined in Section 5.3.2.3.6 and the interrupt service defined in Section 6.3.6. c-2

After the list transfer step, the data loader runs as a TFTP server, without any hypothesis in the sequence of the target hardware request files. The only constraint is that the target hardware loads the header file before associated data files, as data files names are only in the header file. c-1

In the transfer step, only the "UPLOAD THREAD" labeled part may be duplicated to implement a parallel load for one target. Status file exchanges are relative to the whole transfer step, whatever the parallelisation level. c-2

According to its context (floppy disc, hard disc,…), the data load application has to identify if the error "file not available" is an unrecoverable error (inducing a failure of the load process) or if it is a recoverable error (requesting an action of the operator). In the case of request for an operator action, the data loader response will be the waiting message defined in Sections 5.3.2.3.4 and 5.3.2.3.5. c-1

The Exception Timer field, in the status file, should be used when the target needs to re-boot to switch from operational mode to data loading mode. This new mode is required to run data loading operations. During the switch phase, the target hardware should not be able to produce a status file. Using this field, the target hardware may inform the data loader that during "Exception Timer" wait, that it will be silent (no status file), yet the upload operation is still in process. If target hardware responds sooner, the data loader should continue normal operation without waiting for the Exception Timer to elapse. It is up to the target hardware supplier to minimize the "Exception Timer" value to minimize the silent phase. The data loader should receive a new status file before this delay expiration. Otherwise the upload operation is aborted. c-2

During check computations (CRC, Checksum,…) the status files should still be transmitted, even after all data files have been transmitted.

The uploading operation is only achieved (an indicator of 100 percent in the status field) after erasing, engraving and correct programmatic checking using an adequate algorithm (e.g. CRC, checksum, etc).

The activity will be completed only after transmission of a status file with a global status code "Operation Completed" 0003_{16} or fatal error. This status file is the last file sent.

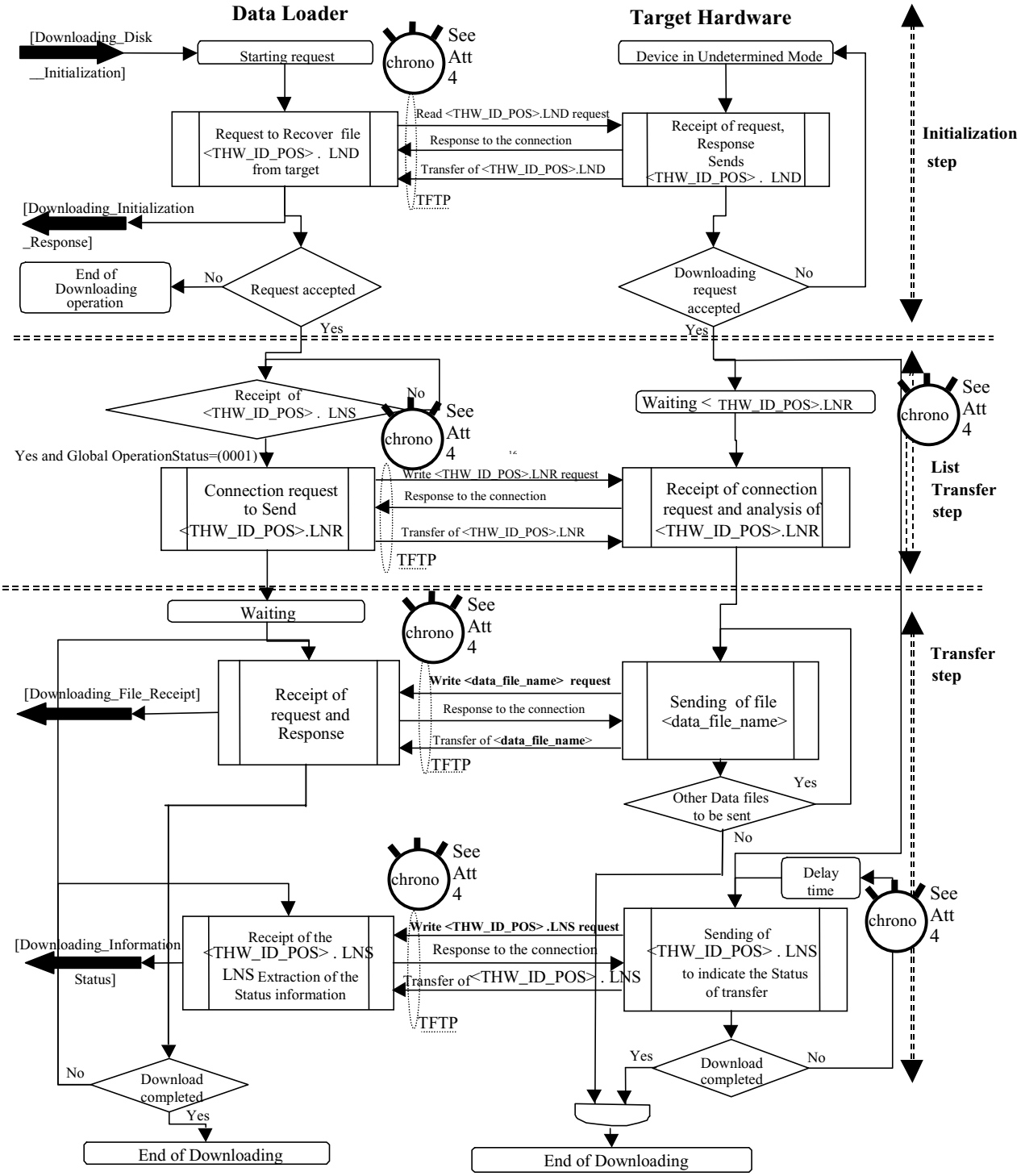
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6.0 LOAD PROTOCOL DEFINITION

6.3.4 Downloading Operation in Media Defined Mode: Initialization, List Transfer and Transfer

The files used are: <THW_ID_POS>.LND, <THW_ID_POS>.LNR and <THW_ID_POS>.LNS

c-1



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6.0 LOAD PROTOCOL DEFINITION

Explanation of diagram in 6.3.4:

The operator should select the media type for the download operation. c-1

Steps following initialization step may have variable duration time. So, to check that both sides (data loader and target hardware) are always in data loading operational mode and in a coherent internal state, status files are frequently sent by the target hardware to the data loader, allowing each one to control each other. c-2

Any invalid structure is considered a fatal error. c-1

As soon as the list transfer step phase begins:

Status files should be sent within a maximum delay in compliance with the DLP time-out requirement defined in Attachment 4. Periodicity transmission is not mandatory, just required within the maximum delay. c-2

Any fatal error detected by the target should induce production of a status file containing adequate status code and status description, in compliance with time-out requirements. c-1

A status file may be sent immediately by the target hardware to support the abort message defined in Section 5.3.2.3.6 and the interrupt service defined in Section 6.3.6. c-2

The Exception Timer field, in the status file, should be used when the target hardware needs to re-boot to switch from operational mode to data loading mode. This mode is required to run data loading operations. During the switch phase, the target hardware should not produce status files. Using this field, the target hardware may inform the data loader that during Exception Timer waits, that it will be silent (no status file), but the download operation is still in process. If the target hardware responds sooner, the data loader should continue normal operation without waiting for the exception timer to elapse. The target hardware supplier should minimize the Exception Timer value to minimize the silent phase. The data loader should receive a new status file before this delay expiration. Otherwise the download operation is aborted. c-1 c-2 c-1 c-2 c-1

The activity will be completed only after transmission of a status file with a global status code “Operation Completed” (0003_{16}) or fatal error. This status file is the last file sent. c-2

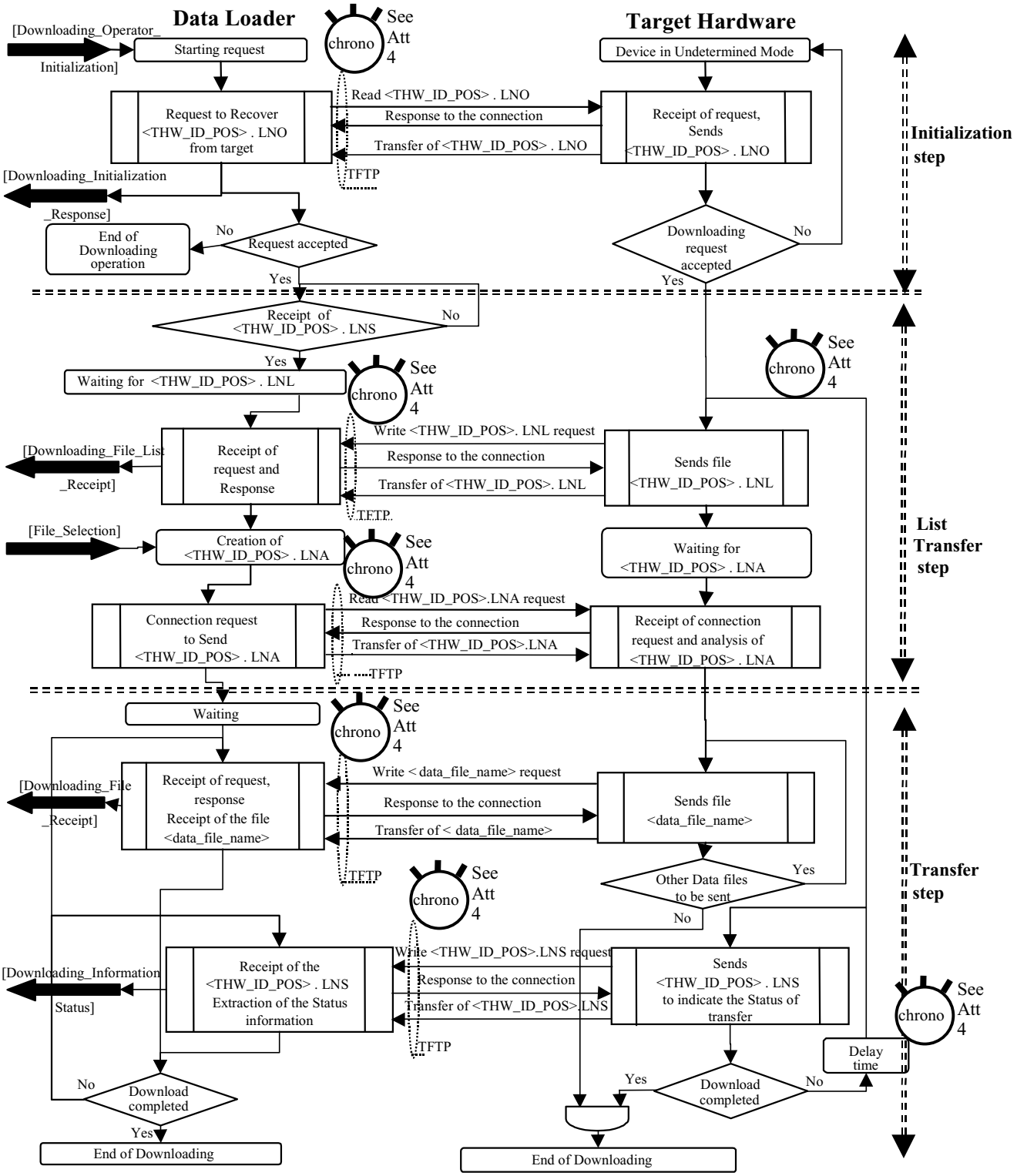
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6.0 LOAD PROTOCOL DEFINITION

6.3.5 Downloading Operation in Operator Defined Mode: Initialization and List Transfer

The files used are: <THW_ID_POS>.LNO, <THW_ID_POS>.LNL, <THW_ID_POS>.LNA and <THW_ID_POS>.LNS

c-1



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6.0 LOAD PROTOCOL DEFINITION

Explanation of diagram in 6.3.5:

The media type should be selectable by the operator for the download operation. | c-2

Steps following initialization step may have variable duration time. To check that both sides (data loader and target hardware) are always in data loading operational mode and in a coherent internal state, status files are frequently sent by the target hardware to the data loader, allowing each one to control each other. | c-1 c-2

Any invalid structure is considered as a fatal error. | c-1

As soon as the list transfer step phase begins:

Status files should be sent within a maximum delay in compliance with the DLP time-out requirement defined in Attachment 4. Periodicity transmission is not mandatory, just required within the maximum delay. | c-2

Any fatal error detected by the target hardware should induce production of a status file containing adequate status code and status description, in compliance with time-out requirements. | c-1

A status file may be sent immediately by the target hardware to support the abort message defined in Section 5.3.2.3.6 and the interrupt service defined in Section 6.3.6. | c-2

The Exception Timer field, in the status file, should be used when the target needs to re-boot to switch from operational mode to data loading mode. This new mode is required to run data loading operations. During the switch phase, the target hardware should not be able to produce status files. Using this field, the target hardware may inform the data loader that during Exception Timer waits, that it will be silent (no status file), but the download operation is still in process. If the target hardware responds sooner, the data loader should continue normal operation without waiting for the Exception Timer to elapse. The target hardware supplier should minimize the Exception Timer value to minimize the silent phase. The data loader should receive a new status file before this delay expiration. Otherwise the download operation is aborted. | c-1 c-2 c-1 c-2 c-1

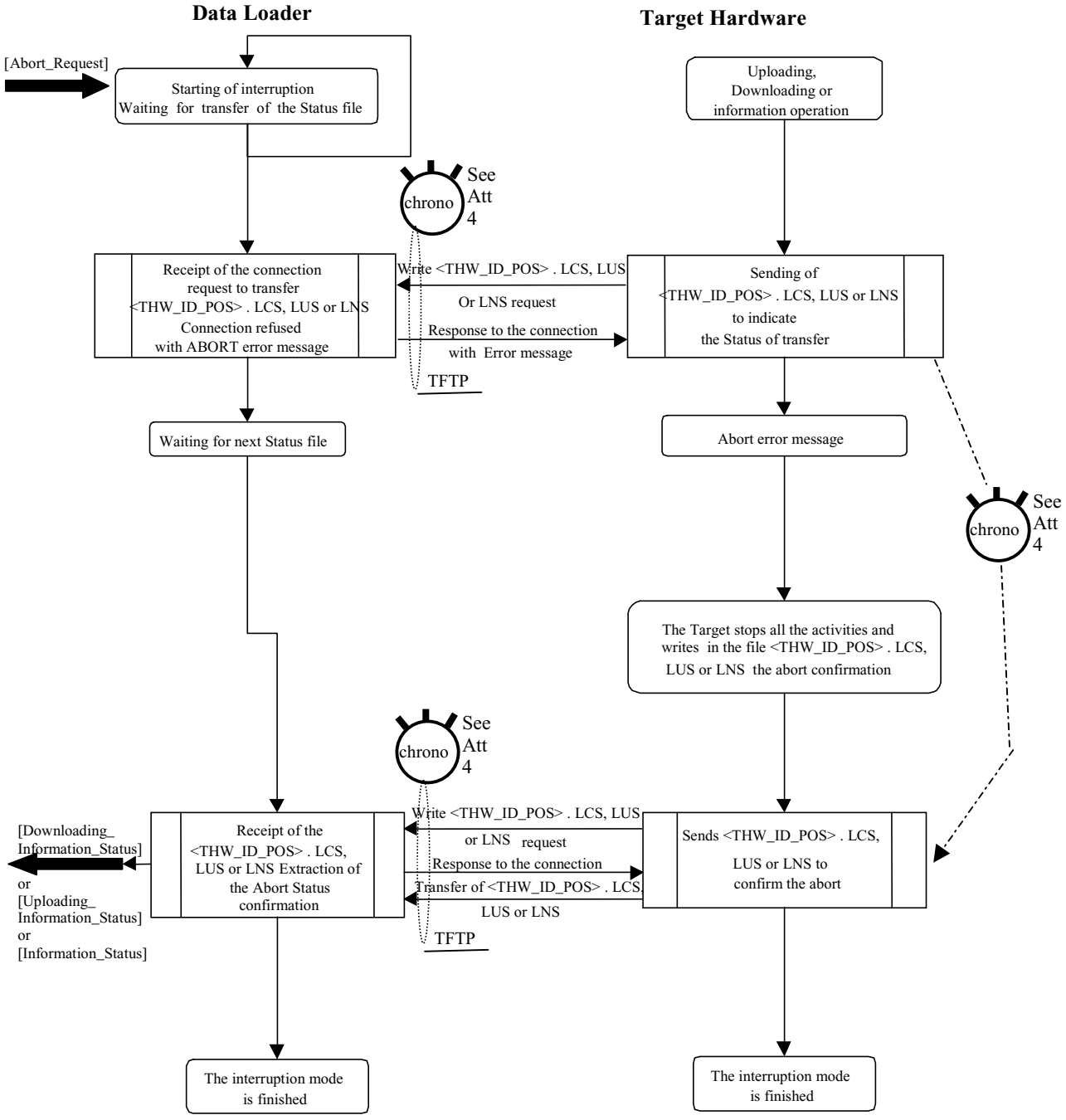
The activity will be completed only after transmission of a status file with a global status code "Operation Completed" (0003_{16}) or fatal error. This status file is the last file sent. | c-2

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6.0 LOAD PROTOCOL DEFINITION

6.3.6 Interruption Service

The files used are: <THW_ID_POS>.LCS, <THW_ID_POS>.LUS and <THW_ID_POS>.LNS



c-1

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6.0 LOAD PROTOCOL DEFINITION

Explanation of diagram 6.3.6:

Interruption of service is supported by the status file. A status file transmission is initiated by the target hardware:

If abort is requested by the target hardware, the target hardware may send one status file indicating the abort request. The complete transfer of this status file indicates the acknowledgment of the interrupt request by the data loader. c-1

If abort is requested by the data loader, the data loader should wait for the next status file write request. from the target hardware, respond to this request with an abort error message defined in Section 5.3.2.3.6, and then wait for the next status file transmission. This last status file from the target hardware supports the target acknowledgment of the abort request from the data loader. The diagram in Section 6.3.6 shows this abort scenario. c-2

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6.0 LOAD PROTOCOL DEFINITION

6.4 Protocol File Description

This Section describes the format of the data files used by the data loading protocol. Table 6.4 summarizes all the files used by the protocol.

c-1 The THW_ID system is represented as THW_ID_POS. The definition for target hardware THW_ID is found in ARINC Report 665 – Loadable Software Standards. The POS is 0 to 4 alphanumeric characters. The THW_ID field and the POS field are separated by a "_".

COMMENTARY

The <THW_ID_POS> name is composed of two elements as follows:

The THW_ID (or EQUIP_ID) described in ARINC Report 665 identifies generic equipment.

The POS (Position) identifies the instantiation of generic equipment.

The THW_ID_POS is obtained by merging the two fields, including an underscore between those two fields (e.g., 02B1_2).

All protocol files are defined using "Big Endian" byte ordering, that is, the most significant eight bits (MS byte) of each 16-bit word are transferred first, followed by the least significant bits (LS byte) in the next eight-bit byte (n+1). The same byte ordering is used to derive field information in all protocol files. Only byte ordering for data files is up to the supplier.

As the first file exchange is produced by the target hardware, the target hardware provides the protocol version that should be used for the ARINC Report 615A operation. According to this information, the data loader is responsible to adapt its functionality to this request
c-2 (specifically, reject if not implemented, select this version if more than this one implemented). On the target hardware side, the protocol version should be the same for all implemented operations.

The structure of the protocol files is referenced by a version of the applicable structure, and supported by a field named "Protocol Version". This Protocol Version parameter allows the target hardware and data loader to use the correct structure to access the data.

The value of protocol version should be the same for all protocol files defined in one revision of ARINC Report 615A.

If the structure changes of even one protocol file, the protocol version value of the new ARINC Report 615A revision should be modified, along with all other files affected by the protocol version change, to reflect the protocol version implemented.

The protocol version is sent by the target hardware in the initialization step. The data loader should adapt to this protocol version or abort the load if this protocol version is not implemented.

In the case of incompatibility of protocol versions, the data loader should abort the operation by sending an abort message to the first status file write request.

The protocol version should not be moved from the second field of a protocol file. c-2

The value of the protocol version is $(A3)_{ASCII}$ for Supplement 2 of ARINC Report 615A.

All textural fields, for example the several description fields, should be zero terminated, shown as 00_{16} , to indicate the end of the data. For emphasis, this statement is repeated in the appropriate field descriptions in the following subordinate sections.

COMMENTARY

The description length field indicates the size of the description field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire description field. For processing efficiency implementers may want to fix the size of the description field (and the associated value of description length) and then vary the message content as necessary. Others will desire to limit the amount of data transmitted and set the description length to zero when no description text is required.

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6.0 LOAD PROTOCOL DEFINITION

Name of File	Operation	File Content	Created by	File type
<THW_ID_POS>.LCI	Information	Used only to initialize an information operation. Load Configuration Initialization	Target Hardware	Protocol
<THW_ID_POS>.LCL	Information	Configuration of the target, it gives the list of the P/Ns in the target. Load Configuration List	Target Hardware	Protocol
<THW_ID_POS>.LCS	Information	Evolution and status of the Information process. Load Configuration Status	Target Hardware	Protocol
<THW_ID_POS>.LUI	Uploading	Used only to initialize an uploading operation.	Target Hardware	Protocol
<THW_ID_POS >.LUR	Uploading	Request for Upload of a List of loads. Load Uploading Request	Data Loader	Protocol
<HEADER_FILE>.LUH *	Uploading	Contains information to permit the Target Hardware to accept the load data Load Upload Header	Suppliers	File
<Data_File_Name> *	Uploading	Name of the uploaded or downloaded file	Supplier or Target Hardware	File
<THW_ID_POS>.LUS	Uploading	Evolution and status of the Uploading process. Load Uploading Status	Target Hardware	Protocol
<THW_ID_POS>.LND	Downloading in Media mode	Used only to initialize a downloading operation. Load dowNloading meDia defined	Target Hardware	Protocol
<THW_ID_POS >.LNR	Downloading in Media mode	Used to give the list of files to be recovered. Load dowNloading Request	Data Loader	Protocol
<THW_ID_POS>.LNS	Downloading in Media and operator mode	Evolution of the Downloading process. Load dowNloading Status	Target Hardware	Protocol
<THW_ID_POS>.LNO	Downloading in Operator mode	Used only to initialize a downloading operation. Load dowNloading Operator defined	Target Hardware	Protocol
<THW_ID_POS>.LNL	Downloading in Operator mode	Used to give the list of files which are potentially recovered. Load dowNloading List	Target Hardware	Protocol
<THW_ID_POS>.LNA	Downloading in Operator mode	Used to give the list of files which are selected by the operator and are to be downloaded. Load dowNloading Answer	Data Loader	Protocol

c-1

* Though these files are not protocol files, they are listed here because the protocol manages them. They are described in ARINC Report 665.

c-1 | 6.4.1 <THW_ID_POS >.LCI, .LUI, .LND and .LNO

Name of Field	Field Size (bits)
File Length	32
Protocol Version	16
Operation Acceptance Status Code	16
Status Description Length	8
Status Description	0 – 2040

Detailed description of each field:

File Length:
Number of 8-bit words in this file including this field.

Protocol Version:
Two bytes (ASCII characters) defining the Protocol Version as explained in Section 6.4, Protocol File Description. c-2

Operation Acceptance Status Code:
Unsigned 16 bit value, XXXX_{16}: Status Code.

Answer to the information operation initialization.

The Status Codes used in this field are:

0001_{16}: The operation is accepted.

1000_{16}: The operation is denied. The reason is described in the status description field.

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6.0 LOAD PROTOCOL DEFINITION

c-1 6.4.1 <THW_ID_POS >.LCI, .LUI, .LND and .LNO (cont'd)

1002_{16}: The operation is not supported by the Target

The Status Codes are summarized in Section 6.4.10 "Data Load Status Code Table."

Status Description Length:
Binary, number of characters in the Status Description field (0 to 255 characters).

Status Description:
Description of the reason for this status code, it is used only for status code 1000_{16}. For Status Codes 0001_{16} and 1002_{16} the content of this field is ignored.

The text should be only printable characters, should not include control characters, and should have a maximum length of 255 characters (i.e., 2040 bits).

The text should be zero terminated 00_{16} to indicate the end of the data.

c-2 COMMENTARY

The Status Description Length field indicates the size of the Status Description field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Status Description field. For processing efficiency implementers may want to fix the size of the Status Description field (and the associated value of Status Description Length) and then vary the message content as necessary. Others will desire to limit the amount of data transmitted and set the Status Description Length to zero when no Status Description Text is required.

Status Description:
Description of the reason for this status code, it is used only for status code 1000_{16}. For Status Codes 0001_{16} and 1002_{16} the content of this field is ignored.

c-2 The text should be only printable characters, should not include control characters, and should have a maximum length of 255 characters (i.e., 2040 bits).

The text should be zero terminated 00_{16} to indicate the end of the data.

c-1 6.4.2 <THW_ID_POS>.LCL

Name of Field	Field Size (bits)
File Length	32
Protocol Version	16
Number of target hardware	16
*Target Hardware Code Length	8
*Target Hardware Code	8-2040
* Serial Number Length	8
* Serial Number	8-2040
* Number of Part Numbers	16
* + Part Number Length	8
* + Part Number	8-2040
* + Amendment Length	8
* + Amendment	0-2040
* + Part Designation Length	8
* + Part Designation Text	8-2040

c-2

+ Fields repeated for each Part Number
* Fields repeated for each Target Hardware

Detailed description of each field:

File Length:
Number of 8-bit words in this file including this field.

Protocol Version:
Two bytes (ASCII characters) defining the Protocol Version as explained in Section 6.4, Protocol File Description.

Number of Target Hardware:
Number of target hardware reported in the table
In fact, one piece of equipment could centralize different equipment identification items within a given perimeter.

Target Hardware Code Length:
Number of characters in the Target Hardware Code field (1 to 255 characters).

COMMENTARY

The Target Hardware Code Length field indicates the size of the Target Hardware Code field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Target Hardware Code field. For processing efficiency implementers may want to fix the size of the Target Hardware Code field (and the associated value of Target Hardware Code Length) and then vary the message content as necessary. c-2

Target Hardware Identifier:
This field corresponds to the "Literal name (text)" field of the FIND answer defined in Attachment 3. The text should be zero terminated, to indicate the end of the data.

Serial Number Length
Number of characters in the Serial Number Length field (1 to 255 characters).

Serial Number:
1 to 255 characters (8 bits ASCII i.e., 2040 bits) defining the Target Hardware Serial Number of the Target Hardware. The text should be zero terminated to indicate the end of the data.

Number of Part Numbers:
Number of part numbers of the current target hardware item.

Part Number Length:
Number of characters in the Part Number (1 to 255 Characters).

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Part Number:
1 to 255 characters (8 bits ASCII i.e., 2040 bits) defining the hardware or any load Part Number. The text should be zero terminated to indicate the end of the data.

Amendment Length:
Number of characters in the Amendment (1 to 255 characters).

If the Amendment Length field value is zero, the Amendment field is omitted and ignored.

Amendment:
1 to 255 characters (8 bits ASCII i.e., 2040 bits) defining the hardware amendment if any.

Amendment is usually related to hardware identification, permitting the tracking of modification for form, fit and function that does not impact interchangeability and interoperability.

Part Designation Text Length:
Number of characters in the Part Designation Text (1 to 255 characters). c-2

COMMENTARY

The Part Designation Length field indicates the size of the Part Designation field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Part Designation field. For processing efficiency implementers may want to fix the size of the Part Designation field (and the associated value of Part Designation Length) and then vary the message content as necessary.

Part Designation Text:
1 to 255 characters (8 bits ASCII) .

Target Hardware name (acronyms or code) for the hardware parts (e.g., FMGEC 1), and function name or designation for the software parts (e.g., FMS 1 OPS), identifying the specific part and occurrence in the aircraft to be displayed.

The text should only be printable characters, should not include control characters, and should have a maximum length of 255 characters (i.e., 2040 bits).

The text should be zero terminated 00_{16} to indicate the end of the data.

6.4.3 <THW_ID_POS>.LCS

Name of Field	Field Size (bits)
File Length	32
Protocol Version	16
Counter	16
Information Operation Status Code	16
Exception Timer	16
Estimated Time	16
Status Description Length	8
Status Description	0 – 2040

Detailed description of each field:

File Length:
Number of 8-bit words in this file including this field.

Protocol Version:
Two bytes (ASCII characters) defining the Protocol Version as explained in Section 6.4, Protocol File Description. c-2

Counter:
Initial value is 0000. c-1

16 bits counter incremented by the Target Hardware before sending the file.

Set to 0000_{16} after $FFFF_{16}$. This allows the Data Loader to detect the duplication of status files or the loss of status files. c-1

Information Operation Status Code:

Unsigned 16 bit value, XXXX_{16}: Status code

Status of the information operation in progress.

The Status Codes used in this field are:

0001_{16}: The target accepts the operation (not yet started).

0002_{16}: The operation is in progress

0003_{16}: The operation is completed without error.

1003_{16}: The operation is aborted by the target hardware, target hardware text is required in the Status Description field to know the reason for this interruption. c-1

1004_{16}: The operation is aborted in the target hardware due to the receipt of an abort error message sent by the Data Loader Protocol. c-2

1005_{16}: The operation is aborted in the target hardware due to the receipt of an abort error message initiated by an operator action. c-2

The Status Codes are summarized in Section 6.4.10, "Data Load Status Code Table."

Exception Timer:

Binary, number of seconds during which the target is expected to be silent. During this delay, the status file is not sent by the target and it does not respond to any request from the data loader (0 to 2^{16} –1 seconds). c-2

This field is used only if the Status Code field is 0002_{16} , for all other status codes, this field must be set to 0000_{16}.

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6.0 LOAD PROTOCOL DEFINITION

6.4.3 <THW_ID_POS>.LCS (cont'd)

Estimated Time:

Binary, estimated number of seconds to complete the operation (-1 to 2^{15} -1 seconds). The value "–1" means the estimated time is not given. Estimated Time is the time remaining to complete the operation as computed by the target hardware. As soon as possible during the operation, the target hardware should provide the estimated time. c-2

This field is used only if the Status Code field is 0002_{16}. For all other Status Codes, this field is set equal to 0000_{16}

If the target hardware does not give this value, the Estimated Time field must be set to $FFFF_{16}$.

Status Description Length:
Binary, number of characters in the Status Description field (0 to 255 characters).

COMMENTARY

The Status Description Length field indicates the size of the Status Description field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Status Description field. For processing efficiency implementers may want to fix the size of the Status Description field (and the associated value of Status Description Length) and then vary the message content as necessary. Others will desire to limit the amount of data transmitted and set the Status Description Length to zero when no Status Description text is required. c-2

Status Description:
Description of the reason for the Status Code. Used for Status Code 1003_{16}. For other Status Codes, the content of this field is ignored, but may contain data (i.e., left over data). Refer to Section 6.4.10 for more information.

The text should only be printable characters, should not include control characters, and should have a maximum length of 255 characters (i.e., 2040 bits). c-2

The text should be zero terminated 00_{16} to indicate the end of the data.

6.4.4 <THW_ID_POS >.LUR

Name of Field	Field Size (bits)
File Length	32
Protocol Version	16
Number of Header Files	16
+ Header File Name Length	8
+ Header File Name	8 – 2040
+ Load Part Number Name Length	8
+ Load Part Number Name	8 – 2040

c-2

+ Fields repeated for each Header File.

Detailed description of each field:

File Length:
Number of 8-bit words in this file including this field.

Protocol Version:
Two bytes (ASCII characters) defining the Protocol Version as explained in Section 6.4, Protocol File Description. c-2

Number of Header Files:
Binary number of Header Files in the Header File List. Any .LUR file should contain at least one Header File.

Header File Name Length:
Number of characters in the Header File Name (1 to 255 characters).

COMMENTARY

The Header File Name Length field indicates the size of the Header File Name field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Header File Name field. For processing efficiency implementers may want to fix the size of the Header File Name field (and the associated value of Header File Name length) and then vary the message content as necessary. c-2

Header File Name:
1 to 255 characters (8-bit ASCII i.e., 2040 bits) as defined by the Header File Name Length field. c-2

The text should be zero terminated 00_{16} to indicate the end of the data. c-2

Load Part Number Name Length:
Number of characters in the Load Part Number (1 to 255 characters). c-2

COMMENTARY

The Load Part Number Length field indicates the size of the Load Part Number field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Load Part Number field. For processing efficiency implementers may want to fix the size of the Load Part Number field (and the associated value of Load Part Number Length) and then vary the message content as necessary. c-2

Load Part Number Name:
1 to 255 characters (8-bit ASCII i.e., 2040 bits) as defined by the Load Part Number Length field.

The text should be only printable characters, should not include control characters, and should have a maximum length of 255 characters (i.e., 2040 bits).

The text should be zero terminated 00_{16} to indicate the end of the data.

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6.0 LOAD PROTOCOL DEFINITION

c-1 | 6.4.5 <THW_ID_POS>.LUS

Name of Field	Field Size (bits)
File Length	32
Protocol Version	16
Upload Operation Status Code	16
Upload Status Description Length	8
Upload Status Description	0 - 2040
Counter	16
Exception Timer	16
Estimated Time	16
Load List Ratio	24
Number of Header Files	16
+ Header File Name Length	8
+ Header File Name	8 – 2040
+ Load Part Number Name Length	8
+ Load Part Number Name	8 – 2040
+ Load Ratio	24
+ Load Status	16
+ Load Status Description Length	8
+ Load Status Description	0 – 2040

+ Fields repeated for each Software Part Number.

Detailed description of each field:

File Length:
Number of 8-bit words in this file including this field.

Protocol Version:
Two bytes (ASCII characters) defining the Protocol Version as explained in Section 6.4, Protocol File Description.

Upload Operation Status Code:
Unsigned 16 bit value, $(xxxx)_{16}$: Status code.

It gives the status of this operation. During the list transfer step, the "Number of Header Files" field must be set to 0.

The Status Codes used in this field are:

0001_{16}: The Target accepts the operation (not yet started).

0002_{16}: The operation is in progress.

0003_{16}: The operation is completed without error.

1003_{16}: The operation is aborted by the target hardware. Target hardware text is required in the status description field to identify the reason for this interruption.

1004_{16}: The operation is aborted in the target hardware due to the reception of an abort error message sent by the data loader protocol.

1005_{16}: The operation is aborted in the target hardware due to the receipt of an abort error message initiated by an operator action.

The Status Codes are summarized in Section 6.4.10 "Data Load Status Code Table."

COMMENTARY

The Upload Operation Status Code defines to the Data Loader the status of the entire load list. The Load Status Field is only an informative data.

Upload Status Description Length:
Binary, number of characters in the Upload Status description field (0 to 255 characters).

COMMENTARY

The Upload Status Description Length field indicates the size of the Upload Status Description field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Upload Status Description field. For processing efficiency implementers may want to fix the size of the Upload Status Description field (and the associated value of Upload Status Description Length) and then vary the message content as necessary. Others will desire to limit the amount of data transmitted and set the Upload Status Description Length to zero when no Upload Status Description text is required.

Upload Status Description:
Description of the reason for the Status Code. Used for Status Code 1003_{16}. For other Status Codes, the content of this field is ignored, but may contain data (i.e., left over data). Refer to Section 6.4.10 for more information.

The text should only be printable characters, should not include control characters, and should have a maximum length of 255 characters (i.e., 2040 bits).

The text should be zero terminated 00_{16} to indicate the end of the data.

Counter:
Initial value is 0000.

16 bit counter incremented by the Target Hardware before sending the file.

Set to 0000_{16} after $FFFF_{16}$. It allows the Data Loader to detect the duplication of files.

Exception Timer:

Binary, number of seconds during which the target is expected to be silent. During this delay the status file is not sent by the target hardware and it does not respond to any request from the data loader (0 to $2^{16} - 1$ seconds).

This field is used only if the Status Code field is 0002_{16}, for the other status code this field must be set to 00000000_2.

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6.0 LOAD PROTOCOL DEFINITION

6.4.5 <THW_ID_POS>.LUS (cont'd)

Estimated Time:
c-2 Binary, estimated number of seconds to complete the operation (-1 to 2^{15} -1 seconds). The value "–1" means the Estimated Time is not given. Estimated Time is the time remaining to complete the operation as computed by the target hardware. As soon as possible during the operation, the target hardware should provide the Estimated Time.

This field is used only if the Status Code field is 0002_{16}. For all other Status Codes, this field is set equal to 0000_{16}

If the target hardware does not provide this value, the "Estimated Time" field must be set to $FFFF_{16}$.

Load List Ratio:
Three characters (8-bit ASCII) representing the percentage of the Load List completed.

c-1 Right adjusted field with leading blanks.

Number of Header Files:
Binary number of Header Files in the Load List

Header File Name Length:
Number of characters in the Header File Name (1 to 255 characters).

Header File Name:
1 to 255 characters (8-bit ASCII) as defined by the Data File Name Length field.

The text should be zero terminated 00_{16} to indicate the end of the data.

COMMENTARY

c-2 The Header File Name Length field indicates the size of the Header File Name field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Header File Name field. For processing efficiency implementers may want to fix the size of the Header File Name field (and the associated value of Header File Name Length) and then vary the message content as necessary.

Load Part Number Name Length:
Number of characters in the Load Part Number (1 to 255 characters).

COMMENTARY

c-2 The Load Part Number Length field indicates the size of the Load Part Number field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Load Part Number field. For processing efficiency implementers may want to fix the size of the Load Part Number field (and the associated value of Load Part Number length) and then vary the message content as necessary.

Load Part Number Name:
1 to 255 characters (8-bit ASCII) as defined by the Load Part Number Length field.
The text should be zero terminated 00_{16} to indicate the end of the data. c-2

Load Ratio: c-1
Right adjusted field with leading blanks.

Three characters (8-bit ASCII) representing the percentage of the specific load completed.

Load Status:
Unsigned 16 bits value, XXXX_{16}: Status Code.

Status for each Header File. It gives the status of load defined by the associated Header File (not only the status of the upload of the Header File.) c-1

The Status Codes used in this field are:

0001_{16}: The operation is accepted (not yet started).

0002_{16}: The operation is in progress.

0003_{16}: The operation is completed without error

1003_{16}: The operation is aborted by the target hardware. Target hardware text is required in the Status Description field to know the reason from this interruption.

1004_{16}: The operation is aborted in the target due to the receipt of an abort error message sent by the Data Loader Protocol. c-2

1005_{16}: The operation is aborted in the target due to the receipt of an abort error message initiated by an operator action.

1007_{16}: The load of this Header File has failed, text is required in the "Status Description" field to explain the failure. c-2

The Status Codes are summarized in the Section 6.4.10 "Data Load Status Code Table".

Load Status Description Length: c-2
Binary, number of characters in the Status Description field (0 to 255 characters).

COMMENTARY

c-2 The Status Description Length field indicates the size of the Status Description field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Status Description field. For processing efficiency, implementers may want to fix the size of the Status Description field (and the associated value of Status Description Length) and then vary the message content as necessary. Others will desire to limit the amount of data transmitted and set the Status Description Length to zero when no Status Description Text is required.

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6.0 LOAD PROTOCOL DEFINITION

Load Status Description:
Description of the reason for the Status Code. Use for the Status Codes 1003_{16} and 1007_{16}. For other status codes, the content of this field is ignored, but may contain data (i.e., left over data). Refer to Section 6.4.10 for more information. c-2

The text should be only printable characters, should not include control characters, and should have a maximum length of 255 characters (i.e., 2040 bits).

The text should be zero terminated 00_{16} to indicate the end of the data.

6.4.6 <THW_ID_POS >.LNR

Name of Field	Field Size (bits)
File Length	32
Protocol Version	16
Number of Files	16
+ File Name Length	8
+ File Name	8 – 2040
User Defined Data Length	8
User Defined Data	0 – 2040

+ Fields repeated for each file to be Downloaded.

Detailed description of each field:

File Length:
Number of 8-bit words in this file including this field.

Protocol Version:
Two bytes (ASCII characters) defining the Protocol Version as explained in Section 6.4, Protocol File Description. c-2

Number of Files:
Binary number of Files for this Downloading.

File Name Length:
Number of characters in the File Name (1 to 255 characters i.e., 2040 bits).

COMMENTARY

The File Name Length field indicates the size of the File Name field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire File Name field. For processing efficiency implementers may want to fix the size of the File Name field (and the associated value of File Name Length) and then vary the message content as necessary. c-2

File Name:
1 to 255 characters (8-bit ASCII i.e., 2040 bits) as defined by the File Name Length field.

The text should be zero terminated 00_{16} to indicate the end of the data. c-2

User Defined Data Length:
Binary, number of bytes in the User Defined Data field (0 to 255 bytes).

User Defined Data:
1 to 255 bytes (8 bit) as defined by the User Defined Data Length field. c-2

6.4.7 <THW_ID_POS>.LNS

Name of Field	Field Size (bits)
File Length	32
Protocol Version	16
Download Operation Status Code	16
Download Status Description Length	8
Download Status Description	0 – 2040
Counter	16
Exception Timer	16
Estimated Time	16
Download List Ratio	24
Number of Files	16
+ File Name Length	8
+ File Name	8 – 2040
+ File Status	16
+ File Status Description Length	8
+ File Status Description	0 – 2040

c-2

+ Fields repeated for each File

Detailed description of each field:

File Length:
Number of 8-bit words in this file including this field.

Protocol Version:
Two bytes (ASCII characters) defining the Protocol Version as explained in Section 6.4, Protocol File Description. c-2

Download Operation Status Code:
Unsigned 16 bits value, XXXX_{16}: Status code.

It gives the status of this operation but only if the "Number of Files" is set to 0.

The Status Codes used in this field are:

0001_{16}: The Target accepts the operation (not yet started).

0002_{16}: The operation is in progress.

0003_{16}: The operation is completed without error.

1003_{16}: The operation is aborted by the target hardware, LRU text is required in the Status Description field to know the reason for this interruption.

1004_{16}: The operation is aborted in the target hardware due to the receipt of an abort error message sent by the Data Loader Protocol. c-2

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6.0 LOAD PROTOCOL DEFINITION

6.4.7 <THW_ID_POS>.LNS (cont'd)

c-2 1005_{16}: The operation is aborted in the target hardware due to the receipt of an abort error message initiated by an operator action.

The Status Codes are summarized in Section 6.4.10, "Data Load Status Code Table."

Download Status Description Length:
Binary, number of characters in the Status Description field (0 to 255 characters).

COMMENTARY

The Download Status Description Length field indicates the size of the Download Status Description field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire Download Status Description field. For processing efficiency implementers may want to fix the size of the Download Status Description field (and the associated value of Download Status Description length) and then vary the message content as necessary. Others will desire to limit the amount of data transmitted and set the Download Status Description Length to zero when no Download Status description text is required.

c-2 Download Status Description:
Description of the reason for the Status Code. Used only for Download Status Codes 1003_{16} and 1007_{16}. For other Status Codes, the content of this field is ignored, but may contain data (i.e., left over data). Refer to Section 6.4.10 for more information.

The text should only be printable characters, should not include control characters, and should have a maximum length of 255 characters (i.e., 2040 bits).

The text should be zero terminated 00_{16} to indicate the end of the data.

c-1 Counter:
Initial value is 0000.

16 bit counter incremented by the Target Hardware before sending the file.

Set to 0000_{16} after $FFFF_{16}$. This allows the Data Loader to detect the duplication of files.

Exception Timer:
c-1 Binary, number of seconds during which the target hardware is expected to be silent. During this delay, the status file is not sent by the target hardware and it does not respond to any request from the Data Loader
c-2 (0 to $2^{16}-1$ seconds).

This field is used only if the Status Code field is 0002_{16}, for the other Status code this field must be set to 0000_{16}.

Estimated Time:
c-2 Binary, estimated number of seconds to complete the operation (-1 to 2^{15} -1 seconds). The value "–1" means the estimated time is not given. Estimated time is the time remaining to complete the operation as computed by the target hardware. As soon as possible during the operation, the target hardware should provide the estimated time.

This field is used only if the Status Code field is 0002_{16}. For all other Status Codes, this field is set equal to 0000_{16}

If the Target does not give this value, the "Estimated Time" field must be set to $FFFF_{16}$.

Load List Ratio:
Three characters (8-bit ASCII) representing the percentage of the Download completed. Right adjusted field with leading blanks. c-1

Number of Files:
Binary, Number of Files in the download selected.

File Name Length:
Number of characters in the File Name (1 to 255 characters).

COMMENTARY

The File Name Length field indicates the size of the File Name field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire File Name field. For processing efficiency, implementers may want to fix the size of the File Name field (and the associated value of File Name Length) and then vary the message content as necessary. c-2

File Name:
1 to 255 characters (8-bit ASCII) as defined by the File Name Length field.

The text should be zero terminated 00_{16} to indicate the end of the data. c-2

File Status:
Unsigned 16 bit value, XXXX_{16}: Status Code.

It gives the status for each file. The status codes used in this field are:

0001_{16}: The download of this file has not yet started. c-2

0002_{16}: The operation is in progress.

0003_{16}: The download of this file has been completed without error. c-2

1003_{16}: The operation is aborted by the target, LRU text is required in the Status Description field to know the reason for this interruption.

1004_{16}: The operation is aborted in the target hardware due to the receipt of an abort error message sent by the data loader protocol. c-2

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6.0 LOAD PROTOCOL DEFINITION

c-2 1005_{16}: The operation is aborted in the target due to the receipt of an abort error message initiated by an operator action.

c-2 1007_{16}: The load of this Header File has failed, text is required in the "Status Description" field to explain the failure.

The status codes are summarized in the Section 6.4.10 "Data Load Status Code Table."

File Status Description Length:
Binary, number of characters in the File Status Description field (0 to 255 characters).

COMMENTARY

The File Status Description Length field indicates the size of the File Status Description field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire File Status Description field. For processing efficiency, implementers may want to fix the size of the File Status Description field (and the associated value of File Status Description Length) and then vary the message content as necessary. Others will desire to limit the amount of data transmitted and set the File Status Description Length to zero when no File Status Description Text is required.

c-2

File Status Description:
Description of the reason for the Status Code. Used only for Status Codes 1003_{16} and 1007_{16}. For other File Status Codes, the content of this field is ignored, but may contain data (i.e., left over data). Refer to Section 6.4.10 for more information.

The text should be only printable characters, should not include control characters, and should have a maximum length of 255 characters (i.e., 2040 bits).

The text should be zero terminated 00_{16} to indicate the end of the data.

c-1 6.4.8 <THW_ID_POS>.LNL

Name of Field	Field Size (bits)
File Length	32
Protocol Version	16
Number of Files	16
+ File Name Length	8
+ File Name	8 – 2040
+ File Description Length	8
+ File Description	8 – 2040

c-2

+ Fields repeated for each file to be Downloaded

Detailed description of each field:

File Length:
Number of 8-bit words in this file including this field.

Protocol Version:
Two bytes (ASCII characters) defining the Protocol Version as explained in Section 6.4, Protocol File Description. c-2

Number of Files:
Binary, Number of Files which can be downloaded.

File Name Length:
Number of characters in the Data File Name (1 to 255 characters).

COMMENTARY

The File Name Length field indicates the size of the File Name field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire File Name field. For processing efficiency implementers may want to fix the size of the File Name field (and the associated value of File Name Length) and then vary the message content as necessary. c-2

File Name:
1 to 255 characters (8-bit ASCII) as defined by the File Name Length field.

The text should be zero terminated 00_{16} to indicate the end of the data.

File Description Length:
Binary, number of characters in the File Description field (0 to 255 characters).

COMMENTARY

The File Description Text Length field indicates the size of the File Description Text field and not the size of its relevant content. The content is terminated by the string terminator 00_{16} and may not fill the entire File Description field. For processing efficiency implementers may want to fix the size of the File Description field (and the associated value of File Description Length) and then vary the message content as necessary. Others will desire to limit the amount of data transmitted and set the File Description Length to zero when no File Description Text is required. c-2

File Description:
The text should be only printable characters, should not include control characters, and should have a maximum length of 255 characters (i.e., 2040 bits).

The text should be zero terminated 00_{16} to indicate the end of the data.

6.4.9 <THW_ID_POS>.LNA

Name of Field	Field Size (bits)
File Length	32
Protocol Version	16
Number of Files	16
+ File Name Length	8
+ File Name	8 - 2040

+ Fields repeated for each file to be Downloaded

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6.0 LOAD PROTOCOL DEFINITION

6.4.9 <THW_ID_POS>.LNA (cont'd)

Detailed description of each field:

File Length:
Number of 8-bit words in this file including this field.

Protocol Version:
Two bytes (ASCII characters) defining the Protocol Version as explained in Section 6.4, Protocol File Description.

Number of Files:
Binary number of Files for this Downloading.

File Name Length:
Number of characters in the File Name (1 to 255 characters i.e., 2040 bits).

6.4.10 Data Load Status Code Table

Table 6.4.10 is the Data Load Status Code Table.

Table 6.4.10, Data Load Status Code Table

Status Code	Meaning	Display Text	Files Used in Operation		
			Information..	Upload..	Download..
0001_{16}	The target hardware accepts the operation. Operation not yet started.	NO DISPLAY Note: for load status, accepted but not started or Global operation Status Code.	.LCI .LCS	.LUI .LUS	.LND .LNO .LNS
1000_{16}	The target hardware doesn't accept the operation.	*"Operation Denied. " Target hardware text required.	.LCI	.LUI	.LND .LNO
1002_{16}	This operation is not supported by the target.	*"Operation not supported by the target." No target hardware text.	.LCI	.LUI	.LND .LNO
0002_{16}	Operation in progress	No Display but Entertain the user.	.LCS	.LUS	.LNS
0003_{16}	Operation completed without error	*"Operation Completed." No target hardware text.	.LCS	.LUS	.LNS
1003_{16}	Operation aborted by the target hardware	*"Operation aborted by the Target Hardware. " Target hardware text required Note: if aborted due to transmission failure, the reason should be noticed.	.LCS	.LUS	.LNS
1004_{16}	Operation aborted by the data loader.	*"Operation aborted by the Data Loader. " No target hardware text.	.LCS	.LUS	.LNS
1005_{16}	Operation aborted by the operator.	*"Operation cancelled by the operator." No target hardware text	.LCS	.LUS	.LNS
1007_{16}	Load Part Number failed	<"Load Part Number name> failed. " Target hardware text required		.LUS	.LNS
	Or Download File failed	<"File name> failed." Target hardware text required.			

* For the Display, the term "operation" must be replaced by the name of the operation: Information, Uploading or Downloading.

COMMENTARY

The reader should assume that a load is composed of a header file and several data files.

The Airborne Data Loader (ADL) should display the text of the previous table upon reception of the status code from the corresponding file. It is the responsibility of the target hardware to validate the load operation. ADL should not have to interpret this Status Code, regarding the status of each data or status files transfer, for end of operation status display.

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ATTACHMENT 1
ARINC 615A ADL AND PDL CONNECTOR SIGNAL ASSIGNMENTS

Function			ADL and PDL	Note	
Reserved for ARINC 615 compatibility			1	6	
Reserved for ARINC 615 compatibility			2	6	
Reserved for ARINC 615 compatibility			3	6	
Reserved for ARINC 615 compatibility			4	6	
Reserved for ARINC 615 compatibility			5	6	
TD + (ARINC 664)			6	5	c-2
TD - (ARINC 664)			7	5	c-2
Reserved for ARINC 615 compatibility			8	6	
Reserved for ARINC 615 compatibility			9	6	
Reserved for ARINC 615 compatibility			10	6	
Reserved for ARINC 615 compatibility			11	6	
Reserved for ARINC 615 compatibility			12	6	
Reserved for ARINC 615 compatibility			13	6	
Reserved for ARINC 615 compatibility			14	6	
Reserved for ARINC 615 compatibility			15	6	
Reserved for ARINC 615 compatibility			16	6	
Future spare			17	1	
Link – A			18	3	
Link – B			19	3	
115 Vac input	Hot		20	4	
Chassis ground / Signal electrical reference		GND	21		
115 Vac input	Cold		22	4	
RD + (ARINC 664)			23	5	c-2
Reserved for ARINC 615 compatibility			24	6	
Reserved for ARINC 615 compatibility			25	6	
ARINC 429 General output bus		A	26	2	
ARINC 429 General output bus		B	27	2	
Shield (ARINC 664)		GND	28		c-2
Reserved for ARINC 615 compatibility			29	6	
Reserved for ARINC 615 compatibility			30	6	
Reserved for ARINC 615 compatibility			31	6	
Reserved for ARINC 615 compatibility			32	6	
Reserved for ARINC 615 compatibility			33	6	
Reserved for ARINC 615 compatibility			34	6	
ARINC 429 General input bus		A	35	2	
ARINC 429 General input bus		B	36	2	
Reserved 28 Vdc input	Hot		37		
Reserved 28 Vdc return	Cold		38		
RD - (ARINC 664)			39	5	c-2
RS-232 Transmit		TXD	40		
RS-232 Receive		RXD	41		
Future Spare			42	1	
Future Spare			43	1	
Future Spare			44	1	
Future Spare			45	1	
Future Spare			46	1	
Future Spare			47	1	
Logic common		(GND)	48		
Logic common		(GND)	49		
Reserved for ARINC 615 compatibility			50	6	
Reserved for ARINC 615 compatibility			51	6	
Reserved for ARINC 615 compatibility			52	6	
Reserved for ARINC 615 compatibility			53	6	

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ATTACHMENT 1
ARINC 615A ADL AND PDL CONNECTOR SIGNAL ASSIGNMENTS

REFERENCE TO NOTES

[1]. Future Spare. Connector pins marked "Future Spare" in the Standard Interwiring list are available for assignment as the airline industry desires. However, if the interchangeability for the system specified in Section 1.4 of this Report is to be retained, any such assignments thought necessary must be coordinated through ARINC and approved by the industry prior to being made. Contact hardware need not be provided in positions labeled "Future Spare" but may be furnished at the equipment manufacturer's discretion. Contact positions labeled "Future Spare (Contact)" will be the first to be selected for use if and when additional contact assignments are needed. Contact positions in equipment mounted service connectors labeled "Future Spare (Contact)" should be furnished with contact hardware (pin or socket as appropriate) and provisions made within the equipment for their easy use.

COMMENTARY

When the definition of a new unit starts with no "Future Spare" signals on the standardized connector, evolutions of this standardized unit will be limited. Consequently, some spare functions on the connector are requested. In ARINC 615 standard, loading a computer by RS 232 link is mentioned but not clearly defined. In this new standard, the RS 232 interface is only mentioned for factory and maintenance tests purpose. For such purpose, hardware flow control is not requested. Discrete input and output signals "RS 232 Baud sel bit 0 to 2" and "select ARINC 429/RS 232" of the previous standard also are not requested for this purpose. Functions 42 to 47 are changed to "Future Spare" functions for this standard. Compatibility with old ARINC 615 loader (PDL and ADL) must be managed if requested by the equipment manufacturer (cut the line in the adaptation cable or extract the pin from the connector).

[2]. Twisted and Shielded Wires. Wires should be shielded or twisted and shielded as indicated with an insulating jacket over the shield. The shield for each data bus should be connected to the appropriate Bus Shield pin (where provided). This should be a single-point connection at the common Bus Shield pin (not "daisy chained").

[3]. Link A,B. The pins Link A and Link B are electrically connected within the data loader. The use of this connection may be defined by the airborne computer designer.

[4]. 115 Vac Shielding. The 115 Vac (H) and 115 Vac (C) should be shielded or twisted and shielded with an insulating jacket over the shield. The shield should be attached to chassis ground.

[5]. Twisted Pair Ethernet (TPE) Signal. Wiring should conform to ARINC Specification 664.

[6]. Reserved for ARINC 615 Compatibility. These functions are used in the ARINC 615 connector. For compatibility purposes, they are not assigned on this connector.

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ATTACHMENT 2
LOAD SCENARIOS

2-1 LOAD SCENARIOS

This Attachment describes different scenarios which must be supported through the Data Load Protocol.

Scenarios	Condition, Hypothesis, Constraints	Actions	Operations
Configuration Equipment Identification	This is the capture of the status and configuration of the Target Hardware (indicated by its EQUIP_ID combined with its occurrence) This scenario can be performed at any time	Operator looks for Target Hardware's configuration Target answers with the up-to-date verification information If needed, the target performs appropriate operations to guarantee P/N validity.	Configuration query from ADL/PDL
Software Modification A new software product is released, all occurrences of Target Hardware must be updated	This is a predicted Airborne operation Incompatibility of the loadable code must be checked beforehand Verification of the configuration must be performed afterward.	Operator uses Service Bulletin and up-date media package to load Target(s) Operator selects (a) load transfer(s) on ADL/PDL Incompatibility is checked by operator after load completion	Upload from ADL/PDL software check by the Target Hardware
Repair A new Target Hardware is installed in place of a failed Target Hardware	This is an unpredicted operation (Operator might not have any Service Bulletin to check P/N!!) Incompatibility of the loaded code must be checked before Verification of the configuration must be performed after.	Operator uses available on-board sources (can be ADL/PDL, can be a file server) Operator selects (a) load transfer(s) on ADL/PDL Incompatibility is checked by operator after load completion	Upload from ADL/PDL software check by the Target Hardware
Cross-load		This activity is not part of 615A	
Automatic Data Extraction data must be down loaded automatically after appropriate media insertion	The Target Operation is not disrupted during the Down Load operation Target may be interrupted if required to be in a data load mode to be downloaded	Operator inserts the pre-programmed media, supporting the file defining the download to perform. Download is performed automatically	Disk Defined Download to ADL/PDL
User Data Extraction A block of data must be down loaded according to the choice proposed by the target	The Target Operation is not disrupted during the Down Load operation Target may be interrupted if required to be in a data load mode to be downloaded	Operator selects the target to download and storage file destination Operator then selects the software product to be downloaded, according to the list proposed by the target Operator activates the download transfer on ADL/PDL	Operator Defined Download to ADL/PDL
Continuous in flight download		This activity is not part of ARINC 615A. The data loader must be able to answer standard TFTP request.	

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ATTACHMENT 2
LOAD SCENARIOS

2-2 CONFIGURATION QUERY FROM ADL/PDL

This operation is made via the ADL/PDL to ease maintenance crew operation.
It may also be performed via the Onboard Maintenance System.

Op#	Originator	Action	Means	Result
1.	Operator	Select destination Target	Network configuration file (see note 1) Destination selection dialog on ADL/PDL	The Data Loader gets the Aircraft Target Name list The Destination Dialog asks for: Target Name Occurrence (1, 2, 3, ...). Each selection level could present selectable choices for next level.
2.	Operator	Request configuration data of the Target	Configuration dialog on ADL/PDL	Load Server Application sends a Configuration request message to Load Device Application. Load Device Application performs appropriate operation to guarantee integrity of identification information. Load Device Application then answers a Configuration message to Load Server Application.

Note 1 The ADL/PDL must get information about the list of installed Target Hardware, their MAC address and their IP address. The file that contains these data must be readable by all ADLs/PDLs. It corresponds to the network configuration file. No broadcast request is used, as the basic principle uses static addresses defined in the network configuration file.

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ATTACHMENT 2
LOAD SCENARIOS

2-3 UPLOAD FROM ADL/PDL

2-3.1 UpLoad Preparation

The aim of the first sub-phase is to define the job to be performed, and to check the availability of the implied elements as far as possible.

Op#	Originator	Action	Means	Result
1.	Operator	Get Service Bulletin (if any) and disk package	Documentation Disk package	The operator knows procedure to be applied
2.	Operator	Select destination Target	A/C target Name list Destination selection dialog on ADL/PDL	The Data Loader gets the Aircraft Target Name list The Destination Dialog asks for: Target Name Occurrence (1,2,3...)
3.	ADL/PDL	Apply Load Request condition to target(s).	Load Request message	The Data Loader checks the presence of the selected target and the availability of the load phase. This initializes the load session for this target. If target response is OK, step 4 is started by the target.
4.	Target	Monitor target status, file transfer and load programming progress	Information Evolution Data Loading monitoring dialog	Target Hardware sends periodic target status and DL updates monitoring dialog with the target status response This step is performed in parallel with other steps until end of load session.
5.	Operator	Select load source media	A/C Target Name list (see note 3) Source media selection dialog on ADL/PDL	The Source Dialog proposes a list of sources : ADL Floppy drive ADL PCMCIA slot File Server (optional) After selection, the Data Loader reads the » LOADS.LUM » file of the selected source The Source Dialog proposes a list of SW load P/N extracted from the « LOADS.LUM » file, in compliance with the EQUIP_ID of the selected destination target

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ATTACHMENT 2
LOAD SCENARIOS

2-3.1 UpLoad Preparation (cont'd)

Op#	Originator	Action	Means	Result
6.	Operator	Select one or more load(s)	Available loads for the target Load selection dialog on ADL/PDL	One or more load(s) are selected by the operator in the load source media. The ADL/PDL produces the appropriate load list. If possible, the ADL/PLD checks the presence of all files involved in the selected load(s) and the correct file size ("if possible" means if files can be accessed, caution with floppies). The ADL/PDL sends the load list to the target The target reads the load list and checks the validity of the request, and answers ACK or NACK.
7.	ADL/PDL	Indicate the pre-check result to the operator.	Information dialog on ADL/PDL	If errors have been detected in the previous step, messages inform the operator and ask him to continue or abort. If answer is "continue", the next sub-phase is activated (Upload process). If no error, the next sub-phase is activated (Upload process). At this step, it's not possible to estimate load duration.

Note 3 If a remote source is to be used, the ADL/PDL must be able to access it as a remote network peripheral, without impact on the Data Load Protocol.

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ATTACHMENT 2
LOAD SCENARIOS

2-3.2 UpLoad Process

The aim of this sub-phase is to perform the effective load, erase and program the memory, and control the correct programming. During this sub-phase, the ADL/PDL is a simple TFTP server, managed by the target.

Op#	Originator	Action	Means	Result
1.	ADL/PDL	Continue the target status, monitoring load process progress.	Information Evolution Data Loading monitoring dialog load estimation time (if available)	Target Hardware sends periodic Target Hardware status and DL updates monitoring dialog with the Target Hardware status response If possible, the target gives the estimate duration of the process.
2.	Target	Send a read request for one file (data file or header file).	Read request message TFTP file transfer	ADL/PDL checks the availability of the file. If not available, message asks the operator to insert appropriate media or indicates the error. Target reads the file and programs appropriate memories These transfers are made as many times as identified in the load list and the associated header files.
3.	Target	Check integrity of the loaded code.	Information Evolution Data Loading monitoring dialog	The target may compute control values of the loaded software The target reports the final result of the load(s) to the ADL/PDL.
4.	ADL/PDL	Report the final result of the load to the operator	Information Evolution Data Loading monitoring dialog	All previous steps are completed, including step 1 : the load session is closed. Result of correct or incorrect load is reported to the operator

2-3.3 UpLoad Termination

The aim of this sub-phase is to ask an operator for acknowledgment of the selected load.

Op#	Originator	Action	Means	Result
1.	Operator	Acknowledge final result of the load	ADL/PDL menu	The Operator can prepare for next phase
2.	Operator	Check new P/N component list	Service Bulletin (if any)	The Operator knows that operation has been completed

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ATTACHMENT 2
LOAD SCENARIOS

2-4 DISK DEFINED DOWNLOAD TO ADL/PDL

Op#	Originator	Action	Means	Result
1.	Operator	Give definition of the download to be performed.	Pre-programmed media Function menu dialog on ADL/PDL	Insertion of the pre-programmed media (floppy or PC-CARD) into the ADL/PDL Selection of the download function on the ADL/PDL Selection of the source media (floppy or PC-CARD).
2.	ADL/PDL	List the automatic download definition files Read the selected automatic download definition file	Pre-programmed media Automatic download definition files	If more than one download definition file exists, the ADL/PDL asks the operator to choose one. ADL/PDL knows the target to be downloaded and the destination media ADL/PDL checks the availability of the destination media
3.	ADL/PDL	Apply Download Request conditions to target(s).	DownLoad Request message	The Data Loader checks the presence of the selected target and the availability of the download phase. This initializes the load session for this target. If target response is OK, step 4 is started by the target.
4.	Target	Monitor target status, file transfer and storage progress	Information Evolution DataLoading monitoring dialog	Target Hardware sends periodic target status and DL updates monitoring dialog with the target status response This step is performed in parallel with other steps until end of download session.
5.	ADL/PDL	Send the automatic download definition file to the target	Transfer of the definition file	Target receives the download definition file and knows the job to be performed After this step, the ADL/PDL is a simple TFTP server, managed by the target.
6.	Target	Write file(s) to the destination media of the ADL/PDL.	Write request message TFTP file transfer Information Evolution Data Loading monitoring dialog	The ADL/PDL receives the files produced by the target and stores them into the destination media

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ATTACHMENT 2
LOAD SCENARIOS

2-4 DISK DEFINED DOWNLOAD TO ADL/PDL (cont'd)

Op#	Originator	Action	Means	Result
7.	ADL/PDL	Report final result of the load to the operator	Information Evolution Data Loading monitoring dialog	All previous steps are completed, including step 4 : the download session is closed. Result of correct or incorrect load is reported to the operator
8.	Operator	Acknowledge final result of the load	ADL/PDL menu	The Operator can prepare for next phase

2-5 OPERATOR DEFINED DOWNLOAD TO ADL/PDL

Op#	Originator	Action	Means	Result
1.	Operator	Give definition of the download to be performed.	Function menu dialog on ADL/PDL	Selection of the download function on the ADL/PDL The Data Loader gets the Aircraft Target Name list The Destination Dialog ask for: Target Name Occurrence (1,2,3...) Selection of the source media (floppy or PC-CARD).
2.	ADL/PDL	Target Hardware selection Destination Media selection	A/C target name list Destination selection dialog on ADL/PDL	The Data Loader checks the presence of the selected media
3.	ADL/PDL	Apply Load Request condition to target(s).	Download Request message	The Data Loader checks the presence of the selected target and the availability of the download phase. This initializes the load session for this target. If target response is OK, the step 4 is started by the target.
4.	Target	Monitor target status, file transfer and storage progress	Information Evolution Data Loading monitoring dialog	Target Hardware sends periodic target status and DL updates monitoring dialog with the target status response This step is performed in parallel with other steps until end of download session.
5.	Target	Send the list of downloadable software parts and approximated sizes	Download menu file	DL receives the list of downloadable parts and proposes it to the operator.

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ATTACHMENT 2
LOAD SCENARIOS

2-5 OPERATOR DEFINED DOWNLOAD TO ADL/PDL (cont'd)

Op#	Originator	Action	Means	Result
6.	Operator	Select one or more download parts	Available downloadable parts for the target Download selection dialog on ADL/PDL	One or more download(s) are selected by the operator The ADL/PDL sends the selected list to the target The target reads the load list and checks the validity of the request, and answers ACK or NACK.
7.	Target	Write file(s) to the destination media of the ADL/PDL.	Write request message TFTP file transfer Information Evolution Data Loading monitoring dialog	The ADL/PDL receives the files produced by the target and stores them into the destination media
8.	ADL/PDL	Report to the operator final result of the load	Information Evolution Data Loading monitoring dialog	All previous steps are completed, including step 4 : the download session is closed. Result of correct or incorrect load is reported to the operator
9.	Operator	Acknowledge final result of the load	ADL/PDL menu	The Operator can prepare for next phase

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ATTACHMENT 3
FIND IDENTIFICATION OF NETWORK DEVICES (FIND) PROTOCOL

1.0 Introduction

c-1 The Find Identification of Network Devices (FIND) Protocol allows a client (hereafter called the initiator) to dynamically obtain:

MAC address

IP address

c-2 Identity and functional characteristics for each available FIND host on the network

c-2 ARINC Report 615A allows the data loader (the initiator) to identify all the 615A loadable target hardware devices (the FIND hosts) present at the time of the request. After registering all the available target hardware devices, the data loader can present the list of loadable devices to the operator for selection.

FIND should be run at least once before any data load operations are performed.

c-2 Before each operation the data loader may run the FIND request and register all the answers in compliance with the FIND definition.

The FIND protocol does not support dynamic address allocation. It assumes that all IP address assignments have been performed by some means, for example, static c-2 configuration and dynamic allocation, prior to the FIND request.

c-2 Since the initiator does not know how many FIND hosts are on the network or their addresses, a broadcast c-2 message is used for the FIND request by the data loader. Each FIND host responds with a unicast answer directed to the data loader address.

c-2 The FIND protocol is implemented on top of the UDP datagram. FIND data is encapsulated in a UDP datagram, using the well-known port number 1001 (in decimal notation). The same port number should be used for request and answer packets.

c-1 MAC address and IP address information are supported by the classical fields within Ethernet frame and IP packet.

2.0 Overview

c-2 The initiator or Data loader sends a broadcast message requesting a response from all FIND hosts on the network, and then waits for responses. The initiator allows a 3-second window for responses. All valid responses are registered as targets for possible loading activity. When the 3 seconds window expires, the registration process is closed.

c-2 The Target hardware or FIND host should respond to the request with an answer message within a 2 seconds time-out or risk not being registered by the Data loader. The c-2 information needed by the initiator may come in the message structure or the FIND packet data. The values for the target hardware MAC and IP addresses are supported by the classical MAC and IP frame fields in a UDP/IP datagram. Other information, such as LRU Identification Position, is supported by UDP payload itself.

COMMENTARY

The difference between the data loader time out and the target hardware time out is due to network propagation delays. c-2

No errors are managed in the FIND protocol. If the target hardware receives an illegal or invalid request, it should not answer. If the data loader receives an invalid answer, the data loader ignores the answer.

3.0 FIND Packets

c-2

There are two types of FIND packets. One is a Information Request packet and the other is a Information Answer packet. Each packet contains a header and a data portion as shown:

2 byte Header	Variable length Data
Opcode	Data List

The FIND packet header contains the opcode associated with that packet: c-2

Opcode	Operation
1	Information ReQuest (IRQ)
2	Information ANswer (IAN)

The data portion of the FIND Information Answer packet contains a list of parameters described in Section 6. List of Parameters, of this attachment. c-2

4.0 Information Request Packet

Source MAC address should be the MAC address of the data loader (or the initiator).

Destination MAC address should be a unicast or the broadcast MAC address.

Source IP Address should be the IP address of the Data loader (or the initiator).

Destination IP address should be a unicast or a broadcast IP address (full broadcast i.e., 255.255.255.255 or class-relevant broadcast e.g., 220.130.48.255 for a class C).

Opcode value should be IRQ.

Data list should be a null ASCII string with terminator: $00_{16.}$ c-1

Data list is terminated by $10_{16.}$ c-2

Section 5.0 Information Answer Packet

Source MAC address should be the MAC address of the Target hardware (or the subscriber).

Destination MAC address should be the source MAC address from the associated request.

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ATTACHMENT 3
FIND IDENTIFICATION OF NETWORK DEVICES (FIND) PROTOCOL

c-2 Source IP Address should be the IP address of the target hardware Identifier (or the subscriber).

COMMENTARY

As one target hardware may act as a proxy or concentrator, one target hardware may produce more than one answer to the same request.

Destination IP address should be the source IP address from the associated request.

Opcode value should be IAN.

Data list is a list of null terminated character strings, each containing the variable value of the structure defined in

Empty field is coded by only one byte containing the ASCII string terminator or null byte.

Available string should not include “null byte” in part of the string.

c-2 A data list should be terminated by 10_{16}.

6.0 List of Parameters

c-2 For each target hardware Identifier, the list of parameters to be obtained by the data loader to perform its initialization is:

c-2 Target hardware Identifier (e.g., “02DA”),

Target Type Name (e.g., “FM”),

Target Position (e.g., “1”, or “L”),

Literal Name (e.g., “FM LEFT”).

Manufacturer code (e.g., “3K7”).

COMMENTARY

c-1 The manufacturer should be identified by the method defined in ARINC Report 665.

For the Information Answer packets, the values of above parameters, are encoded in null terminated strings, in the same order as the above list, with the size limits defined below (not including the ASCII string terminator):

Target hardware Identifier: 15 characters maximum, c-2

Target Type Name: 8 characters maximum,

Target Position: 8 characters maximum,

Literal name: 20 characters maximum.

Manufacturer Code: 3 characters. c-1

7.0 Examples

7.1 Packet Example

Data loader Request

Opcode (IRQ)	0001_{16}
ASCII string terminator	00_{16} c-1
Packet Terminator	10_{16} c-2

Target hardware identifier answer c-2

Opcode (IAN)	0002_{16}
Target hardware identifier (text)	02DA c-2
String terminator (ASCII NULL)	00_{16}
Target hardware Type Name (text)	FM
String terminator (ASCII NULL)	00_{16}
Target hardware position (text)	L
String terminator (ASCII NULL)	00_{16} c-1
Literal Name (text)	FM LEFT
String terminator (ASCII NULL)	00_{16}
Manufacturer code	3K7
String terminator (ASCII NULL)	00_{16}
Packet terminator	10_{16} c-2

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ATTACHMENT 4
TIME-OUT AND RETRY NUMBER DEFINITION

1. Introduction

ARINC Report 615A protocol is supported by two layers:

A TFTP layer to transfer files (protocol files and data files). This layer defines one exchange by one request and its answer (regardless of the type of packet)

A DLP layer to manage general process and coordinate activities between the data loader and the target hardware.

There are two levels where time-out and retry should be defined. One level is managed by the TFTP layer in case of a failed TFTP exchange, and another level is managed by the DLP layer due to a failed TFTP file transfer. (See Figures A4-3, A4-4 and A4-5).

TFTP level: The TFTP layer is activated to transfer one file, and manages objects corresponding to "one exchange". The TFTP layer does not manage links between file transfers.

Inside one TFTP file transfer, each exchange should be acknowledged within the "TFTP time-out" delay window. Time-out expiration without acknowledgement should induce one error for this exchange. Such an error is not a fatal error.

c-2 Inside one TFTP file transfer, each exchange may be retried if a recoverable error has been detected (CRC error, time-out error, etc.). After the number of retries equals the TFTP VALUE (TFTP retry number) for the same exchange and an error still results, the TFTP file transfer should result in an error. The DLP upper layer should be informed of such an error.

DLP level: The DLP layer is activated for each data load operation (upload, download, etc.) for each target hardware, and manages objects corresponding to "one TFTP file transfer". The DLP layer has no knowledge of the "TFTP exchanges".

When one error has been signaled by the TFTP layer, the DLP layer should retry this failed TFTP file transfer (the entire file). The number of retries allowed at the DLP layer is noted as "DLP Retry Number". After the DLP Retry Number has been exceeded for the same TFTP file transfer and an error still exists, this DLP file transfer should be declared as a fatal error. The upper layer (application) should be informed of such a fatal error. For the ARINC 615A protocol definition this implies the end of the data load operation. Between any two TFTP file transfers, the maximum delay is defined and noted as "DLP Time-Out". This Time-Out expiration, between the end of one TFTP exchange and the beginning of another TFTP exchange, induces a DLP error for the data load operation. This Time-Out applies only to some peer of a TFTP file transfer. Peers are noted as initial and secondary TFTPs. The "initial and Secondary" meanings are chronological (the initial TFTP file transfer precedes the secondary one).

This error is classified as a fatal error, without any allowed retries. If the actual DLP Time-Out Value is too constraining, it should be increased. A DLP Retry Number is not applicable if a DLP Time-Out error, a fatal error, occurs. c-2

Protocol reliability is dependent on accurate definition of the Time-Out (TO) and Retry Number. It is up to the client to activate the iterations of the Trivial File Transfer Protocol (TFTP) file transfer in accordance with the Data Load Protocol (DLP) Retry Number. The purpose of this attachment is to define Time-Out and Retry Numbers for the TFTP and the DLP protocols. c-2

2. Overview

For each protocol, two classes of elements are defined, with different characteristics for each class:

The Time-Out (TO) is characterized by its definition, value, constraints and a chronological illustration.

The Retry Number is characterized by its definition, value and constraints.

3. TFTP

3.1 TFTP Time Out

Definition:
For each end-system (data loader or target hardware) the TFTP time-out is the time measured between the emission of one TFTP packet (whatever the type of TFTP packet) and the reception of the associated answer packet for the same TFTP exchange (whatever the type of TFTP packet). The generic notation is TFTP-TO.

Value:
The TFTP Time-Out (TFTP-TO) value is indicated in seconds. The time-out value should be 2 seconds. c-2

Constraints:
It is necessary to split this constraint into two parts: the network constraint and the subscriber constraint:

Network constraint: the transmission duration of one TFTP packet should not be longer than TFTP-TO / 4. This is to account for the sending and receiving bus transfer time. c-2

Subscriber constraint: the processing duration between the reception of one TFTP packet and the emission of the associated packet should not be longer than TFTP-TO / 2. This is the time left over after subtracting out the sending network constraint and subtracting out the receiving network constraint. c-2

See Figure A4-1.

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ATTACHMENT 4
TIME-OUT AND RETRY NUMBER DEFINITION

3.1 TFTP Time Out (cont'd)

COMMENTARY:

Using a large TFTP block size, the TFTP-TO Value would be sufficient in a general case. For a particular case, the block size parameter would be

c-2 reduced to obtain compliance with this time-out value.

No minimal Value is given for the duration between the reception of one packet and the emission of the associated answer packet. According to the policy of the target hardware, this delay may be very short.

3.2 Retry Number

Definition:
For each end-system (data loader or target hardware) the TFTP Retry Number is the number of times a TFTP exchange is attempted after one initial failed exchange (inside the same TFTP session), which may be recoverable. It is the number of allowed retries before declaring a TFTP failure (via an error code) to the DLP. The Retry Number value is in addition to the first exchange, e.g., a retry number value of 2 implies that the maximum number of exchange tries is 3. That is 1 original attempt plus 2 retries for a total of 3 tries. The
c-2 Retry Number is applicable for each exchange inside one TFTP. The retry number has no cumulative notion between various exchanges in the same TFTP. All errors relative to the same exchange are accumulated to compare the number of tries to the TFTP Retry Number. The algorithm applies to any TFTP exchange (initial request, acknowledge, data transfer, last acknowledge, etc).

Value:
The TFTP Retry Number value is indicated as an integer value. The required value is 1.

Constraints:
Any TFTP should use the same Retry Number value. Setting specific retry numbers according to the file type (protocol file, data file, etc.) could increase the complexity of the implementation without any real improvement of reliability.

COMMENTARY

As TFTP is based on the UDP stack, and as the UDP cannot ensure a correct transfer of any block, it is mandatory to provide an error recovery capability to prevent a UDP transfer error (This principle is defined in the TFTP RFC).

A value of 1 is based on an estimation of the reliability on a Standard Ethernet Network.

Note that this Retry Number is linked to the reliability of the network itself (communication aspect), and is not sized to support application errors (such as insufficient CPU allocation, packet lost in the End System Layer, insufficient delay to compute CRC, etc.). The assumption is that the protocol is correctly implemented.

The recommendation is to improve the reliability of the network itself, rather than to increase the Retry Number. c-2

3.3 Sorcerer's Apprentice Syndrome

According to the TFTP specification, as described in the RFC 1123, the implementation should contain the fix for a potential Sorcerer's Apprentice problem. The sender, that is, the side originating the data packet, should never re-send the current data packet on receipt of a duplicate acknowledge. c-1

4. Data Load Protocol (DLP)

4.1 Time Out

Definition:
Applicable to a data loader only, the DLP Time-Out is the time between the emission of the last packet of the initial TFTP (acknowledge) and the reception of the first packet of the secondary TFTP (write request). The Generic notation is DLP-TO.

This DLP Time-Out (DLP-TO) should apply to the sequences of initial TFTP file transfer and the secondary TFTP file transfer listed below. TFTP is referenced using the extension of the protocol file used during the associated TFTP. c-2

Initial TFTP file transfer	Secondary TFTP file transfer
.LCI	.LCS
.LCS	.LCS
.LUI	.LUS
.LUS	.LUS
.LNI	.LNS
.LNS	.LNS

A DLP Time-Out error is classified as a fatal error, without any allowed retry. If an actual DLP Time-Out Value is too constraining, it should be increased. A DLP Retry Number is not applicable if a DLP Time-Out error, a fatal error, occurs. c-2

If the .LCI-.LCL sequence is initiated before expiration of the DLP-TO, the target hardware should not produce the status file (.LCS). The operation is intended to be performed rapidly without the proliferation of status files being generated. c-2

Value:
The DLP Time-Out value is indicated in seconds. The time-out value should be 13 seconds, based on the allowed refresh rate of operator information, balanced with the workload induced by such a process (on both sides, the data loader and target hardware). c-2

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ATTACHMENT 4
TIME-OUT AND RETRY NUMBER DEFINITION

4.1 Time Out (cont'd)

Constraints:
To conform to the DLP-TO time-out, the target hardware should have a Duration Time, the time between the reception of the last packet of the previous TFTP file transfer and the emission of the first packet of the next TFTP file transfer, in compliance with the following equation:

Duration_time < DLP_TO
- [DLP_retry x (TFTP_retry + 1) x TFTP_TO]
- [TFTP_retry x TFTP_TO]
- 2 x TFTP_TO / 4

Figure A4-2 illustrates DLP TimeOut for example values as follows:

DLP_retry = 2
TFTP_retry = 2
TFTP_TO = 2 seconds

As an example of application of the equation, assuming parameters of:

DLP_retry = 1
TFTP_retry = 1
TFTP_TO = 2 seconds

c-2

and assuming a duration time, DLP_TO = 6 seconds, when applied to the equation, yields:

DLP_TO = 6
+ (1 x (1 +1) x 2)
+ (1 x 2)
+ 2 x (2 / 4)

= 13 seconds.

Therefore, according to the equation and for the desired maximum duration time of 6 seconds, the DLP-TO should be set at 13 seconds.

4.2 Retry Number

Definition: For each end-system (data loader or target hardware) the DLP Retry Number is the number of times a TFTP file transfer is attempted again, after one initial failed TFTP file transfer (the entire file), which may be recoverable. It is the number of allowed retries at the DLP level before declaring a DLP failure (via an error code). The DLP Retry Number Value is in addition to the first TFTP file transfer, e.g., a DLP Retry Number value of 2 implies that the maximum number of TFTP file transfer tries is 3. That is 1 original attempt plus 2 retries for a total of 3 tries. A DLP Retry Number applies to each file to be transferred (data files and protocol files). The retry number has no cumulative notion between various TFTP file transfers in the same ARINC 615A operation.

c-2

Values:
The DLP Retry Number value is indicated as an integer value. The required value is detailed in the following table.

c-2

TFTP is referenced using the extension of the protocol file used during the associated TFTP.

TFTP Items	Retry Number
.LCI	1
.LCL	1
.LCS	1
.LUI	1
.LUR	1
.LUH	1
Data File	1
.LUS	1
.LND	1
.LNR	1
.LNS	1
.LNO	1
.LNL	1
.LNA	1

Constraints:
Today the DLP uses the same Retry Number Value for all files. Setting a specific Retry Number according to the file type (protocol file, data file, etc.) could increase the complexity of the implementation, without real improvement of the reliability.

c-2

COMMENTARY

The DLP, just as any layer protocol definition, should include the Retry Number and Time-Out definition.

As a data loader function may be supported by COTS products (for a shop Data loader, for example), with the associated COTS Operating system, a value of 1 is based on an estimation of the reliability on TFTP COTS stacks (such as a temporary error due to file access, etc.).

Note that this Retry Number is linked to the reliability of the communication stack, and is not sized to support application errors. The assumption is that the protocol is correctly implemented.

c-2

The recommendation is to improve the reliability of the network itself, rather than to increase the Retry Number.

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ATTACHMENT 4
TIME-OUT AND RETRY NUMBER DEFINITION

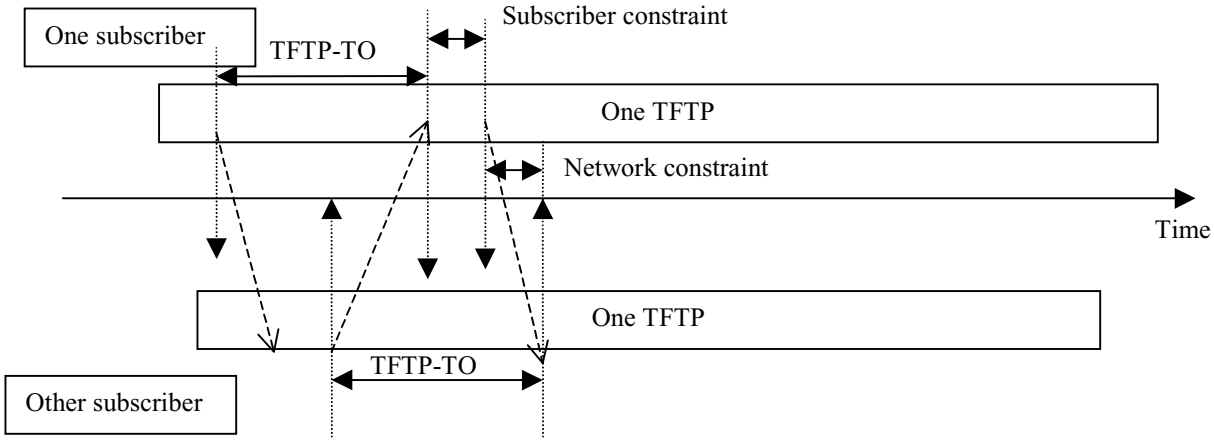


Figure A4-1 - TFTP Time-out

Example with : DLP_retry = 2
TFTP_retry = 2
TFTP_TO = 2 s

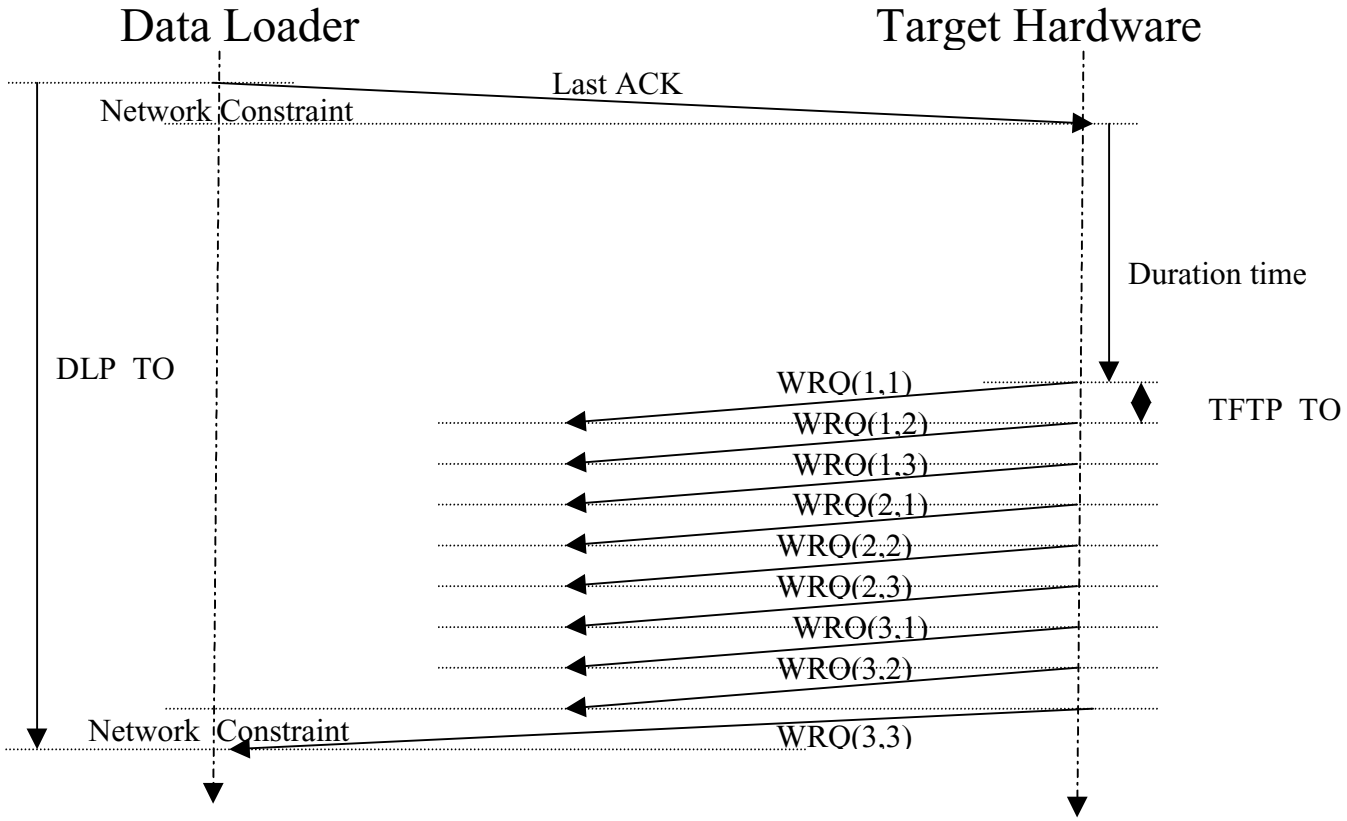


Figure A4-2 - DLP Time-out

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ATTACHMENT 4
TIME-OUT AND RETRY NUMBER DEFINITION

NOTATIONS for Figures A4-3, A4-4 and A4-5:

Send (y): Transmission of one TFTP packet, corresponding to the TFTP exchange number "y", whatever the type of TFTP packet.

Receive (y): Reception of one TFTP packet, corresponding to the TFTP exchange number "y", regardless of the type of TFTP packet. This packet acknowledges the send (y) packet. The send (y+1) will acknowledge the receive (y).

Such notation allows the same representation on the client or server side, without differences due to the equipment function (data loader or target hardware).

As a reminder, for TFTP, the WRQ and DATA packets are acknowledged by an ACK or ERROR packet, while RRQ and ACK are acknowledged by a DATA or ERROR packet.

FIGURES:

c-2

Figure A4-3 represents the usage and meaning of:
- TFTP Retry Number,
- TFTP Time-Out,
- DLP Retry Number,

on the side of the client with the request.

Figure A4-4 represents the usage of the same parameters on the side of the server with the request.

Figure A4-5 represents the usage of a DLP Time-Out on the data loader side.

COMMENTARY

For the DLP, each subscriber (data loader and target hardware) may play the part of a client and server.

The DLP Time-Out applies only to the data loader side, while the other parameters apply to both sides.

- After a fatal error, the data loader operation (Upload, download, etc.) should be completed.

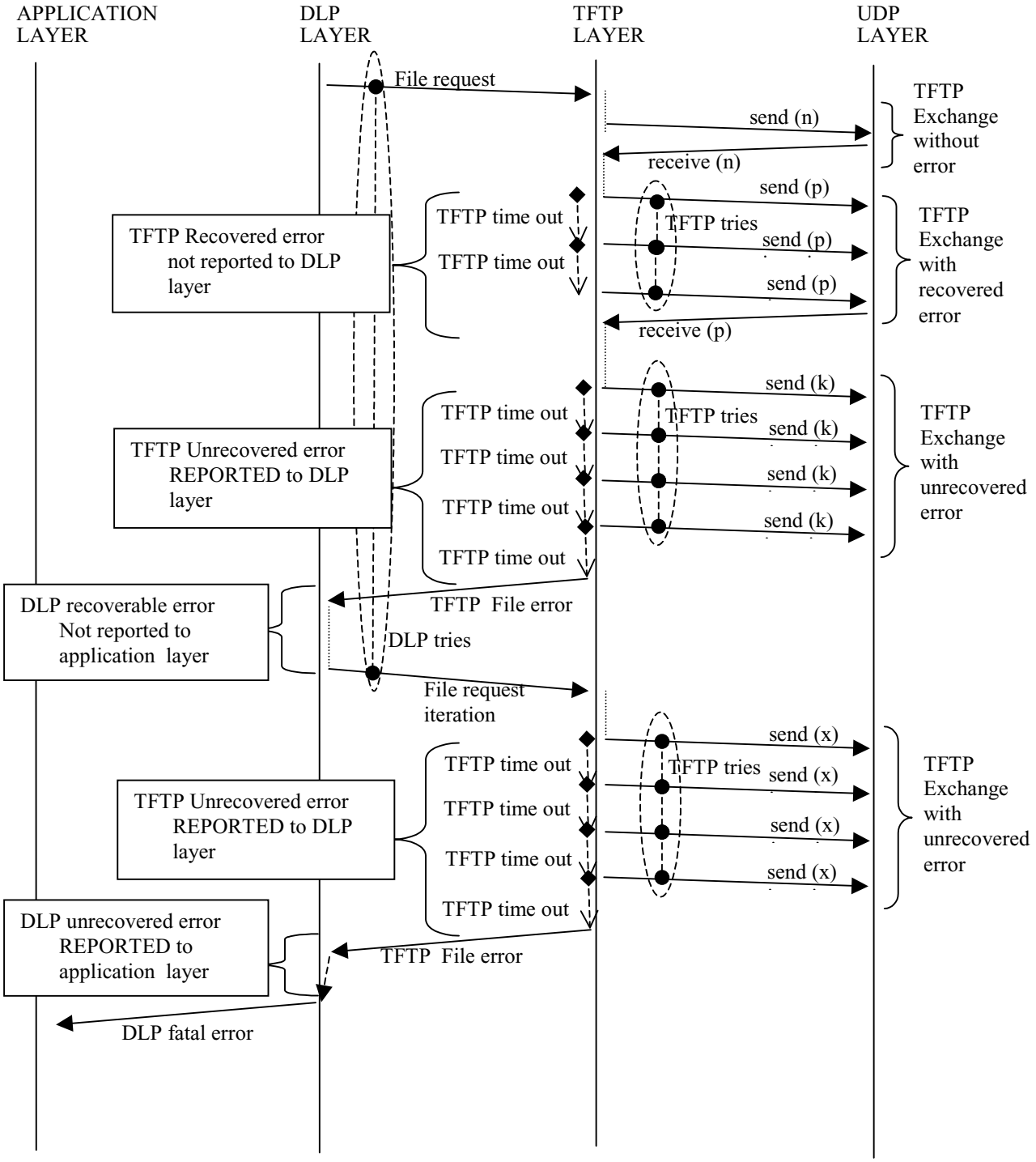
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ATTACHMENT 4
TIME-OUT AND RETRY NUMBER DEFINITION

FIGURE A4-3: CLIENT SIDE
(INITIATOR OF THE FILE REQUEST)

The Example below is with:

- A TFTP Retry Number set to 3 (a total of 4 possible tries for one exchange).
- A DLP Retry Number set to 1 (a total of 2 possible tries for one file transfer).



c-2

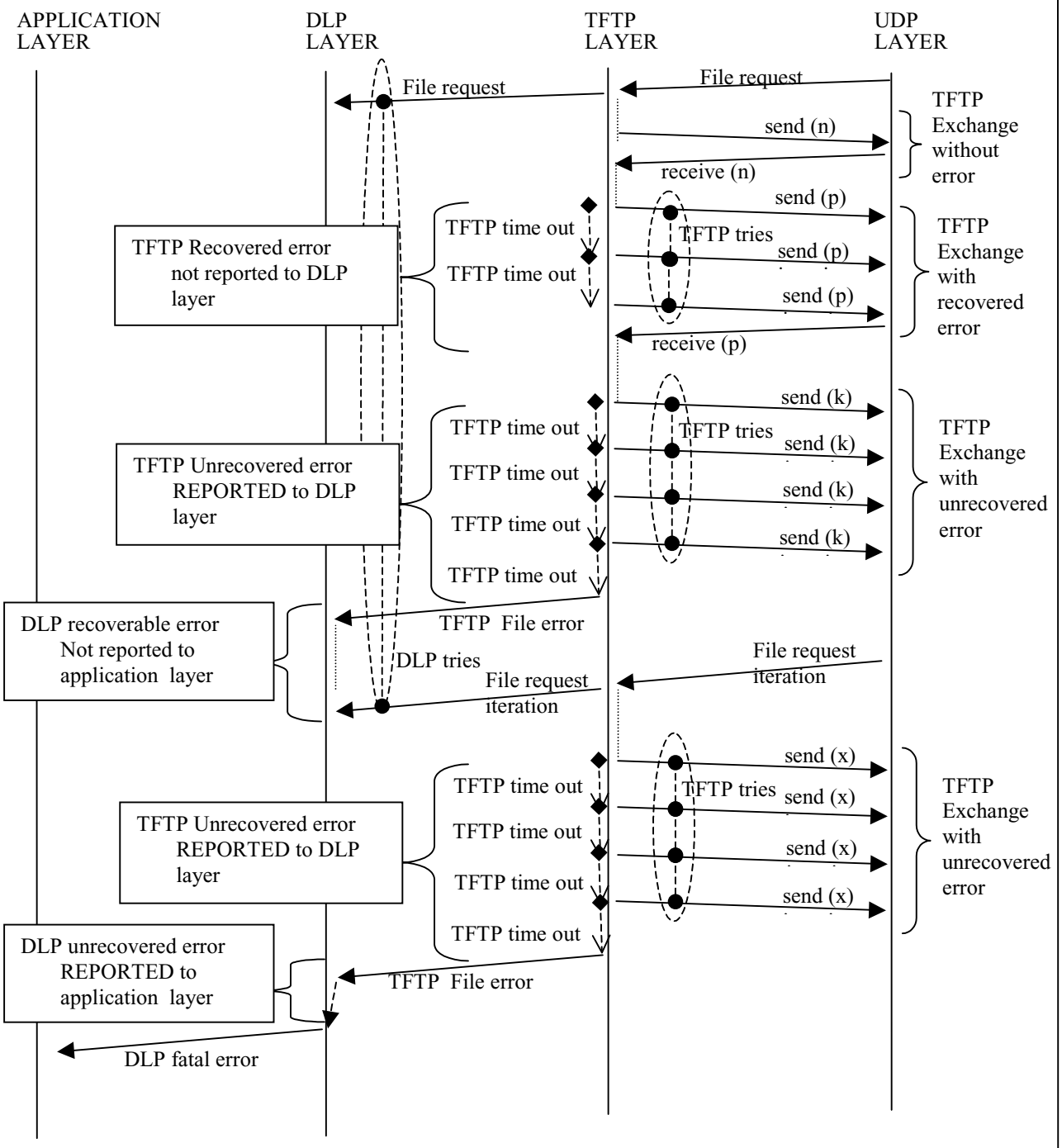
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ATTACHMENT 4
TIME-OUT AND RETRY NUMBER DEFINITION

FIGURE A4-4: SERVER SIDE
(SERVER OF THE FILE REQUEST)

The Example below is with:

- A TFTP Retry Number set to 3 (a total of 4 possible tries for one exchange).
- A DLP Retry Number set to 1 (a total of 2 possible tries for one file transfer).



c-2

COMMENTARY:

It is up to the client to activate the iterations of the TFTP file transfers.

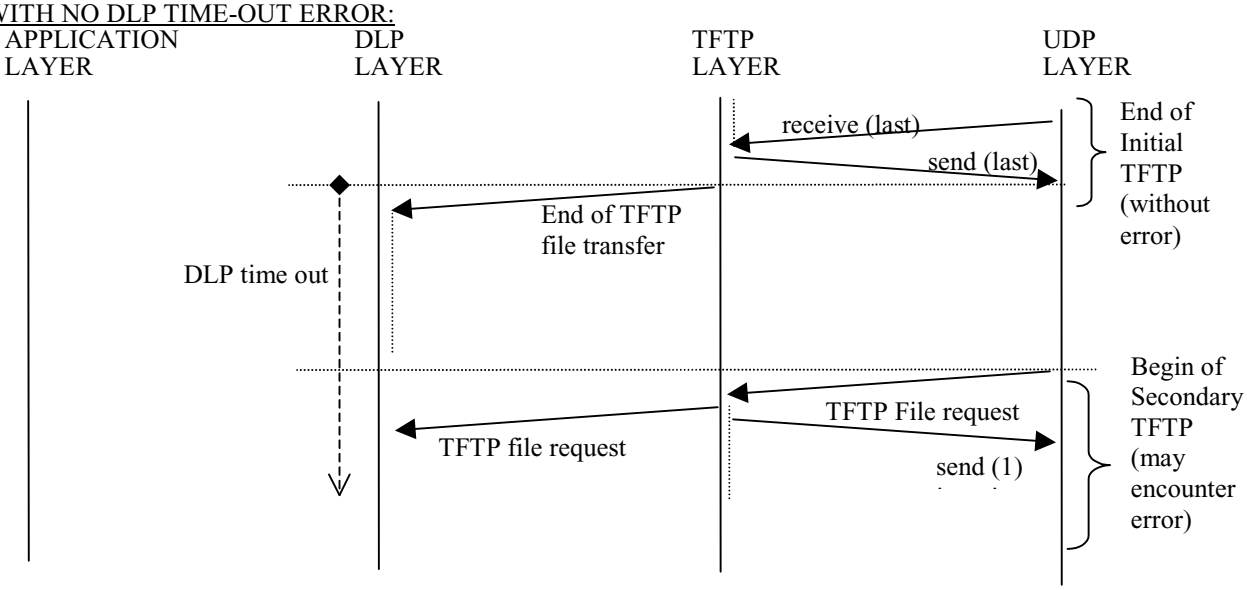
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ATTACHMENT 4
TIME-OUT AND RETRY NUMBER DEFINITION

FIGURE A4-5: DLP TIME-OUT

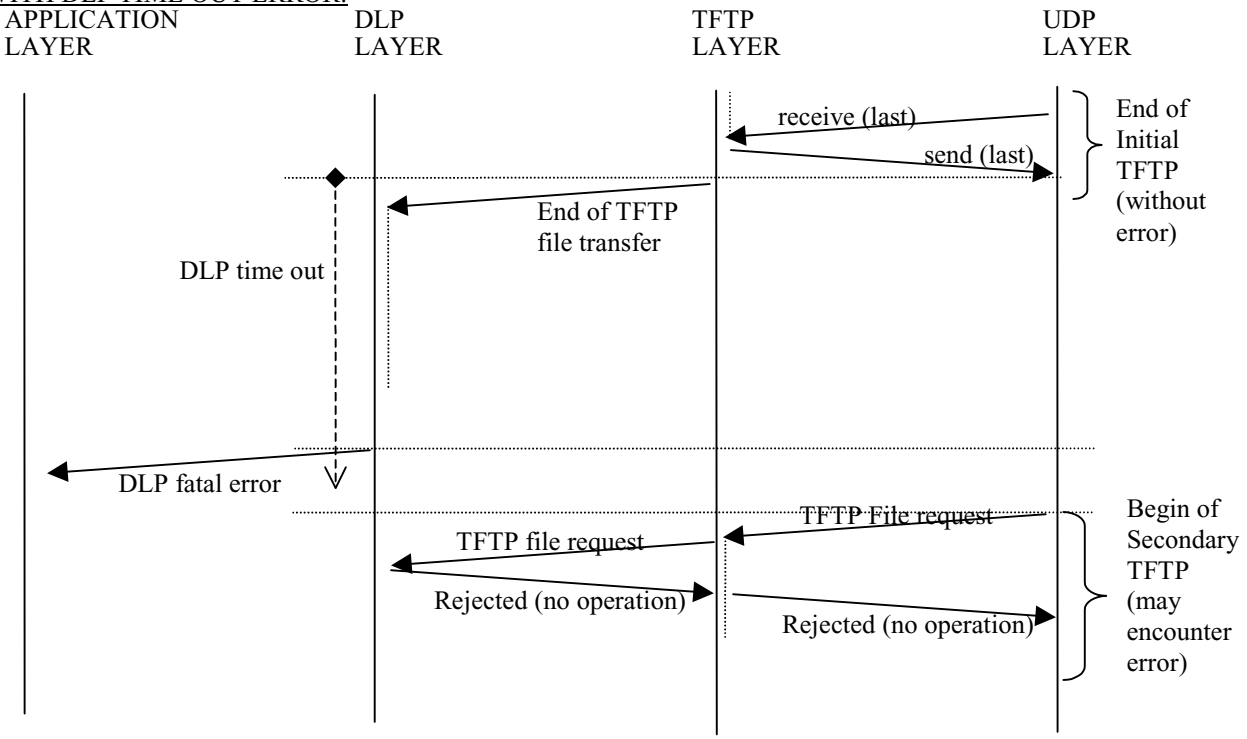
(ONLY ON DATA LOADER)

CASE WITH NO DLP TIME-OUT ERROR:



CASE WITH DLP TIME OUT ERROR:

c-2



COMMENTARY

A DLP Time-Out applies only to the peer of a TFTP file transfer [Initial TFTP, Secondary TFTP] referenced in table of Section 4.1 of Attachment 4. A DLP time out is classified as fatal error.

--`,,`--`,,`,`,,`,`---
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ATTACHMENT 5
ENVIRONMENTAL TEST CATEGORIES

The following example RTCA DO-160 categories apply to the environmental specifications of the data loader. The latest verison of RTCA DO-160 applies to new designs.

Environment	DO-160 Section	PDL Enclosure Open	PDL Inside Enclosure	ADL	ADL with Optical Media Device(Note 7) c-1
Temperature & Altitude	4	CAT B3		CAT A1	
Temperature Variation	5	CAT B		CAT C	
Humidity	6	CAT A	CAT B	CAT A	
Shock	7	YES	YES	YES	
Vibration (see Notes 1 & 2)	8	CAT X		CAT K	
Explosion	9	CAT X		CAT X	
Waterproofness (see Note 2)	10	CAT W	CAT R	CAT X	
Fluid Susceptibility	11	CAT X	CAT F	CAT X	
Sand and Dust	12	CAT X	CAT D	CAT X	
Fungus	13	CAT X	CAT F	CAT X	
Salt Spray	14	CAT X	CAT S	CAT X	
Magnetic Effect	15	CAT X		CAT A	
Power Input	16	CAT A		CAT A	
Conducted Voltage Transient	17	CAT A		CAT A	
Audio Frequency Conducted Susceptibility	18	CAT A		CAT A	
Induced Signal Susceptibility	19	CAT A		CAT A	
Audio Frequency Susceptibility (Radiated and Conducted)	20	CAT A		CAT A	
Spurious Radio Frequency Emission	21	CAT A		CAT A	

1. The use of alternative categories may be necessary if the installation is to be made in other than turbine-powered fixed-wing aircraft. Refer to RTCA DO-160 directly.

2. Allows the alternative already provided by DO-160 (sine wave or random).

3. Lightning protection should be provided. When considering lightning protection, the provisions of Section 22 of RTCA DO-160 should be used as a guide.

4. Rack-mounted and cockpit-mounted units should withstand spillages of liquids (beverages).

5. The values specified in this Report have precedence over the values specified in DO-160 where the two values differ. The references here are given to clarify the testing procedures needed.

COMMENTARY

Temperature and altitude category A1 is designed for the rack and cockpit mounted equipment because the installation should provide a temperature environment of less than +55ºC. However, the users feel a major cause of high maintenance costs for avionics equipment is operation on the ground when temperatures in excess of +55ºC may be encountered. The users, therefore, strongly urge the equipment designers to accommodate operation in ambient temperatures up to +71ºC for the purpose of achieving the highest possible operating life of the equipment. Regarding the decompression test, the equipment is required to operate after (but not during) the exposure to abnormally high altitudes.

6. In some cases, the criteria for the PDL are more stringent when the cover is in place for transport. Categories are given here only when they differ from the category applicable for the PDL with its protective cover removed. These assignments apply only to a non-operating condition.

7. Category specifications for the ADL with an optical media device, which may differ from the criteria applicable for the ADL, may emerge as hardware implementations of the software data loading protocol are developed. c-1

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APPENDIX A
HUMAN INTERFACE GUIDELINES

Manufacturers of software loader equipment, which is compliant with ARINC Report 615A - Part 1, should follow the human interface guidelines presented in this appendix. These guidelines have been developed from airline recommendations to mitigate current data loading problems reported from field operations. Adherence to these guidelines should provide a consistent, unambiguous and informative user interface for the software-loading task.

1. Physical Interface for Data Loading Devices

1.1 A display should be capable of providing the operator with written status and operating instructions. The display should be readable in daylight and in night conditions.

1.2 Buttons, cursor control, and all other manual control devices should be of appropriate size, shape and placement to allow an operator to operate easily the loader while wearing gloves in cold conditions.

1.3 Simple and secure power “ON/OFF” buttons should be provided, especially, on portable devices.

2. Operations and Procedures

2.1 Procedures should be menu driven to guide the operator through the necessary actions and give sufficient information on choices that must be made. Provided this menu-driven criterion could be met, differences in procedures for different loaders or for target systems with the same loader should be acceptable.

2.2 The software configuration loaded in the target system should be capable of being provided at any time.

2.3 Pre-requisite tasks should be clearly and unambiguously presented before starting the data transfer operation.

2.4 Positive confirmation should be required for all selections and button activation; for example, changing a cursor shape to an hourglass is a common and acceptable method.

2.5 Progress of the load operation should be provided continuously to the operator. If a failure occurs, the data loader should indicate the stage of the load operation. During the load-operation transfer-and-verification phases, status bars indicating the percentage of completion should be presented.

2.6 When the loader is performing functions that may take more than a few seconds to complete, an appropriate indication should be presented to the operator. When the task may take more than a minute, an indication of the expected or normal task duration should be provided also.

2.7 Designs should avoid situations in which the operator is presented with a blank or frozen screen for extended periods with no status indications.

2.8 When it is necessary to change the media to the next unit of a set, suitable prompts should be given. These instructions should identify the positive action to be taken, and not simply indicate that the current condition is incorrect. For example, “Wrong Disk Inserted” does not indicate that the normal operation requires insertion of the next disk in sequence.

2.9 Following a successful load, indication should be given that the load is complete and correct.

2.10 Simplified English should be used to accommodate operators who are not native English speakers.

3. Mass Storage Devices

For data loaders that contain mass storage devices, the operator should be able to view the contents of the mass memory, load new software parts, and delete unwanted software parts, without the need for the loader to be connected to the aircraft or to an Ethernet network. Display of the remaining storage capacity and any necessary hard drive maintenance utilities should be provided.

4. Error Conditions and Recovery

4.1 The data loader should indicate when a load is unsuccessful and display any corrective action or verification steps.

4.2 If the target device detects a failure during the load process, it should set the failure indication on the data loader and display failure information. Simply resetting and restarting should not be acceptable because it results in endless loops without an error indication.

4.3 The data loader should provide the operator with the ability to reinitialize a load in the event of an error. Be aware that often both the target system and the data loader must have power cycled from off to on before a load should be restarted.

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APPENDIX A
HUMAN INTERFACE GUIDELINES

4.4 The data loader should have the ability to verify via a standard cyclic redundancy check (CRC) that the loading media has valid data. Be aware that independent verification that the loading media is bad should eliminate the loader and the target system as the source of a load failure.

4.5 Indication of all error conditions should be provided in simple English.

4.6 Use of codes requiring special expertise or knowledge should be avoided.

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APPENDIX B
REFERENCE GUIDE

This Reference Guide lists the references in ARINC Report 615A, "Software Data Loader Using Ethernet Interfaces." The references are categorized by their importance to the Portable Data Loader (PDL) developer, Airborne Data Loader (ADL) developer, and Target HardWare (THW) developer. The following numbers identify the categories:

1 - Reference document required to implement ARINC Report 615A.
2 - Reference document with sufficient information in ARINC Report 615A.
3 - Reference document that provides additional information, but not required for implementation of ARINC Report 615A.

Reference Documents	PDL	ADL	THW
American Standard Code Information Interchange (ASCII)	2	2	2
ARINC Project Paper 664, Aircraft Data Network, Part 2 – Ethernet Physical and Data Link Layer Specification	1	1	1
ARINC Project Paper 664, Aircraft Data Network, Part 3 – Internet-based Protocols Suites and Services	1	1	1
ARINC Project Paper 664, Aircraft Data Network, Part 4 – Internet-based Address Structures and Assigned Numbers	1	1	1
ARINC Report 604, Guidance for Design and Use of Built-in Test Equipment (BITE)	2	2	2
ARINC Report 609, Design Guidance for Aircraft Electrical Power Systems	1	1	1
ARINC Report 615, Airborne Computer High Speed Data Loader	3	3	3
ARINC Report 624, Design Guidance for Onboard Maintenance Systems	2	2	2
ARINC Report 644A, Portable Maintenance Access Terminal (PMAT)	2	3	2
ARINC Report 646, Ethernet Local Area Network	3	3	3
ARINC Report 665, Loadable Software Standards	1	1	1
ARINC Specification 429, Mark 33 Digital Information Transfer System (DITS), Part 1, Functional Description, Electrical Interface, Label Assignments and Word Formats [Equipment ID for THW ID only] c-1	1	1	1
EUROCAE ED-12, Software Consideration in Airborne Systems and Equipment Certification	3	1	1
EUROCAE ED-14, Environmental Conditions and Test Procedures for Airborne Equipment	3	1	1
European Computer Manufacturer's Association (ECMA) Standard 267, 120 mm DVD-Read Only Disc	2	1	3
European Joint Aviation Authority (JAA) Regulatory Requirements	3	1	1
Federal Aviation Authority (FAA) Regulatory Requirements	3	1	1
Federal Communications Commission (FCC) Regulatory Requirements	3	1	1
Internet Assigned Numbers Authority (IANA)	2	2	2
MS25212, ADL Size	3	1	3
MS27473T-18A-53P, Airborne Data Loader interface connector aircraft/airborne computer end	2	1	1
MS27473T-18A-53S, Data Loader interface termination at Data Loader End	1	1	3
MS27473T-18B-53P, Airborne Data Loader interface connector aircraft/airborne computer end	2	1	1
MS27508E-18A-53P, PDL and ADL Connector	1	1	2
PC Card Slot Standard	2	1	3
PC Card Standard	2	1	3
Phillips - Sony Orange Book recordable compact disc physical format	2	1	3
Phillips - Sony Yellow Book compact disc physical format	2	1	3
Request for Comment (RFC) 1350, TFTP Trivial File transfer Protocol	2	2	2
RFC 1123, Requirements for Internet Hosts Application and Support	2	2	2
RFC 1785, TFTP Option Negotiation Analysis	2	2	2
RFC 2347, TFTP Option Extension	2	2	2
RFC 2348, TFTP Blocksize Option	2	2	2
RFC 2349, TFTP Timeout Interval and Transfer Size Option	2	2	2
RFC 768, UDP User datagram Protocol	2	2	2
RFC 791, IP Internet Protocol	2	2	2
RS232	1	1	1
RTCA DO-160, Environmental Conditions and Procedures for Airborne Equipment	3	1	1
RTCA DO-178, Software Considerations in Airborne Systems and Equipment Certification	3	1	1
Simple Network Identification Protocol (SNIP) - (See ARINC Report 615A, Attachment 3)	1	1	1

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SUPPLEMENT 1
TO
ARINC REPORT 615A©
SOFTWARE DATA LOADER USING
ETHERNET INTERFACES,
PHYSICAL STANDARDS AND PROTOCOLS

Published: January 12, 2001

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: November 14, 2000

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A. PURPOSE OF THIS DOCUMENT

This Supplement introduces various changes and additions to ARINC Report 615A. This includes the addition of a third media type of an optical disc drive capable of reading Compact Disc – Read Only Memory (ROM) and Digital Video Disc – ROM for loading data onto airliners.

B. ORGANIZATION OF THIS SUPPLEMENT

Changes introduced by Supplement 1 are too extensive to make the integration of replacement pages from a separate supplement into Specification 615A practical for our readers.

The changes introduced by Supplement 1 have been identified using change bars and are labeled in the margin by a "c-1" indicator.

C. CHANGES TO ARINC REPORT 615A INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Report introduced by this Supplement. Each change or addition is identified either by the section number and title currently employed in the Report or by the section number and title that is employed in the supplement. In each case there is a brief description of the addition or change. A replacement white page document is included in the second part of this document, as noted in B above. In this way an accurate record of the development of the Report is preserved.

Title: Initially ARINC Report 615A, Software Data Loader Using Ethernet Interfaces, was developed in two parts: Part 1 – Physical Standards and Protocols, and Part 2 – Loadable Software Standards. Supplement 1 of ARINC Report 615A, Part 1 is published with the title: Software Data Loader Using Ethernet Interfaces. Supplement 1 of ARINC Report 615A, Part 2 is published as Supplement 1 of ARINC Report 665, Loadable Software Standards.

1.3 Function of the Equipment

For clarity, a sentence was added to state that the protocol file names are case sensitive and to provide a reference.

1.6 Interoperability

A reference was changed to ARINC Report 665, "Loadable Software Standards."

1.10 Related Documents

An ECMA document was added for reference.

2.2.6 Data Bus Interface

For clarity, a COMMENTARY was added to state the data loader capability for 10BaseT connection.

4.1 Media Interchangeability

A third media type of "Optical Disc" was added to the list of specified media types. The statement that a mix of any of the media types for a single data load was not supported by the data loader was expanded to include the third media type. Cautionary statements were added to the COMMENTARY to emphasize this point. A reference was changed to ARINC Report 665.

References were revised to "disc" instead of "disk" for consistency.

4.1.2 PC Card Device

The definition of "ATA" was deleted to correct an error. A reference to the PC Card Slot standard was added for accuracy.

An added sentence stated that the use of PC Card for uploads is not supported by this standard.

An added COMMENTARY stated that airlines do not want PC Cards to be used for uploads. However, non-commercial operators could perform certain optional functions that are not defined in ARINC Report 615A or ARINC Report 665.

4.1.3 Optical Medium

This section and its subordinate sections were added to specify the storage media for the optical medium and define various methods of recording and their capacities.

5.0 Load Protocol

Section 5.0 has been revised in its entirety and reproduce for clarity and readability. These changes include:

5.1 Introduction

A sentence was added to clarify protocol file name format. A reference was changed to ARINC Report 665.

5.2 Scope

This section was modified to add scope to the Load Protocol specification to obtain scriber information using the Simple Network Identification Protocol (SNIP).

5.3.1 General Description

This section was modified to add an explanation of the role of the SNIP and a reference to Attachment 3, which defines SNIP for ARINC Report 615A. Also, for User Datagram Protocol (UDP), Internet Protocol (IP) and Medium Access Control (MAC) / Physical Layer, references were added for definition of these protocols and layers to ARINC Project Paper 664, Parts 3 and 2.

An added COMMENTARY stated that ARINC Report 615A does not specify MAC or IP addresses and referenced where address rules were defined.

5.3.2 TFTP Protocol

This section was modified to define maximum file size and block numbering.

5.3.2.1 TFTP Reference Documents

"TFTP" was added to the title of this section for clarity.

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5.3.2.2 TFTP Options

This section was modified to include a statement about the status of UDP and IP data after the last TFTP option.

5.3.2.2.1 Block Size

This section was changed to correct errors in size descriptions.

5.3.2.2.2 Transfer Size of the File

This section was changed for clarity.

5.3.2.2.3 Timeout Interval

This section was changed for clarity.

5.3.2.3 TFTP Adaptation to Data Loading Protocol

The title of this section was changed for clarity.

5.3.2.3.1 TFTP Super–set

This section was added for clarity.

5.3.2.3.2 Dedicated Port Number

This section was added for clarity.

5.3.2.3.3 Error Message General Description

The title of this section was changed and the section was re-numbered. A sentence was added to clarify the use of the error code message field.

5.3.2.3.4 Definition of the Waiting Message

A maximum value for wait delay was defined.

5.3.2.3.6 Definition of the Abort Message

This section was changed for clarity to complete the definition of the ABORT error message.

A statement was added to clarify the maximum value for the wait delay.

5.3.2.3.8 Blocksize Option Implementation

This section was added for clarity.

5.3.2.3.9 TFTP Block Number

This section was added for completeness.

5.3.2.3.10 File Transfer Size

This section was added for clarity.

5.3.2.3.11 Timeout Interval

This section was added for clarity.

5.4 Operations

The term “LRU” for Line Replacement Unit was changed to “THW” for Target HardWare to avoid ambiguity. This resulted in revisions from <LRU_ID_POS >.LNR to <THW_ID_POS >.LNR in many places within section 5.4.

6.0 Load Protocol Definition

Section 6.0 has been revised in its entirety. These changes include:

“LRU_ID” was changed to “THW_ID” to avoid ambiguity.

New figures and explanations for the message sequence charts were added.

“Time Out” feature and the new option of “Exception timer” was added.

Added a definition of an initial value for “Counter.”

Added “Big Endian” byte ordering for all protocol files.

Deleted “initial” from all prior references to “initial value.”

Added a COMMENTARY on Airborne Data Loader displays.

ATTACHMENT 3 – SIMPLE NETWORK IDENTIFICATION PROTOCOL (SNIP)

This Attachment was changed for clarity and completeness in its entirety. Changes include:

Added a statement about MAC address and IP address.

Amplified a COMMENTARY on SNIP.

Added Manufacturer code to parameter list.

Added an Exchange Example.

Added a COMMENTARY on manufacturer identification.

Deleted two COMMENTARIES and an example on MIB syntax, which are no longer required.

Changed a reference to ARINC Report 665 for accuracy.

ATTACHMENT 4 - TIME-OUT AND RETRY NUMBER DEFINITION

This attachment was changed as follows:

Added a sentence in Section 3.2, “Retry Number,” to clarify the definition of retry number.

Added “Section 3.3 Sorcerer’s Apprentice Syndrome” to eliminate this problem for retry numbers.

ATTACHMENT 5 – ENVIRONMENTAL TEST CATEGORIES

A column and note were added to the table for “ADL with Optical Media Device.”

APPENDIX B – REFERENCE GUIDE

This entire appendix was added to guide implementers through the references cited in the ARINC Report 615A.

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SUPPLEMENT 2
TO
ARINC REPORT 615A©
SOFTWARE DATA LOADER USING
ETHERNET INTERFACES,
PHYSICAL STANDARDS AND PROTOCOLS

Published: May 10, 2002

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: April 8, 2002

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A. PURPOSE OF THIS SUPPLEMENT

This Supplement introduces various changes and additions to ARINC Report 615A. The revisions include clarification of the following: length of the IP layer, operations that should be performed for a successful up-load operation, configuration information to include fields for the number of target hardware and associated parameters, definition of the value for packet terminator for the target hardware answer table, and a new equation for calculation of Data Load Protocol (DLP) timeout.

B. ORGANIZATION OF THIS SUPPLEMENT

The first part of this document, printed on goldenrod-colored paper, is the Supplement itself. It contains descriptions of the changes introduced into the Report and, where appropriate, extracts from the original text for comparison purposes. The second part consists of the Report, modified as required by the Supplement.

The changes introduced by Supplement 2 have been identified using change bars and are labeled in the margin by a "c-2" indicator.

C. CHANGES TO ARINC REPORT 615A, INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Report introduced by this Supplement. Each change or addition is identified either by the section number and title currently employed in the Report or by the section number and title that will be employed when the Supplement is eventually incorporated. In each case there is a brief description of the addition or change. Sections that are modified by this Supplement are reproduced. In this way an accurate record of the development of the report is preserved.

3.2.2 Selector Controls

"Target Hardware Identification" is an added function desired of all data loaders. This function should be implemented by Find Identification of Network Devices (FIND) and that reference should be made to Attachment 3.

4.1.2.1 PC Card Interface

For clarity, the first paragraph is replaced with an explanation of "AT Attachment (ATA)."

5.3.1 General Description

To avoid potential conflict with another avionics protocol with the acronym SNIP, the SNIP protocol used in ARINC Report 615 A is renamed the Find Identification of Network Devices (FIND) protocol.

5.3.2.2 TFTP Options

The term "no-option implementation" is changed to "non-implemented option" for clarity.

5.3.2.2.1 Block Size

In this section, the value for the length of the Internet Protocol (IP) layer is corrected from "64" to "60" bytes.

5.3.2.2.3 Timeout Interval

The term "PDU" is revised to "packet" for accuracy.

5.3.2.3.1 TFTP Super-set

Trivial File Transfer Protocol, (TFTP), super-set adapts the standard TFTP, which is defined by the communication industry, to the specific environment of aircraft data loading. Additional text in this section clarifies the need for the TFTP super-set.

5.3.2.3.3 Error Message General Description

The explanation of Error Code Value 3 is revised for clarity.

5.3.2.3.8 TFTP Option Implementation

This section and its three subordinate sections:

5.3.2.3.8.1 Blocksize Option Implementation
5.3.2.3.8.2 File Transfer Size
5.3.2.3.8.3 Timeout Interval,

provide a new definition for a minimum implementation for equipment to claim conformance to this specification.

5.3.2.3.9 TFTP Block Number

For accuracy and clarity, the last sentence is changed to "If overflow occurs due to the number of transmitted blocks, the next value after 65535 should be 1."

5.3.2.3.10 File Transfer Size

This section is deleted and the material is reorganized as Section 5.3.2.3.8.2. For accuracy, the verb "be" is inserted in "should not be used" and the reason for not using this feature is added.

5.3.2.3.11 Timeout Interval

This section is deleted and the material is reorganized as Section 5.3.2.3.8.3. For accuracy, the clause "since timeout is defined in ARINC 615A, Attachment 4…" is deleted and the explanation of Timeout Interval is expanded.

5.4.1 Introduction

For accuracy, the acronym "SNIP" is changed to "FIND."

In this section, the verb is changed from "should" to "may" to change from a recommended FIND request to an optional FIND request before each operation in the load protocol. This provides more flexibility to data loaders.

5.4.5.3 Transfer Interruption

This section clarifies the ability of data loader and target hardware to interrupt the transfer phase during certain operations.

6.2.17.2 Effect on Receipt

The Status Code word is revised to $1004_{16.}$

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6.3.2 Information Operation: Initization and Transfer Step

Clarification is added for recommended actions after the transfer step begins. A statement is added to clarify when a status file may be sent during this operation.

6.3.3 Uploading Operation: Initialization, List Transfer and Transfer

In this section at the end of the explanation of diagrams, clarification on which operations should be accomplished for a successful up-loading operation and an explanation of the exception timer field are added.

6.3.4 Downloading Operation in Media Defined Mode: Initialization, List Transfer and Transfer

In this section at the end of the explanation of diagrams, clarification on which operations should be accomplished for a successful down-loading operation is added.

6.3.5 Downloading Operation in Operator Defined Mode: Initialization, List Transfer and Transfer

In this section at the end of the explanation of diagrams, clarification is provided on which operations should be accomplished for a successful down-loading operation is added.

6.3.6 Interrupt Service

In this section at the end of the explanation of diagrams, clarification is provided on the role of the data loader during an interruption of service.

6.4 Protocol File Description

The "THW_ID system" is clarified.

For accuracy, added text specifies when the Protocol Version Value should be set and used.

An added statement clarifies that all textural fields should be zero terminated.

An added commentary provides guidance on the Description Length field and the Description field.

6.4.1 <THW_ID_POS>.LCI,.LUI,.LND and .LNO

For clarity in the Protocol Version field description, a reference is added to the Protocol Version described in Section 6.4, Protocol File Description.

The maximum length of the text of 255 characters (i.e., 2040 bits) is clarified. The allowed length of 80 characters is clarified, and text termination is specified.

In the commentary, clarity is provided by expanding the explanation of the role of the Status Description Length field parameter.

6.4.2 <THW_ID_POS>.LCL

Clarity is provided in the table of configuration information to include fields for number of Target Hardware and for associated parameters.

For clarity in the Protocol Version field description, a reference is added to the Protocol Version described in Section 6.4, Protocol File Description.

For consistency, the max length of the text of 255 characters (i.e., 2040 bits) is clarified, the allowed length of 80 characters is clarified, and text termination is specified. Detailed descriptions of all fields are provided for completeness.

In the commentaries, clarity is provided by expanding the explanation of the role of the THW Code Length field parameter and the Part Designation Length field parameter.

6.4.3 <THW_ID_POS>.LCS

For clarity in the Protocol Version field description, a reference is added to the Protocol Version described in Section 6.4, Protocol File Description.

For accuracy, Status Codes 0004_{16} and 0005_{16} are changed to 1004_{16} and 1005_{16}, respectively.

For consistency, the max length of the text of 255 characters (i.e., 2040 bits) is clarified, the allowed length of 80 characters is clarified, and text termination is specified.

For accuracy, the definition of Estimated Time is expanded.

In the commentary, clarity is provided by expanding the explanation of the role of the Status Description Length field parameter. In the first paragraph, the text is revised for clarity and a reference is added to Section 6.4.10 for more information.

6.4.4 <THW_ID_POS>.LCR

For clarity in the Protocol Version field description, a reference is added to the Protocol Version described in Section 6.4, Protocol File Description.

For consistency, the max length of the text of 255 characters (i.e., 2040 bits) is clarified, the allowed length of 80 characters is clarified, and text termination is specified.

In the commentaries, clarity is provided by expanding the explanation of the role of the Header File Name Length field and the Load Part Number Length field parameters.

The field Load Part Number Name in the table of configuration information is harmonized with the Load Part Number Name description.

6.4.5 <THW_ID_POS>.LUS

For clarity in the Protocol Version field description, a reference is added to the Protocol Version described in Section 6.4, Protocol File Description.

For consistency, the max length of the text of 255 characters (i.e., 2040 bits) is clarified, the allowed length of 80 characters is clarified, and text termination is specified.

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In the commentaries, clarity is provided by expanding the explanation of the role of the Upload Status Description Length, Header File Name Length, Load Part Number Length, and the Status Description Length field parameters.

For clarity, a definition of Estimated Time is provided.

In the first paragraph, the text is revised for clarity and a reference is added to Section 6.4.10 for more information.

6.4.6 <THW_ID_POS>.LNR

For clarity in the Protocol Version field description, a reference is added to the Protocol Version described in Section 6.4, Protocol File Description.

For consistency, the max length of the text of 255 characters (i.e., 2040 bits) is clarified, the allowed length of 80 characters is clarified, and text termination is specified.

In the Commentary, clarity is provided by expanding the explanation of the role of the File Name Length field parameter.

6.4.7 <THW_ID_POS>.LNS

For clarity in the Protocol Version field description, a reference is added to the Protocol Version described in Section 6.4, Protocol File Description.

The max length of the text of 255 characters (i.e., 2040 bits) is clarified, the allowed length of 80 characters is clarified, and text termination is specified.

In the Commentary, clarity is provided by expanding the explanation of the role of the Download Status Description Length, File Name Length, and File Status Description Length field parameter.

In the first paragraph, the text is revised for clarity and a reference is added to Section 6.4.10 for more information.

6.4.8 <THW_ID_POS>.LNL

For clarity in the Protocol Version field description, a reference is added to the Protocol Version described in Section 6.4, Protocol File Description.

In this section for consistency, the max length of the text of 255 characters (i.e., 2040 bits) is clarified, the allowed length of 80 characters is clarified, and text termination is specified.

In the commentaries, clarity is provided by expanding the explanation of the role of the File Name Length and the Part Description Length field parameter.

6.4.9 <THW_ID_POS>.LNA

For clarity in the Protocol Version field description, a reference is added to the Protocol Version described in Section 6.4, Protocol File Description.

The max length of the text of 255 characters (i.e., 2040 bits) is clarified, the allowed length of 80 characters is clarified, and text termination is specified.

In the commentaries, clarity is provided by expanding the explanation of the role of the File Name Length field parameter.

6.4.10 Data Load Status Code Table

A reference to Table 6.4.10 is added for clarity.

In Table 6.4.10 the following changes are made:

The "Operation aborted by Data Loader" message is redefined as Status Code 1004_{16}.

The "Operation cancelled by operator" message is redefined as Status Code 1005_{16}.

For Status Code 1007_{16} under "Meaning" the term "Or Download File failed" is added for accuracy.

For accuracy, the acronym "LRU" is replaced by "THW."

ATTACHMENT 1 – ARINC 615A ADL ANDPDL CONNECTOR SIGNAL ASSIGNMENTS

For accuracy, references are revised to ARINC Specification 664, Aircraft Data Network from "ARINC 646."

ATTACHMENT 3 - FIND IDENTIFICATION OF NETWORK DEVICES (FIND) PROTOCOL

The title of this attachment and the name of the protocol itself are changed from "Simple Network Identification Protocol" to "Find Identification of Network Devices (FIND)" Protocol," to avoid a potential conflict with an avionics company that is producing an aviation product with a communication protocol whose acronym is "SNIP." The Acronym SNIP is replaced by FIND throughout the attachment.

1.0 Introduction

Adding guidance as to when to perform a FIND request provides clarity. For clarity, an added commentary provides guidance on the minimum implementation of a FIND request and on good practice when target hardware is changed.

2.0 Overview

In the second bullet, the phrase: "MIB compatible structure" is deleted for clarity.

An added Commentary explains the difference between the Data Loader Time Out and the Target Hardware Time Out.

3.0 FIND Packets

The acronym SNIP is replaced by FIND.

4.0 Information Request Packet

The termination for Data List is revised.

5.0 Information Answer Packet

The termination for Data List is revised.

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7.1 Examples

For accuracy in the target hardware answer table, the value for Packet terminator (ASCII EOF) is revised to "10_{16}."

ATTACHMENT 4 -TIMEOUT AND RETRY NUMBER DEFINITION

For clarity and accuracy, this attachment is revised with an explanation of the composition of the ARINC 615A protocol in two layers, their operations, and their interactions. This attachment includes a revised equation and figure for Data Load Protocol (DLP) Time-out and three new figures that illustrate operations of time-out and retry numbers.

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